

MAGNETOM

Replacement of Parts System

Gradient

Harmony syngo MR
Symphony syngo MR
Sonata syngo MR
Trio
Upgrades to syngo MR:
from Impact/Expert
from Vision

Document Version

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1.1 General

The following describes replacement of the "Field Replaceable Units" (FRUs) in the Gradient System K2209 (GPA), K2217 (KAS) and the Gradient Coil components of the MR systems Harmony/Symphony/Quantum/Sonata.

This document also provides references for the required checks and adjustments.



Protective earth connection test.

If a protective earth connection has to be disconnected for replacing a component, measure the protective earth connection resistance $\leq 100 \text{ m}\Omega$ with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the "sub-component" (e.g. front cover) and the protective earth wire connected to the "main component" (e.g. GPA frame).

Document the successful measurement in the service report (measured value and measured equipment) as required by IEC 62353.

1.2 GPA K2209

Tab. 1 GPA K2209

GPA K2209					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
GSSU/ D1 DAC (4768003)	20 min	- Check jumper/switch position - Check offset Run TuneUp: - Phantom Shim - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Phantom Shim Check - Calc Artefacts	(→ DAC D1 / Page 22)	Field service grounding kit (ESD) Set of standard tools Digital voltmeter
GSSU/ D2 Regulator (The new D2 is required for the current sensors type IT600S) (4768029 old, 4769084 new)	20 min	- Check jumper/switch position - Check SENS offset Run TuneUp: - Regulator Adjustment - Phantom Shim - Grad Delay - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check - Phantom Shim Check - CTC - ECC - Grad Delay Check - Calc Artefacts	(→ Regulator D2 / Page 24)	Field service grounding kit (ESD) Set of standard tools Digital voltmeter
GSSU/ D3 Synchronization (4768045)	20 min	Check jumper setting	Perform Quality Assurance: Coil Check (SN DIP)	(→ Synchronization D3 / Page 25)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D4 Modulator (4768060)	20 min	Check jumper setting	Perform Quality Assurance: Coil Check (SN DIP)	(→ Modulator D4 / Page 26)	Field service grounding kit (ESD) Set of standard tools

GPA K2209					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
GSSU/ D5 Turbo Modulator (4768086)	20 min	Check jumper setting	Perform Quality Assurance: Coil Check (SN DIP)	(→ Turbo Modulator D5 / Page 26)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D6 Service/CAN (4768102)	20 min	Check jumper setting	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check	(→ Service/CAN D6 / Page 27)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D7 FOC Interface (4768128)	20 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	(→ FOC D7 / Page 27)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D8 Backplane (4768144)	40 min	No action necessary	QA: SN DIP	n.a.	Field service grounding kit (ESD) Set of standard tools
Driver Supply D10 (4769050)	20 min	Check jumper setting	Perform Quality Assurance: Coil Check (SN DIP)	n.a.	Field service grounding kit (ESD) Set of standard tools Digital voltmeter
GSSU/ Small signal Power Supply E4A (3859482)	20 min	Check/Adjustment of output voltages	Perform Quality Assurance: Coil Check (SN DIP)	(→ Small Signal Power Supply E4A / Page 29)	Set of standard tools Digital voltmeter
D9 Supervision (47 69 126) (installed in the patient table control K2202)	20 min	Check jumper/switch position	Perform Quality Assurance: Coil Check (SN DIP)	(→ Supervision D9 / Page 30)	Field service grounding kit (ESD) Set of standard tools

GPA K2209					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Resonant Converter (RC/MK2) (11 41 097) replaces old RC/MK1 (47 69 605) upgrade kit to RC/MK2 (57 26 158) if required	120 min	Measurement/Adjustment of output voltage	Run the duty cycle sequence	(→ Resonant Converter / Page 32)	Set of standard tools Forklift or equivalent lifting device to position the resonant converter at the correct installation height of approx. 16 cm (if the rollers of the cabinet are installed). Torx driver bit CR-VT20 Torx driver bit TX 25
Power Stages (PS) (47 69 258) or (59 39 447) (Important for Ultra Gradients: Do NOT install different types of final stages within one axis!)	60 min	Run TuneUp: - Regulator Adjustment - Phantom Shim - Grad Delay - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check - Phantom Shim Check - Grad Delay Check - Calc Artefacts - Image Orientation	(→ Power Stages / Page 44)	Set of standard tools

GPA K2209					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Current Sensors Type IT600S with LA205-S (11 41 311) This sensor type requires the modified regulator board D2 (47 69 084)	<180 min	- Check SENS offset adjustment on D2, refer to D2 replacement procedure. Run TuneUp: - Regulator Adjustment - Phantom Shim - Grad Delay - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check - Phantom Shim Check - CTC - ECC - Grad Delay Check - Calc Artefacts - Image Orientation	(→ Current Sensor Assembly / Page 53)	Set of standard tools Socket Allen for M10 (5,6 mm) Allen screws Driver bit Torx X10
Chokes (3058278)	always requires replacement of the Power Stage(s) of the affected axis				

1.3 GPA K2217

Tab. 2 GPA K2217

GPA K2217					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
GSSU/ D112 DAC (73 69 973) replaces D111 (59 41 534) for GPA Light and D11 (57 25 507)	20 min	- Check jumper/switch position - Check offset Run TuneUp: - Phantom Shim - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Phantom Shim Check - Calc Artefacts	(→ DAC D112/111/11 / Page 65)	Field service grounding kit (ESD) Set of standard tools Digital voltmeter
GSSU/ D12 Regulator (57 26 349) D123 (59 41 617) for Allegra D124 (73 69 767) for GPA Light	20 min	- Check jumper/switch position - Check SENS offset Run TuneUp: - Phantom Shim - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check - Phantom Shim Check - CTC - ECC - Grad Delay Check - Calc Artefacts	(→ Regulator D12/123/124 / Page 67)	Field service grounding kit (ESD) Set of standard tools Digital voltmeter

GPA K2217					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
GSSU/ D14 Modulator (57 26 364) D142 (59 39 173) for GPA Light	20 min	Check jumper setting	Perform Quality Assurance: Coil Check (SN DIP)	(→ Modulator D14/142 / Page 68)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D16 CAN-Service (57 25 515)	20 min	Check jumper setting	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check	(→ CAN-Service D16 / Page 69)	Field service grounding kit (ESD) Set of standard tools
GSSU/ D17 I/O (57 26 430)	20 min	Check jumper/switch position	Perform Quality Assurance: Coil Check (SN DIP)	(→ I/O D17 / Page 69)	Field service grounding kit (ESD) Set of standard tools
GSSU/ Assembly Carrier (73 69 874) including backplane but without GSSU boards	40 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	n.a.	
GSSU/ Small signal Power Supply E4A (30 39 124)	20 min	Check/Adjustment of output voltages	Perform Quality Assurance: Coil Check (SN DIP)	(→ Small Signal Power Supply E4A / Page 71)	Set of standard tools Digital voltmeter
D9 Supervision (47 69 126) (installed in the patient table control K2202)	20 min	Check jumper/switch position	Perform Quality Assurance: Coil Check (SN DIP)	(→ Supervision D9 / Page 72)	Field service grounding kit (ESD) Set of standard tools

GPA K2217					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Soft Start Unit D31 (57 26 414)	20 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	(→ Power Stage Supply Components / Page 74)	Set of standard tools
Rectifier D32 (57 27 016)	10 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	(→ Power Stage Supply Components / Page 74)	Set of standard tools
Line Filter of GPA K2217 (57 25 671)	60 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	(→ Power Stage Supply Components / Page 74)	Set of standard tools
Main Contactor K2 (46 73 653)	60 min	No action necessary	Perform Quality Assurance: Coil Check (SN DIP)	(→ Power Stage Supply Components / Page 74)	Set of standard tools Torx 25 driver bit
Power Stage (59 38 142) Power Stage (59 38 720) for GPA Light	60 min	Run TuneUp: - Regulator Adjustment - Phantom Shim - Grad Delay - Grad Sens	Perform Quality Assurance: - Coil Check (SN DIP) - Regulator Check - Phantom Shim Check - Grad Delay Check - Calc Artefacts - Image Orientation	(→ Power Stages / Page 86)	Set of standard tools

GPA K2217					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Current Sensors (57 25 655) old material number (10743890) new material number	345 min	- Check SENS offset adjustment on D12, refer to D12 replacement procedure. Run TuneUp: - Regulator Adjust - Table Adjustment - Phantom Shim - CTC - ECC - Grad Delay - Grad Sens	Perform Quality Assurance: - Image Orientation - Regulator Check - Phantom Shim Check - CTC Check - ECC Check - Gradient Delay Check - Grad Sens Check - Calc Artifacts - Spike Check - Stability Check - FatSat Check	(→ Current Sensor Assembly / Page 93)	Set of standard tools Torque wrench 20-50 Nm Special tools for gradient connection (material number 10563214) Digital voltmeter
Chokes (57 25 887)	120 min	No action necessary	Perform Quality Assurance: Image Orientation	(→ Chokes (LF-Filters) / Page 107)	Set of standard tools
Fan (30 63 476)	40 min	No action necessary	Check fan function Perform Quality Assurance: - Coil Check (SN DIP) - Image Orientation	(→ Fans / Page 110)	Set of standard tools

GPA K2217					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Water Distributor (57 25 960)	20 min	No action necessary	- Check tightness of cooling water connections and search for leaks Perform Quality Assurance: - Coil Check (SN DIP)	(→ GPA Water Distributor / Page 112)	Set of standard tools
Cabling	10 min	No action necessary	Perform Quality Assurance: - Coil Check (SN DIP) - Image Orientation	N/A	Set of standard tools

1.4 Gradient Coil Components

Tab. 3 Gradient Coil Components

Gradient Coil Components					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Gradient coil cooling repair kit (100 48 056)	4-5 hr	n.a.	- Visually inspect for water leakages Perform Quality Assurance: - Coil Check (SN DIP) - Phantom Shim Check - Spike Check	Refer to (→ Gradient Coil Water Manifolds / Page 114).	- Non-magnetic tools - Water pail (10 ltrs.) - Plenty of paper towels - Distilled water, at least 10 ltrs. - Alcohol or similar cleaning agent (125 ml) - Water-soluble lubricant (purchase locally, e.g. drugstores)
Gradient cable kit Sonata (11 41 923)	4-5 hr	All connections must be tightened with the proper torque, using the listed non-magnetic torque wrenches	- Double check and ensure that all connections are tight Perform Quality Assurance: - Coil Check (SN DIP) - Spike Check - Image Orientation	Refer to (→ Gradient Cables (Harmony, Sonata) / Page 131).	- Non-magnetic standard tool set (7757706) - Non-magnetic torque wrench ranging from 5 - 20 Nm (8395829) - Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803) - Non-magnetic socket assortment (8396116)

Gradient Coil Components					
FRU / Material number	Time	Adjustments	Function test	Instructions	Tools
Gradient cable kit Vision Upgrades (73 65 815)	4-5 hr	All connections must be tightened with the proper torque, using the listed non-magnetic torque wrenches	- Double check and ensure that all connections are tight Perform Quality Assurance: - Coil Check (SN DIP) - Spike Check - Image Orientation	Refer to (→ Gradient Coil Connection Plate (Vision Upgrades) / Page 144).	- Non-magnetic standard tool set (7757706) - Non-magnetic torque wrench ranging from 5 - 20 Nm (8395829) - Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803) - Non-magnetic socket assortment (8396116)
Gradient cable kit Trio (73 67 910)	3-4 hr	All connections must be tightened with the proper torque, using the listed non-magnetic torque wrenches	- Double check and ensure that all connections are tight Perform Quality Assurance: - Coil Check (SN DIP) - Spike Check - Image Orientation	Refer to (→ Gradient Cables (Trio) / Page 153).	- Non-magnetic standard tool set (7757706) - Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803) - Non-magnetic socket assortment (8396116)

**WARNING**

Troubleshooting and repairs, which involve Current sensors, gradient filters, gradient coil, gradient cables might lead to problems with the image orientation of the system.

The effect could be a wrong diagnosis.

- » Perform the "Image Orientation Check" after all repair procedures when you have to disconnect or open the gradient coil power circuit, e.g. replacement of:
- » Current Sensors
- » Gradient Filters
- » Gradient Coil
- » Gradient Cables

**WARNING**

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch off the gradient cabinet
- » Old gradient cabinet K2209: The circuit breakers F9/3*50A and F14 in the CCA must be switched off.
- » New gradient cabinet K2217: The cutter switch F3/3*80A in the CCA mains box or mains box 2 and F14/CCA must be switched off.

**WARNING**

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.



Never unplug the quick connectors of a water-cooled component when the water pump in the RCA is in operation (because of high impact pressure).

Switch off the water pump via the circuit breaker F1 in the RCA cabinet.



Before returning the replaced Power Stage, drain the water from the pipes by opening both ends.



Before returning the replaced rectifier board D32, drain the water from the pipes by opening both ends.



For replacing GPA components, perform the adjustments and checks as stated in the following descriptions:

2.1 GSSU Components

2.1.1 Safety Information

- Switch-off the circuit breakers F9 (3*50A) first, then F14 (4A) in the CCA.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the following procedures.

2.1.2 Required Tools

- Grounding kit (ESD)
- Set of standard tools
- Digital voltmeter

2.1.3 Common Procedure for all GSSU Components

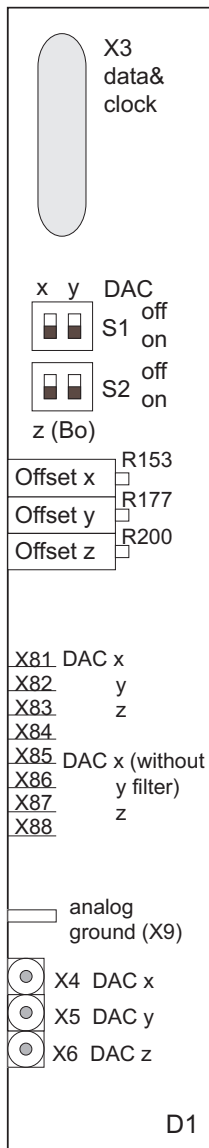


For the exchange of boards of the small signal unit, refer to the section of ESD rules included in (→ General Safety Notes / TD00-000.860.01)

- Switch-off the circuit breakers F9 (3*50A/GPA 2209) first, then F14 in the CCA.
- Remove the signal lines. The front cover plate of the small signal unit (GSSU/K2209) can be folded down by removing the upper two fixing screws. Replace the defective board after checking the existing jumpers on the new board. As an aid to pull the boards you should insert a cable tie or a wire forming a loop through holes at the front edge of the board.
If you are also replacing Power Supply E4A or board D7, remove the protective cover.
- Reconnect all connectors (FOC's, signal/power lines).
- Switch-on the fuses, reset the GPA (System/Reload All) and perform the required adjustments.
- Establish the original configuration.
- Ensure the E4A fan is functioning properly.

2.1.4 DAC D1

Fig. 1: Front of DAC D1



Descriptions:

X3:

18 bit parallel gradient data and 4MHz clock signal from MMC

S1/S2:

The DAC output can be switched on/off for every axis; S2/B₀ is not used

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V) 10 V corresponds to 300 A

Check Jumpers and Switch Positions:

- Jumper X16: 2-3 (sign of gradient data, 2-Complement)
- DIP switch S1: both in lower position
- DIP switch S2: both in lower position
- Rotary switch S3: Position 7
- Rotary switch S4: Position 2 (this configuration results in a clock signal FSYNC = 50 kHz)

Offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes.



Prerequisite is the Regulator Adjustment (Tune-up).

- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D4 in down position.
- Connect a DVM to the desired actual value output X3 on D2. Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D1 to adjust the offset for the X, Y and Z axes if necessary.
 - » If the offset was changed, perform the Tune-up steps Phantom Shim and Gradient Sensitivity again

Run TuneUp:

- Phantom Shim
- Grad Sens

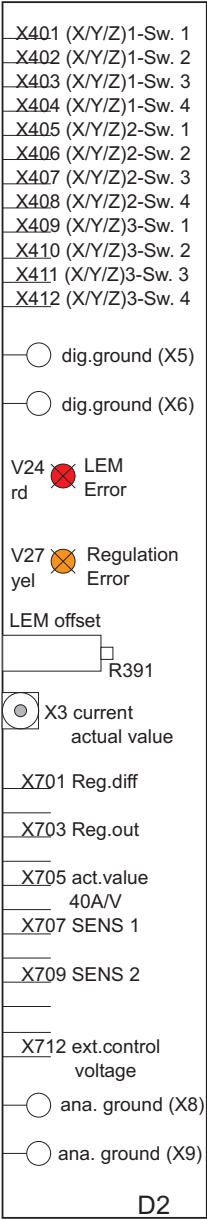
Perform Quality Assurance:

- Coil Check (SN DIP)
- Phantom Shim Check
- Calc Artefacts

2.1.5 Regulator D2

Fig. 2: Front of Regulator D2

Descriptions:



R391 SENS-Offset adjustment

X3 Actual current value, QLA (1V=40A)

X701 Regulator difference signal

X703 Regulator output signal

X705 Actual current value (1V=40A)

X707 SENS1 current value

X709 SENS2 current value

Default Regulator Setting by DIP Switches:

- Check the DIP switches according to the following table:

Switch	S1 (P)	S2 (I)	S1 adjusts the proportional portion of each regulator axis. S2 sets the integral regulator portion of each axis.
1	ON	ON	
2	off	off	
3	off	off	
4	off	off	
5	off	off	
6	off	off	
7	off	off	
8	off	off	

SENS Offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X3 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the modulator board D4 in upward position ("Modulator off").
- Check the offset: Offset < +/- 0.05 mV; use R191 to adjust the offset if necessary.
 - » If the offset was changed, perform the Tune-up steps regulator adjustment, Phantom Shim, Grad Delay and Grad Sens

Run TuneUp:

- Regulator Adjustment
- Phantom Shim
- Grad Delay
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check
- Phantom Shim Check
- CTC
- ECC
- Grad Delay Check
- Calc Artefacts

2.1.6 Synchronization D3

Check Jumper Settings:

- Jumper X3: 3A - 3B, 6A - 6B, 9A - 9B

Perform Quality Assurance:

- Coil Check (SN DIP)

2.1.7 Modulator D4**Check of DIP Switch Configurations:**

- DIP switch S1: both in lower position
- DIP switch S2: both in lower position
- DIP switches S3 and S4: Check setting according to the following table:

Tab. 4 DIP setting Modulator D4

	DIP switch S3	Function	DIP switch S4	Function
1	ON	deadtime1	ON	modsemo x
2	off	deadtime2	ON	modsemo y
3	off	mode1	ON	modsemo z
4	off	mode3	ON	sunuse x
5	off	fqa	ON	sunuse y
6	off	mode2	ON	sunuse z
7	ON	fqb	ON	swloss1
8	off	mode4	off	swloss2

Perform Quality Assurance:

- Coil Check (SN DIP)

2.1.8 Turbo Modulator D5**Check of DIP Switch Configurations:**

- Check the DIP switches according to the following table:

Tab. 5 DIP setting Turbo Modulator D5

	DIP switch S1	Function	DIP switch S2	Function
1	ON	deadtime1	ON	modsemo x
2	off	deadtime2	ON	modsemo y
3	ON	mode1	ON	modsemo z
4	off	mode3	ON	sunuse x
5	ON	not used	ON	sunuse y
6	ON	mode2	ON	sunuse z

	DIP switch S1	Function	DIP switch S2	Function
7	ON	not used	ON	swloss1
8	ON	not used	off	swloss2

Perform Quality Assurance:

- Coil Check (SN DIP)

2.1.9 Service/CAN D6

Check Jumper Settings:

- Jumper X601: 1- 2

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check

2.1.10 FOC D7



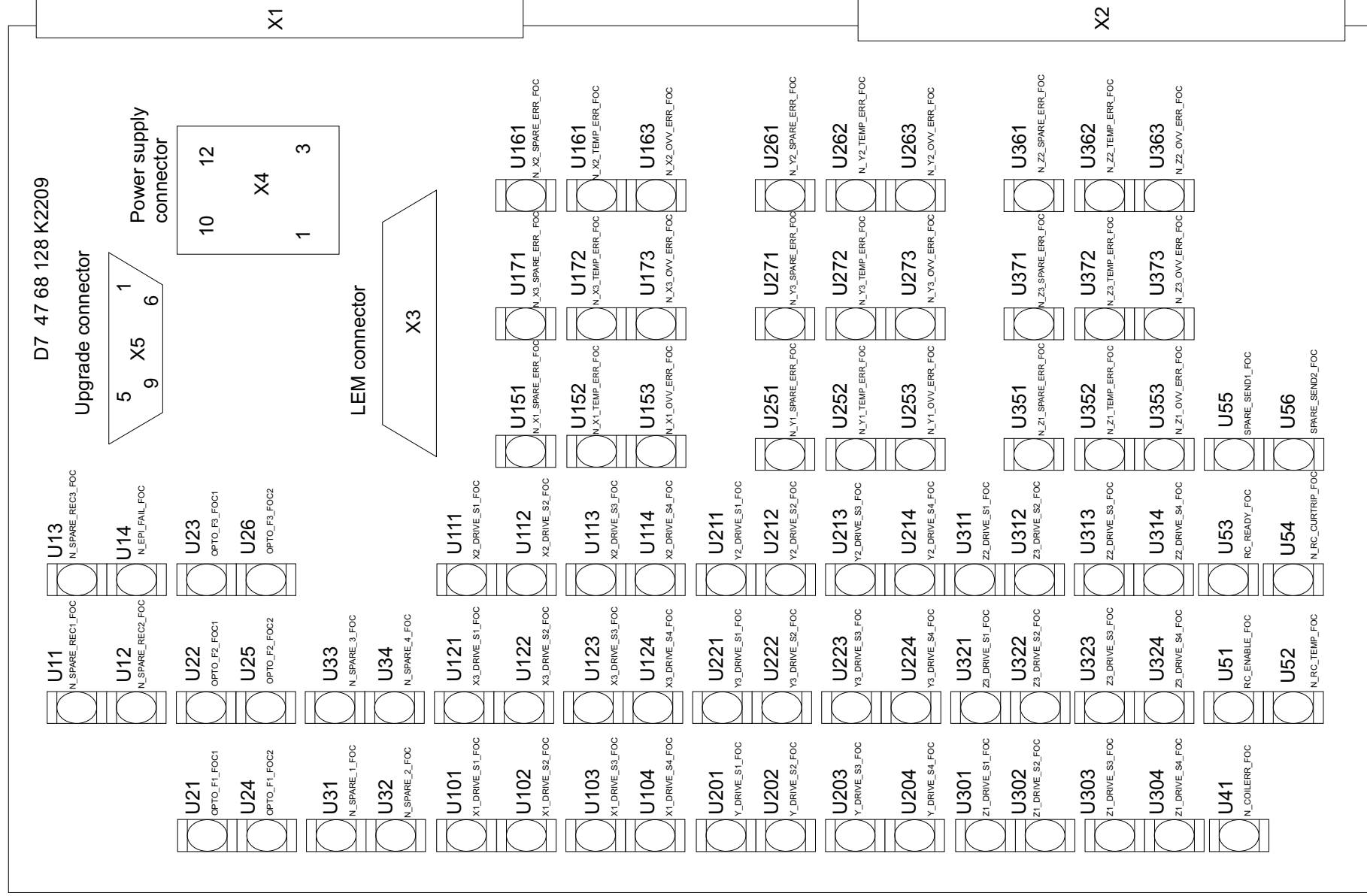
The D7 PCB is the interface from GSSU to Power Sages, as well as to current measurement and supervision assemblies.

The D7 is mounted at the GSSU left-side and converts electric signals into FOCs and vice versa.

Connect cables and FOCs:

- (→ Fig. 3 Page 28) shows the configuration of the connections for the fiber optic cables.

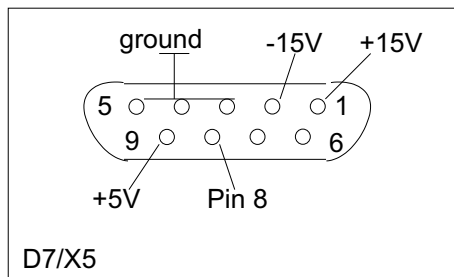
Fig. 3: D7 K2209 Connector positions



Upgrade Interface Connector:

- If there is no upgrade component installed, the pin 8 must be connected to ground.

Fig. 4: Upgrade interface connector at D7

**Perform Quality Assurance:**

- Coil Check (SN DIP)

2.1.11 Driver Supply D10

Check Jumpers:

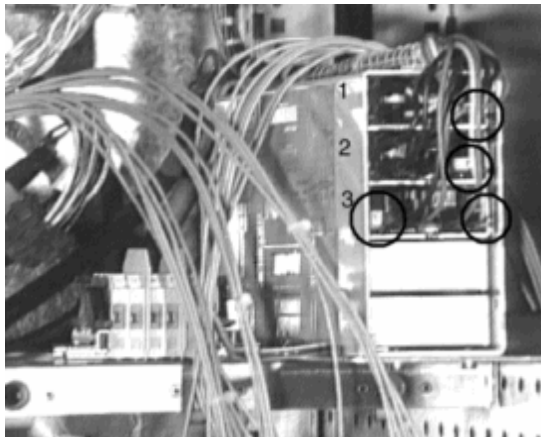
- Jumper X4: external clock 1-2 (default)

Perform Quality Assurance:

- Coil Check (SN DIP)

2.1.12 Small Signal Power Supply E4A

Fig. 5: Adjustment of GSSU power supply voltages

**Adjustment of Supply Voltages:**

- Adjust the following voltages:

Slot 1: +5V Vcc

- Measure the voltage at the terminals of E4A or on D7/X5, Pin 9-5

Slot 2: +15V for D10 Driver Supply:

- Measure the voltage at the terminals of E4A

Slot 3: +/- 15V:

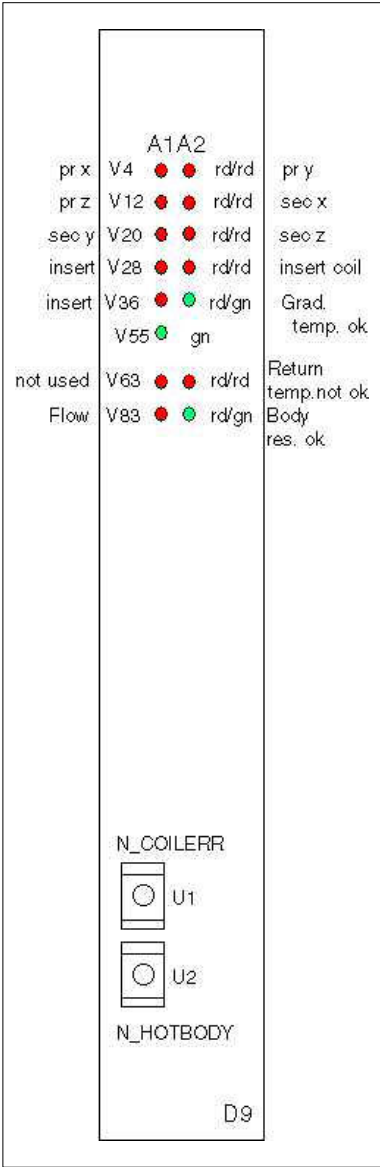
- Measure the voltage at the terminals or on D7/X5, Pin 1-5 and 2-5.

» Adjust the voltages with the marked potentiometers
(→ Fig. 5 Page 29)

2.2 Supervision D9

i The coil supervision board D9 is located in the patient table control electronics rack, at left-side.

Fig. 6: Front of Supervision D9

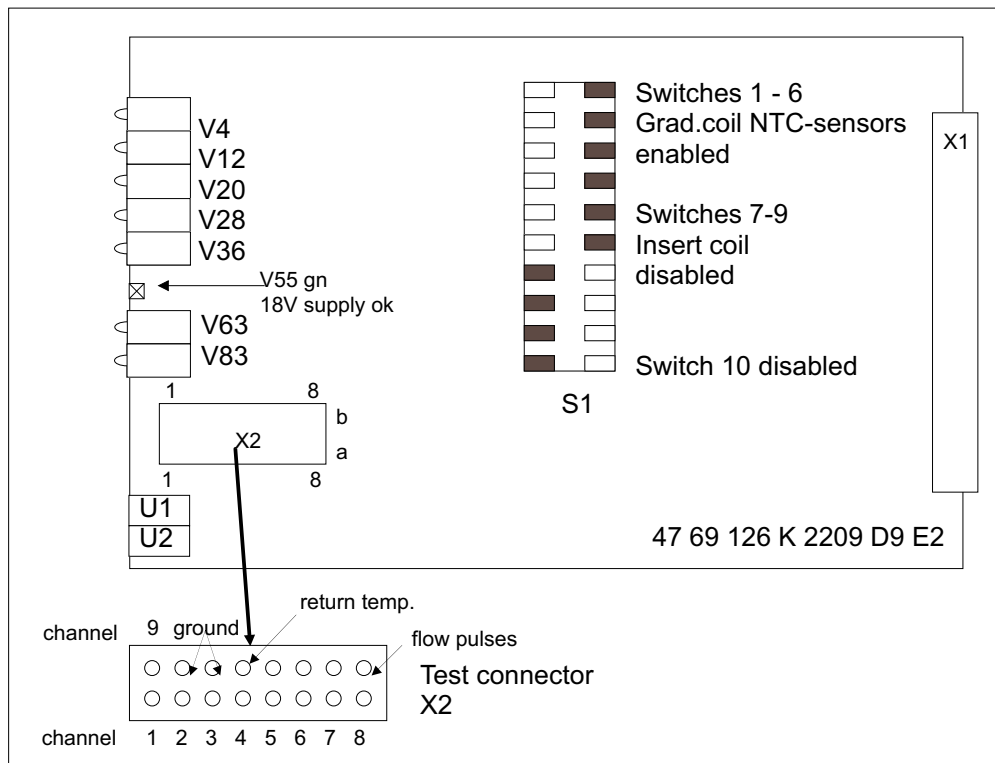


Descriptions:

- LED's:
- pr x,y,z:
NTC-sensor status at primary side of the gradient coil
- sec x,y,z:
NTC-sensor status at secondary side of the gradient coil
- V55gn: 22V supply voltage ok
- V63 rd right:
Return temperature status of cooling water for the body resonator (Temperature too high)
- V83 rd left:
Water flow supervision for the body resonator (Flow not ok).
- V83 gn right: Body coil ok

- N_COILERR:
FOC connection to D7/U41 (Light off or glowing weakly: => Error)
- N_HOTBODY
FOC connection to BTB1/U5 (Light off => Error/ Body Coil not ok)

Fig. 7: Position of DIP switches at D9

**Check Jumpers and Switch Positions:**

- Switches 1-6 = ON
 - Switches 7-10 = off
- (Switch 1 => pr x, switch 2 => pr y, a.s.o.)

Perform Quality Assurance:

- Coil Check (SN DIP)

2.3 Resonant Converter

2.3.1 Resonant Converter (MK1)

The component is no longer delivered and is replaced by RC (MK2) (material number 11 41 097)

2.3.1.1 Safety Information

- Switch-off the circuit breakers F9 (3*50A) first, then F14 (4A) in the CCA.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the following procedure.



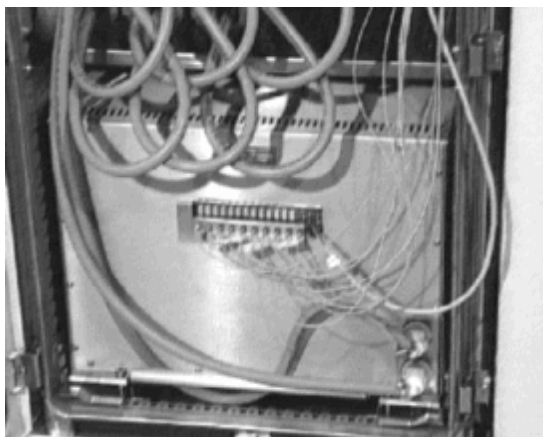
The old RC is very heavy (approx. 80kg).
Two persons are required to replace the Resonant Converter.

2.3.1.2 Required Tools

- Set of standard tools
- Forklift or equivalent lifting device to position the resonant converter at the correct installation height of approx. 16 cm (if the rollers of the cabinet are installed).

2.3.1.3 Parts Location

Fig. 8: Resonant converter assembled in the GPA cabinet



(→ Fig. 8 Page 32) shows the resonant converter (RC) installed in the lower half of the GPA cabinet. It also shows the interfaces of the RC to the peripheral systems:

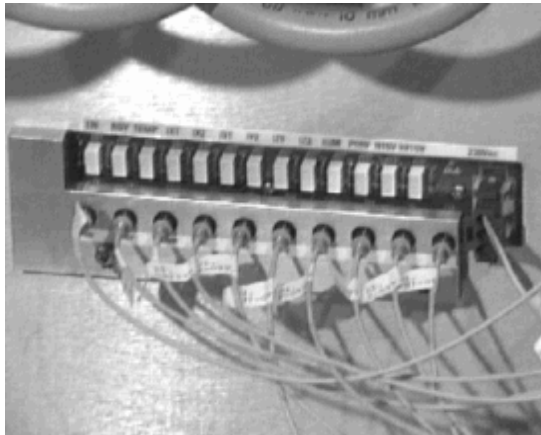
- Cooling water hoses
- fiber optic cables (for controlling/transmitting error messages)
- Power supply unit for control circuit

The inputs/outputs of the power circuits are located to the left on the rear panel of the RC. They connect automatically when the RC is pushed into the GPA cabinet (refer to (→ Fig. 13 Page 34) and (→ Fig. 14 Page 35)).

The RC is equipped with rollers (→ Fig. 13 Page 34) and sits on guide rails. The unit is secured with two locking screws (refer to (→ Fig. 12 Page 34)).

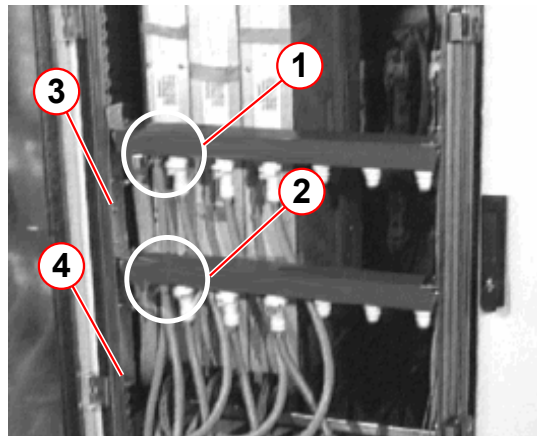
2.3.1.4 Removing the connected Cables

Fig. 9: FOC's and power supply of RC control



- Pull all fiber optic cables as well as the supply voltage cable for the control circuit (→ 3/ Fig. 9 Page 33).
- Secure the fiber optic cables (with e.g. adhesive tape) to prevent damage when the RC is pulled.

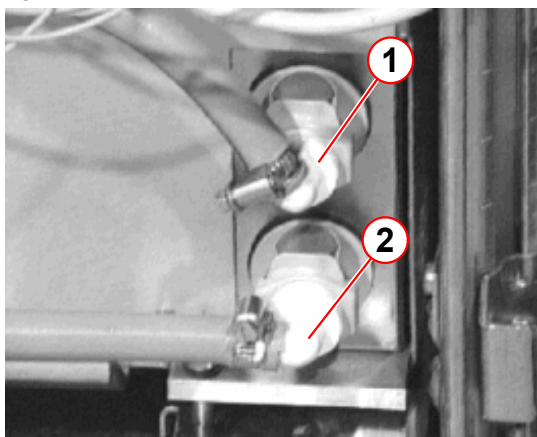
Fig. 10: Water distributor in normal position



(→ Fig. 10 Page 33) shows the inlet and outlet connections for the chilled water hoses of the RC to the water distribution system of the GPA cabinet.

- (1) Inlet side
- (2) Outlet side
- (3) Upper screw
- (4) Lower screw

Fig. 11: Connections of water hoses at RC



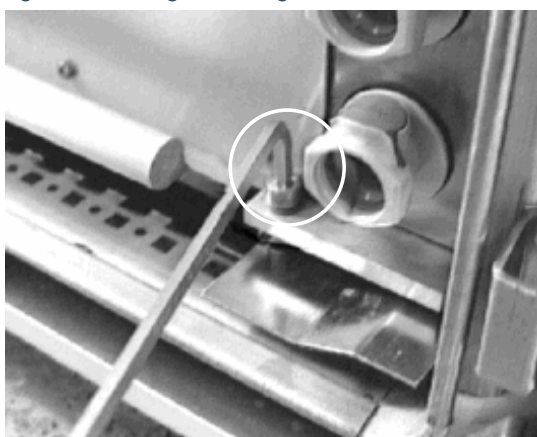
- (1) Outlet
- (2) Inlet

- Remove the connections for the chilled water hoses at the lower edge of the RC (→ Fig. 11 Page 34).

The water connections shut off automatically. However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

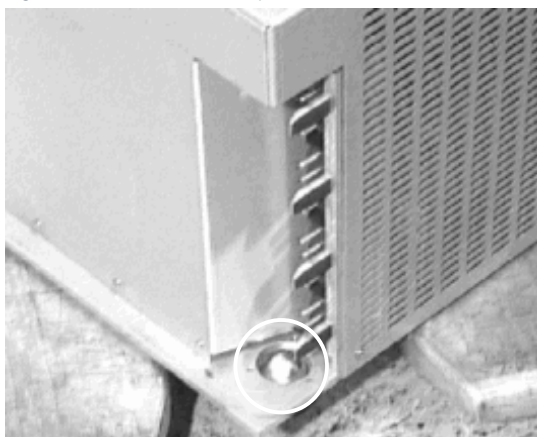
2.3.1.5 Removing the Resonant Converter

Fig. 12: Locking screw (right side)



- Remove the two Allen locking screws to the right and left at the lower edge of the RC.

Fig. 13: Roller ball and power contacts at the RC



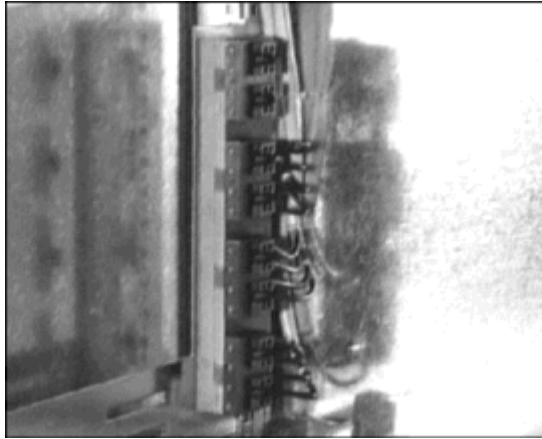
- Position an e.g. pallet jack matching the size of the RC in front of the GPA cabinet. Raise it until it is level with the lower edge of the RC.
- Pull the RC at the handle. Gently remove it from the GPA cabinet and slide it onto the pallet jack.

The RC is rather heavy. It weighs approx. 80 kg.



The RC is equipped with rollers (→ Fig. 13 Page 34). Use appropriate means to hold the RC in position.

Fig. 14: Power contacts inside the cabinet



(→ Fig. 13 Page 34) and (→ Fig. 14 Page 35) show the power contacts which contact automatically when you slide the RC into position inside GPA cabinet.



Prior to installing the replacement RC, verify that the power contacts at the RC and the GPA cabinet are in good condition.



Before returning the replaced Resonant Converter, drain the water from the pipes by opening both ends.

2.3.2 Replacement of Resonant Converter (MK1) with (MK2)

This document also includes the extra procedure which is necessary to retrofit a GPA from RC (MK1) to RC (MK2) (i.e. modification of the power line connections and reassembling of the new Resonant Converter (RC).

For this upgrade, please order upgrade-kit (material number 57 26 158) together with the new RC (MK2) (material number 11 41 097).



There are two types of GPA cabinets:

- **Willsher & Quick (W&Q)** cabinet (used in system no.: Harmony < 10175, Symphony < 13131)
- **Schroff cabinet** (used in system no.: Harmony > 10175, Symphony > 13131)

2.3.2.1 Safety Information

- Switch-off the MR system.
- Switch-off the circuit breakers F9 (3*50A) first, then F14 (4A) in the CCA.
- Adhere to the (→ General Safety Notes / TD00-000.860.01), when performing the following procedure.

- Before working at the connector terminal in the GPA, ensure that there is no dangerous voltage at the connectors.
 - » Measure the voltage at the contacts with a DVM.



The old RC is very heavy (approx. 80kg).
Two persons are required to replace the Resonant Converter.

2.3.2.2 Required Tools

- Set of standard tools.
- Torx driver bit (CR-VT20) (for removing the connectors from the connector terminal in the W&Q GPA cabinet).
- Torx driver bit (TX 25) (for removing the connector terminal in the Schroff GPA cabinet).
- Forklift or equivalent lifting device to remove the old RC.
(Installation height of approx. 16 cm, if the rollers of the cabinet are installed).

2.3.2.3 Content of the upgrade-kit (Material Number 57 26 158)

- Complete power cabling (with fitted connector rails at GPA cabinet end and RC end).
- 2 Screws (head socket hex screw, refer to (→ Fig. 22 Page 41)).
(for fixing the connector rail at the rear of RS).
- 2 short Allen screws (for locking the RS in the gradient cabinet).
- Cable ties (for fixing the connector terminal in the GPA cabinet).
- Washers



Order the Resonant Converter MK2 separately!
New Resonant converter (MK2):
Material number 11 41 097 K2209

Fig. 15: New Resonant converter and mounted power cabling at backside of RC



2.3.2.4 Preparation

- Disassemble the old resonant converter as described in this document.



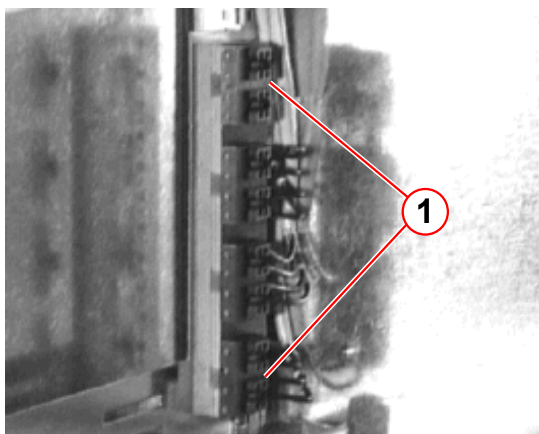
In the worst case, the GPA cabinet can only be accessed from the front. Otherwise, the following procedure can be much more simplified by removing the left cover of the GPA cabinet.

2.3.2.5 Repositioning the Connector Terminal in the (W&Q) GPA Cabinet

To connect the new RC power cabling with the connectors in the GPA cabinet, the fabric-based laminated block (connector terminal) in the GPA cabinet has to be shifted:

To loosen the connector terminal, the upper and lower connectors have to be removed first.

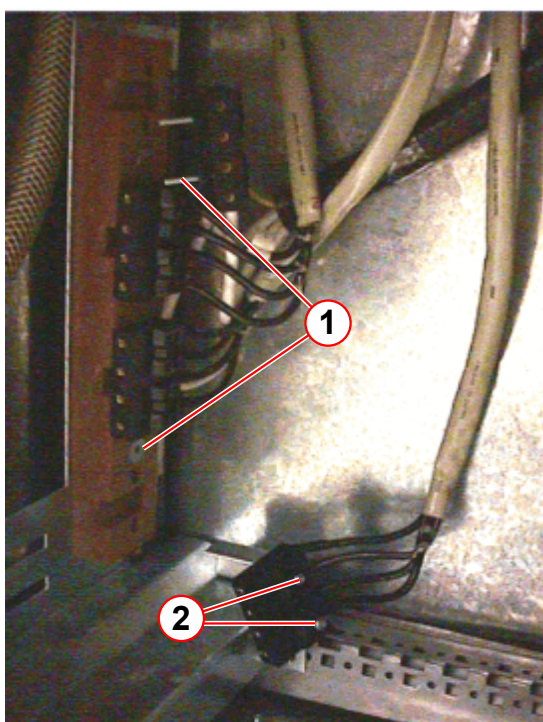
Fig. 16: Connector block mounted in the Willsher & Quick cabinet



(1) Upper and lower connector, mounted on the connector block, each with two torx (20 screws)

- Remove the two Torx-screws at the upper and the lower black connector (refer to (→ Fig. 16 Page 38), (→ Fig. 17 Page 38)).

Fig. 17: Upper and lower connector removed from the fabric-based laminate block (connector block)



(1) Upper and lower countersunk screws, fixing the connector block
(2) Torx (25) screws at connector

When the upper and lower black connectors have been removed, you will get access to the two countersunk screws (5 mm hex socket head screws), which fix the connector terminal.

- Remove the countersunk screws at the connector terminal.

Fig. 18: Removed connector block - all black connectors are mounted again

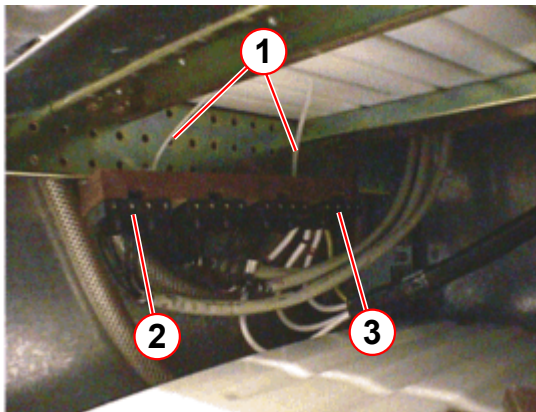


The connector terminal is now loose.

- Mount the two black connectors again on the connector terminal with the Torx screws.

2.3.2.6 Mounting the Connector Terminal in the (W&Q) GPA Cabinet

Fig. 19: Connector block, fixed at the cabinet frame with cable ties - connector bushings in horizontal position (W&O)



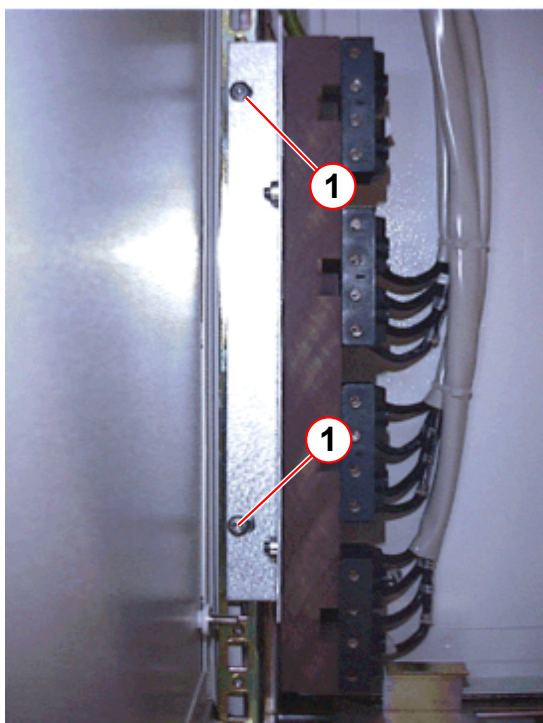
- (1) Cable ties
- (2) X-connection to RC output X
- (3) Power input RC

The connector terminal will be now fixed at the left cabinet frame by cable ties, passed through the holes in the cabinet frame and the two screw holes in the connector terminal (refer to (→ Fig. 19 Page 39)).

- Pass cable ties (corresponding to the distance of the screw holes in the connector terminal, about 16 cm apart) through the lowest holes in the cabinet side frame. (The connector terminal should be positioned approx. in the middle of the cabinet side frame).
- Pull the connector terminal in the position shown in (→ Fig. 19 Page 39) (connector bushings horizontally; X,Y,Z-connections to RC output in front, power input connections to RC at the back of GPA cabinet).
- Pass the cable ties through the screw holes in the connector terminal and fasten the connector terminal.

2.3.2.7 Repositioning the Connector Terminal in the Schroff GPA Cabinet

Fig. 20: Connector block mounted in the Schroff cabinet



(1) Torx (25) screws

- Remove the two Torx-screws (→ Fig. 20 Page 40).
 - » The connector terminal together with the angle plate is now loose.

2.3.2.8 Mounting the Connector Terminal in the Schroff GPA Cabinet

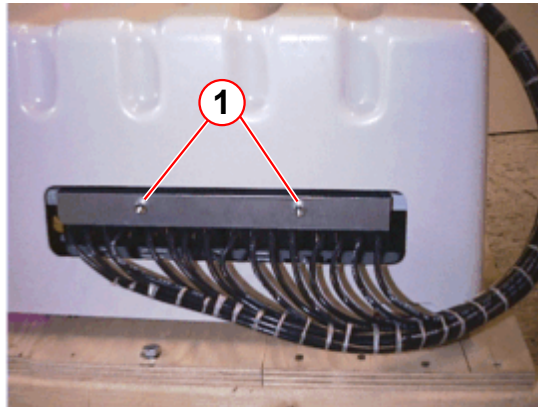
Fig. 21: Connector block mounted in the Schroff cabinet



- Pass cable ties (corresponding to the distance of the screw holes in the angle plate) through the lowest holes in the cabinet side frame. (The connector terminal should be positioned approx. in the middle of the cabinet side frame).
- Pull the connector terminal in the position shown in (→ Fig. 21 Page 40) (connector bushings horizontally; x,y, connections to RC output in front, power input connections to RC at the back of GPA cabinet).
- Pass the cable ties through the screw holes in angle plate and fasten the connector terminal.

2.3.2.9 Mounting the Connector rail at the RC

Fig. 22: Connector rail at the back of the RC



(1) Fixing screws

- Plug in the connector rail at the back of the RC and fix it with two screws.

2.3.2.10 Installing the new RC in the GPA Cabinet

Fig. 23: RC installed in the Willsher & Quick GPA cabinet



(1) Locking Allen screws

- Slide the RC (two persons are recommended) into the GPA cabinet.
- Lock the RC with two Allen screws.

- Plug in the connector rail into the bushings of the connector terminal and secure the connection with cable ties.

Fig. 24: RC cabling connected via connector rail to the connector block (Willsher & Quick cabinet)

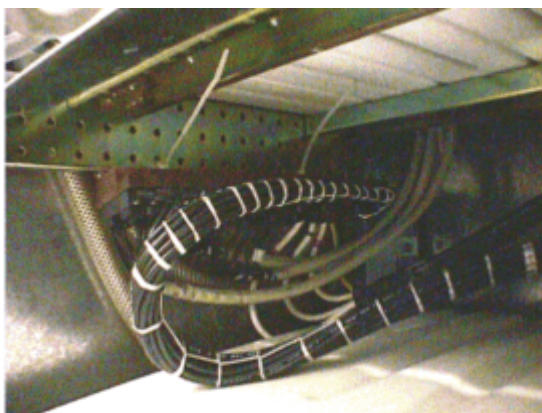


Fig. 25: RC cabling connected via connector rail to the connector block (Schroff cabinet)



- Reconnect the water pipes (→ Fig. 23 Page 41).
- Reconnect the fiber optic lines and the AC input line (→ Fig. 25 Page 42).

Fig. 26: Reassembled RC in the Schroff cabinet

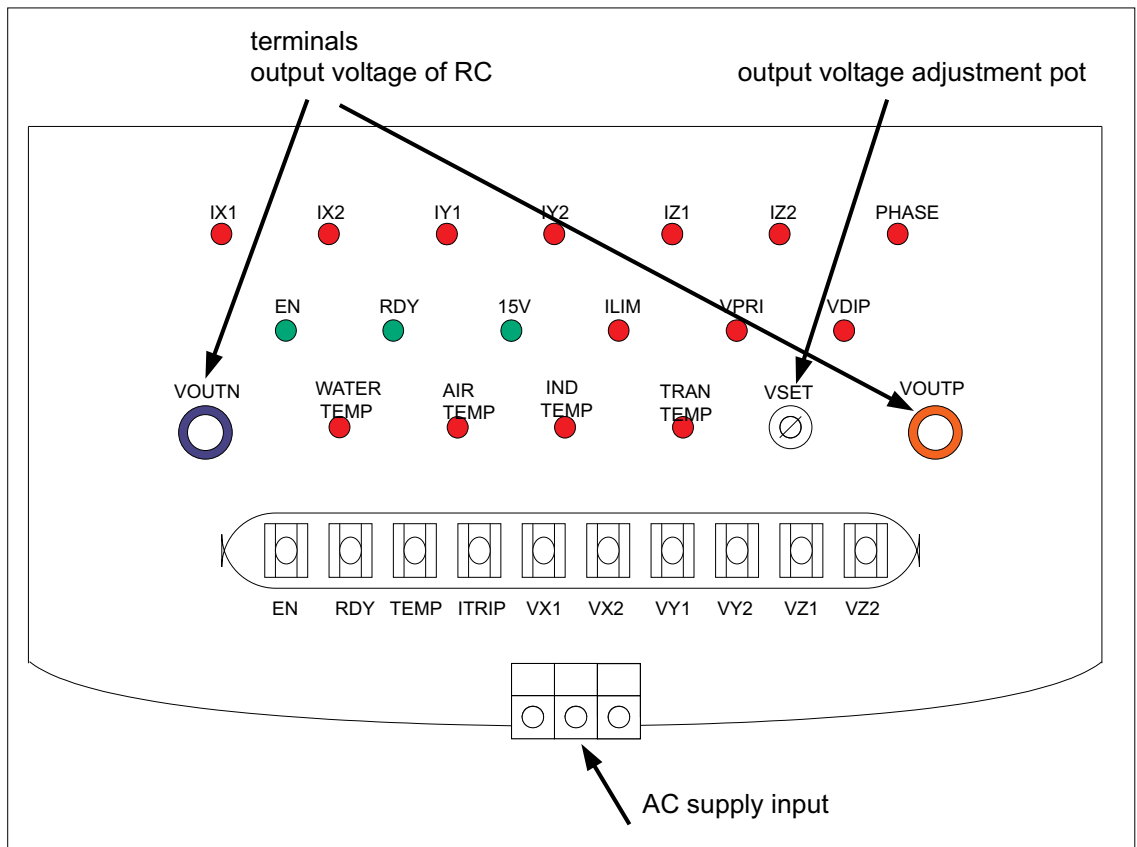


2.3.2.11 Start-up of the RC

- Switch-on the MR system
 - Connect a DVM (range 1000 VDC) to VOUTN and VOUTP terminals at the RC.
 - Switch-on the service mode (S1 right side at D4 in upwards position).
 - Switch-off the modulators (D4, switches S1 left and S2 in upwards position).
 - Switch-on the RC (first F14 and then F9, CCA) The resonant circuit of the RC can be switched on/off manually by the push-buttons S402 ("on") and S403 ("off") at D6.
 - Read the voltage at the DVM; The voltage should be between 4,05 and 4,10 VDC. (4 VDC $\Leftrightarrow$ 400 VDC at RC output)

- If out of tolerance, adjust the voltage with pot VSET at RC to 407 VDC.

Fig. 27: Error LED's, FOC connectors and AC supply input at RC



Don't forget to deactivate the "service mode" and switch-on the modulators for normal operation of the MR system.



Before returning the replaced RC, drain the water from the pipes by opening both ends

2.4 Power Stages

(Power Stage material number 47 69 258 K2209 or Papillon 400 Power Stage 5939447 K2209)

When replacing the older model Power Stage (47 69 258) with a Papillon 400 Power Stage (59 39 447) a slightly different procedure applies. (→ Replacement of Power Stage (4769258) with Papillon 400 (5939447) / Page 51).

NOTICE

Important for Ultra Gradients: Do NOT mix different types of final stages within one axis!

Risk of component malfunction or damage.

- » However, for different gradient axes, mixing of different types of Power Stages is allowed.

2.4.1 Safety Information



WARNING

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch-off the gradient cabinet
- » Gradient cabinet K2209: The circuit breakers F9/3*50A and F14 in the CCA must be switched off.



WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy both the PS and Resonant Converter!

- » Use a towel to catch any spilled water when handling the water quick connectors.

2.4.2 Required Tools

- Set of standard tools

2.4.3 Identifying the defective Power Stage

A defective Power Stage is often indicated by a “non-ready” RC in conjunction with the error message RC_CURTRIP and one or more red error LED's IX1...IZ2, especially if the modulators are disabled.

The LED of the RC also indicates a defective Power Stage.

For verification, you can either measure each Power Stage with a DVM or shake it and listen for loose parts inside (i.e. residuals of cracked semiconductors).

2.4.4 Removal Procedure

Fig. 28: View of GPA cabinet with two power stages inserted. The water supply is not connected or folded down



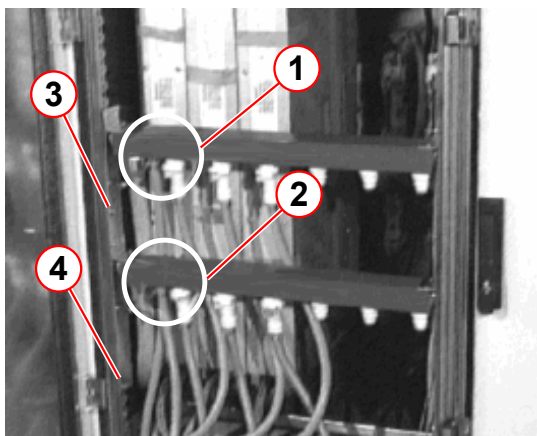
- Switch-off the circuit breakers F9 (3*50A) and F14 (4A) in the CCA.
- If only three Power Stages are installed, remove the protection cover in front of the additional three plug-in positions first.

The configuration of the Power Stages in the cabinet is as follows:

X1 - Y1 - Z1 - (X2 - Y2 - Z2).

2.4.4.1 Removing the Water Hoses

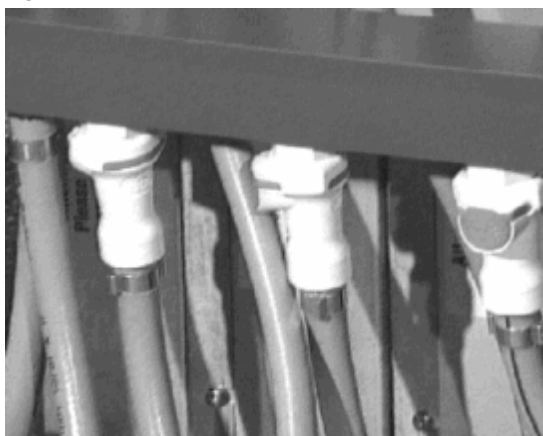
Fig. 29: Water distributor in normal position



- (1) Inlet side
- (2) Outlet side
- (3) Upper screw
- (4) Lower screw

- To fold down the water distributor, loosen the lower two screws (→ 4/Fig. 29 Page 46) at both sides and remove the upper two screws (→ 3/Fig. 29 Page 46) at both sides at the distributor frame (→ Fig. 29 Page 46).
- Fold down the water distributor.

Fig. 30: Quick connections at the water distributor



- Remove the quick connections of the water hoses from the inlet and outlet sides.

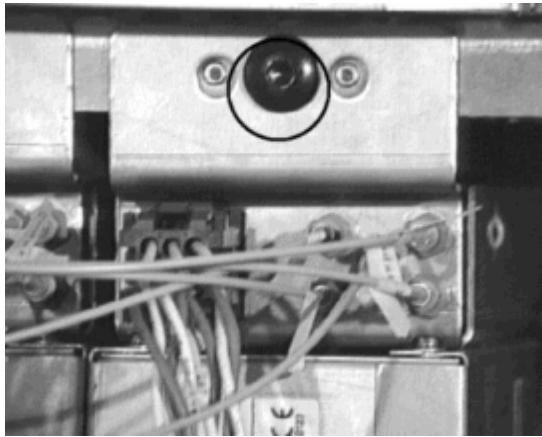


The water connections shut off automatically.

However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

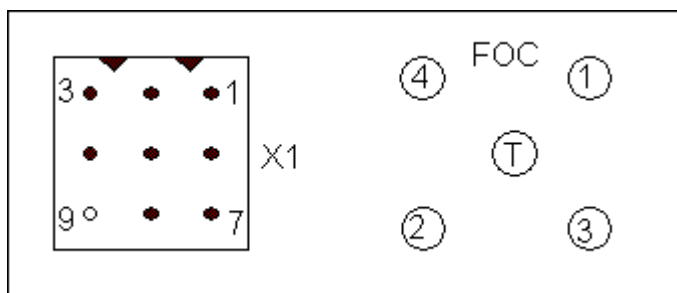
2.4.4.2 Removing the Power Stage

Fig. 31: Quick locking screws and connectors on the PS



- Disconnect the five FOC's and the connector X1 from the Power Stage (→ Fig. 31 Page 47) and (→ Fig. 32 Page 47).

Fig. 32: Connector X1 and FOC'S at power stage



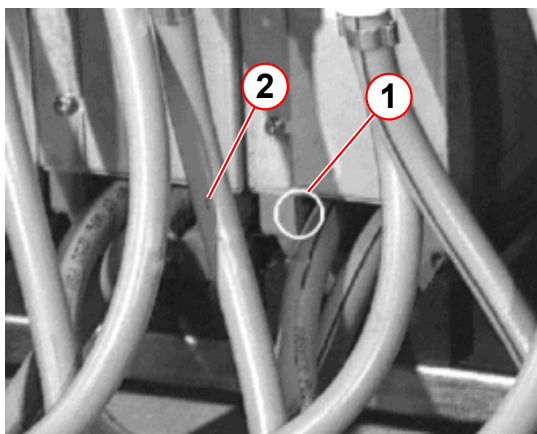
- Open the quick locking screw by backing it up 1/4 turn counterclockwise (→ Fig. 31 Page 47).
- Remove the Power Stage by pulling it straight back. It should come out approx. 2 cm. This removes the contact blades from the connectors (fuse holders). Do not tilt the Power Stage while pulling it out of the cabinet.
(Weight of the PS is approx. 20 kg).

NOTICE

If there are six Power Stages installed, attach a piece of wire or a cable tie forming a loop through a hole on the bottom of the PS, shown in (→ Fig. 33 Page 48).

- » Using the loop you can pull the Power Stage out.
- » Do not use the water hoses to pull the Power Stage out.

Fig. 33: Position of the hole for attaching a cable tie top



- (1) Inlet side
(2) Hole for a cable tie loop

Fig. 34: Contact blades at back of power stage



(→ Fig. 34 Page 48) and (→ Fig. 35 Page 48) show the back of one of the removed Power Stages, including the inputs/outputs of the power circuits.

Fig. 35: Power input/output connectors back of PS

<u>UB+</u>	<u>UB-</u>	<u>out2/4</u>	<u>out1/3</u>
0V	400V		

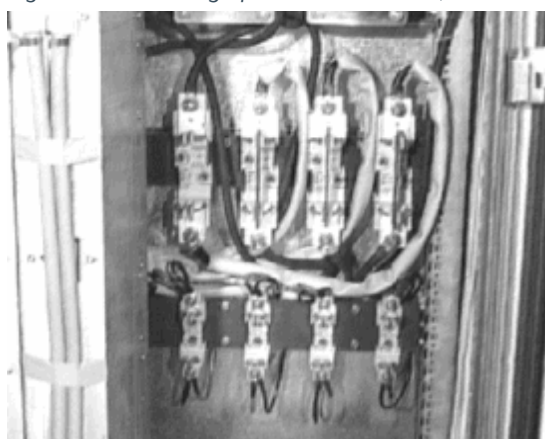
Inputs PS:

UB+/UB- , max. 425 VDC from RC

Outputs PS:

out 1/3 to the gradient coil, out 2/4 from the gradient coil (max. 300 A)

Fig. 36: Power stage power connectors (fuse holders)



- After removing the Power Stage, check the power contactors in the GPA cabinet. If you notice any damage, replace the respective fuse holders:

Inputs:

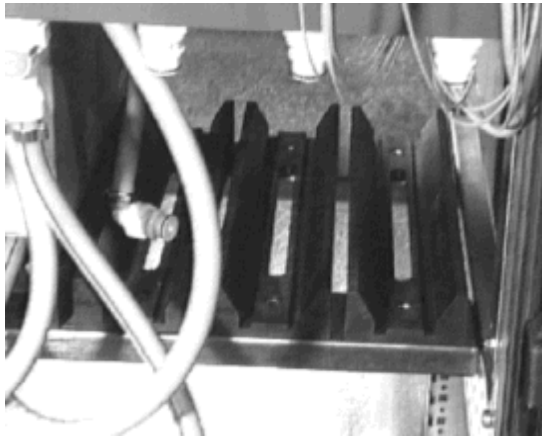
Fuse holder NH2 (150A), material number 37 59 995 K2149

Outputs:

Fuse holder NH3 (160A), material number 37 59 987 K2149

2.4.4.3 Reassembly Procedure

Fig. 37: Prismatic guiding rails



- Check the slide-in contactors in back of the Power Stage. They have to run vertically to the back of the Power Stage without bends or warps.
- Position the Power Stage on the guide rail. Ensure that the Power Stage does not tilt, while sliding it up to the end stop in the GPA cabinet.

Fig. 38: Third power stage slightly pushed in

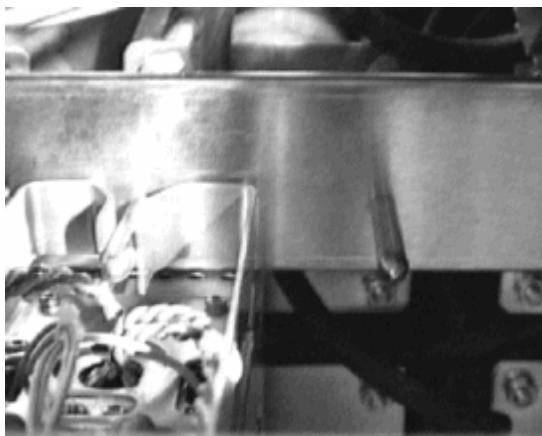


The guide pin on top in the cabinet centers and keeps the Power Stage from tilting forward (→ Fig. 39 Page 49).



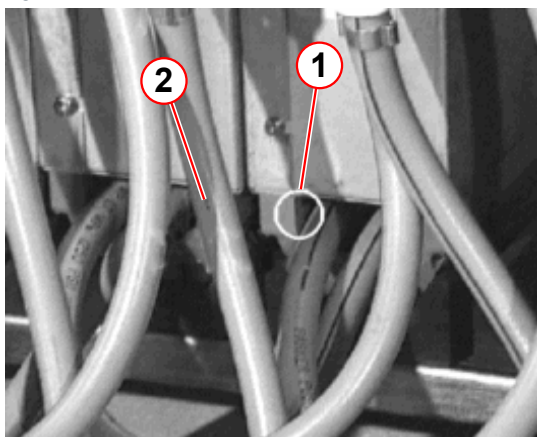
If not all six Power Stages have been installed, visually inspect the Power Stages to ensure that they make satisfactory contact.

Fig. 39: Guide pins centering and securing the power stage (top)



- Secure the Power Stage with the quick locking screws. Tighten them by 1/4 turn in clockwise direction (→ Fig. 31 Page 47).
- Reconnect connector X1 and the five fiber optic cables (→ Fig. 31 Page 47) and (→ Fig. 32 Page 47)). Verify that the fiber optic cables are connected correctly. If not, the Power Stage may be destroyed.

Fig. 40: Water inlet: left water hose



- (1) Inlet side
(2) Hole for a cable tie loop

- Connect the water hoses of the Power Stage with the water distributor as shown in (→ Fig. 29 Page 46) and (→ Fig. 30 Page 46).

The long, left water hose of the Power Stage is connected with the inlet of the water distributor.

- Fold the water distributor back and up. Secure it with screws (→ Fig. 29 Page 46).



Before returning the replaced Power Stage, drain the water from the pipes by opening both ends.

2.4.5 Final Steps

Run TuneUp:

- Regulator Adjustment
- Phantom Shim
- Grad Delay
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check
- Phantom Shim Check
- Grad Delay Check
- Calc Artefacts
- Image Orientation



Before returning the replaced Power Stage, drain the water from the pipes by opening both ends.

2.5 Replacement of Power Stage (4769258) with Papillon 400 (5939447)



The connector X1 and the Fibre optics "1-4" and "T" have different locations for Power Stages ((→ Fig. 41 Page 51) or (→ Fig. 42 Page 51)).

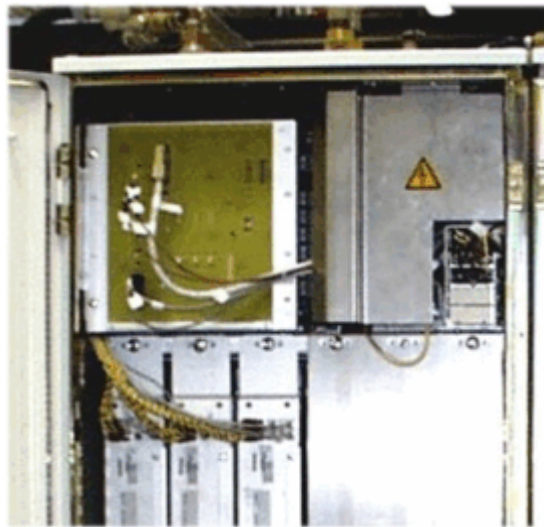
Fig. 41: Power stage 4769258



For this reason the wire harness inside the cabinet has been provided with enough slack to accommodate for both Power Stages.

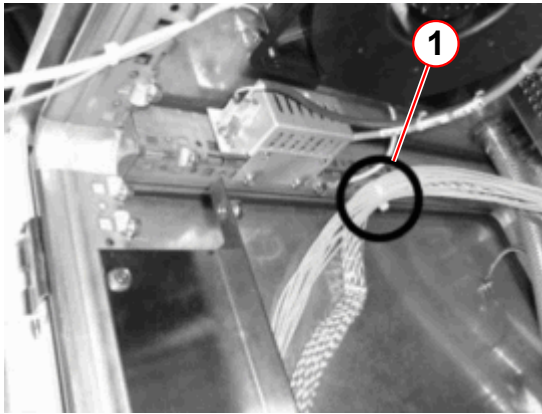
To obtain the full available slack, the GSSU has to be removed, see (→ Removing the GSSU / Page 55).

Fig. 42: Power stage 5939447



Once the GSSU has been removed a cable tie which fastens the Power Stage harness to the cabinet chassis will be visible (→ Fig. 43 Page 52).

Fig. 43: Tie wrap



(1) Tie wrap

Cut this cable tie and pull out the cable harness just enough to connect the wire harness leads to the Power Stage without stressing the wires (→ Fig. 42 Page 51).

Reinstall the GSSU.

Finished.

2.6 Current Sensor Assembly

Type LB200-S/SP4, material number 47 69 118 K2209 is not available as spare part anymore.

Order new material number 73 69 882 (complete current sensor unit with new type sensors IT600S & LA205-S and three modified regulator boards).

In case the new current sensor type IT600S & LA205-S is already installed, individual sensors can be replaced, order material number 11 41 311 then.

2.6.1 Safety Information

- Switch-off circuit breakers F9 (3*50A) and F14 (4A) in the CCA.

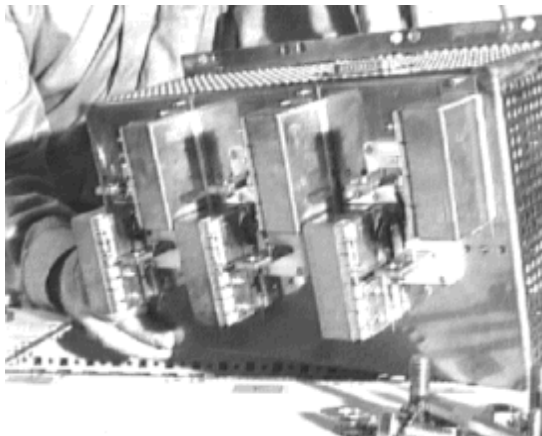
Adhere to (→ General Safety Notes / TD00-000.860.01) in the Safety chapter when performing the procedure described below.

2.6.2 Required Tools

- Set of standard tools
- Socket Allen for M10 (5,6 mm) Allen screws
- Driver bit Torx X10

2.6.3 Removing/installing the Current Sensors

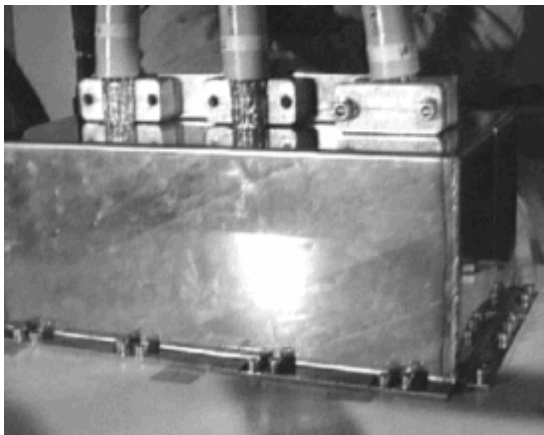
Fig. 44: Current sensor assembly



(→ Fig. 44 Page 53) shows the complete current sensor assembly after it has been removed. The assembly consists of three pairs of current sensors, each pair for one of the gradient axes (plus/minus direction). In addition, filter capacitors are installed in the assembly.

2.6.3.1 Removing the upper Cover and the Gradient Cables

Fig. 45: Gradient cables on top of the GPA cabinet



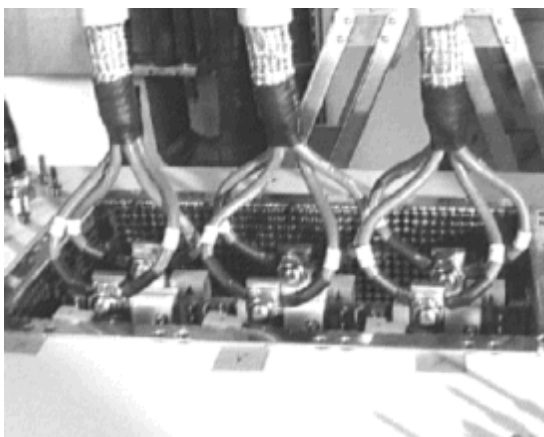
- Remove the strain relief for the gradient cables as well as the screws for the cover (two sections in this case) on top of the GPA cabinet ((→ Fig. 45 Page 54) and (→ Fig. 46 Page 54)).



The type of gradient cable and cable cover depends on the installation position of the GPA with respect to the RF-filter panel.

In the instruction, only the long (shielded) set of gradient cables is shown.

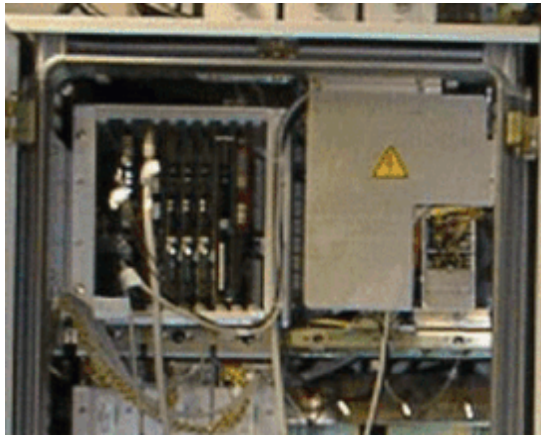
Fig. 46: Upper connections of gradient cables with the SENS via M12 bolts



- Remove the screw connections of the gradient cable at the upper connections of the SENS (→ Fig. 46 Page 54). You will need two crescent wrenches to remove the nuts at the M12 bolts.
To ensure correct reinstallation, note the order of the cable clamps and washers and/or star washers.

2.6.3.2 Removing the GSSU

Fig. 47: View on the Gradient Small Signal Unit (GSSU)



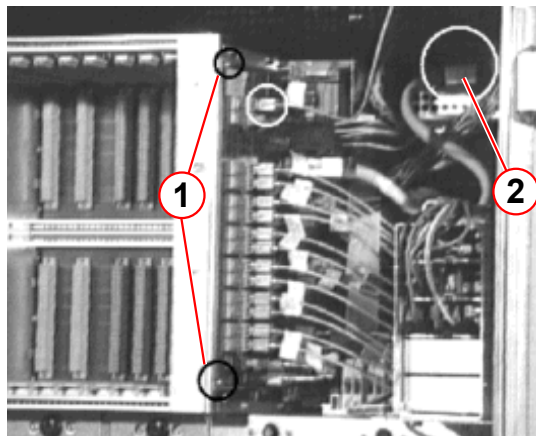
You have to remove the GSSU completely to access the lower cable connections for the SENS.



Before performing the next steps, cover the Power Stages with a piece of cardboard to avoid screws from falling into the cabinet. This avoids possible damage later on.

- Remove the protection cover from the right side of the GSSU (→ Fig. 47 Page 55).

Fig. 48: Removing the FOC board D7

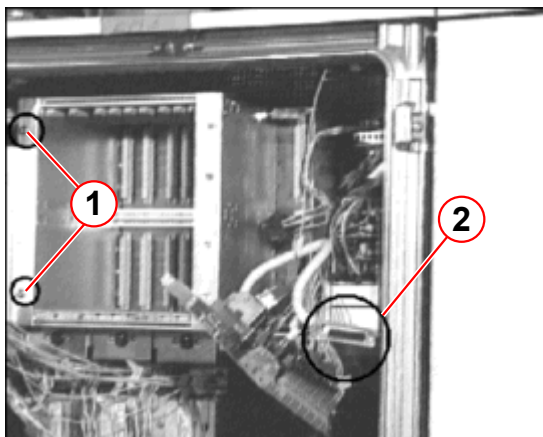


- (1) Torx 10
- (2) X1

Now remove the D7 as follows:

- Pull connector X1 from the connection in the SENS rack (→ Fig. 48 Page 55).
- At U21/D7, disconnect the fiber optic cable "OPTO_F1_FOC" to the driver current supply. Secure the FOC so that it is out of the way.
- Remove the two screws (Torx 10) (→ 1/Fig. 48 Page 55) .
- Pull D7 from the backplane. The other fiber optic cables remain connected.

Fig. 49: Removed D7 board and SENS-connector X1



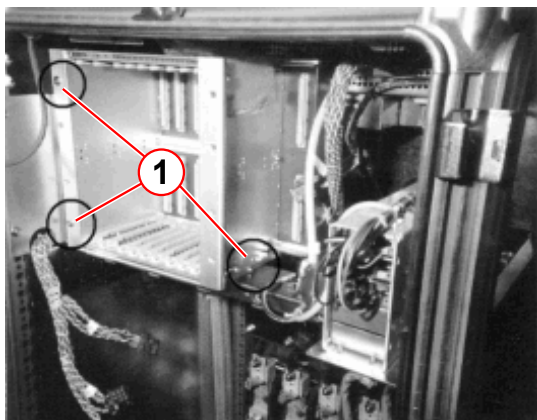
(1) D7
(2) X1

- Put D7 aside. (→ 1/ Fig. 49 Page 56) shows the removed D7 as well as connector X1 pulled from the SENS assembly.
Ensure that the fiber optic cables on D7 do not become damaged during the work.
- Pull the connectors X500, X501, X401, X402 from D6/GSSU and X3 from D1/GSSU.



You do not have to pull any assemblies (except D7) of the GSSU in order to remove it.

Fig. 50: Four quick locking screws secure the GSSU in the GPA cabinet

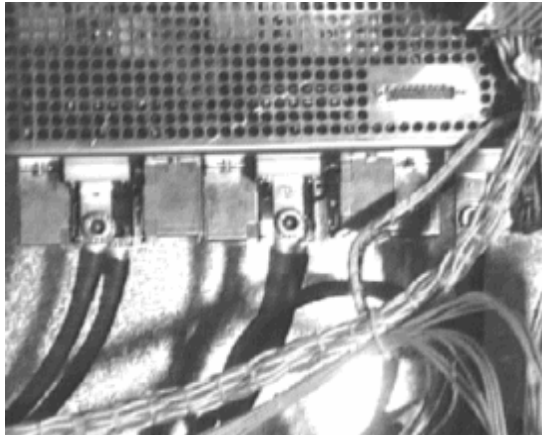


(1) Locking screws

- Remove the two quick lock screws along the lower right side and then the two quick locking screws on top and the lower left front edge of the GSSU rack. Back up the Allen key by 1/4 turn counterclockwise (→ 1/ Fig. 50 Page 56).
- Remove the GSSU slowly and carefully from the GPA cabinet.

2.6.3.3 Disconnecting the Gradient Cables and removing the SENS

Fig. 51: Lower screw connections (M10) of gradient cables

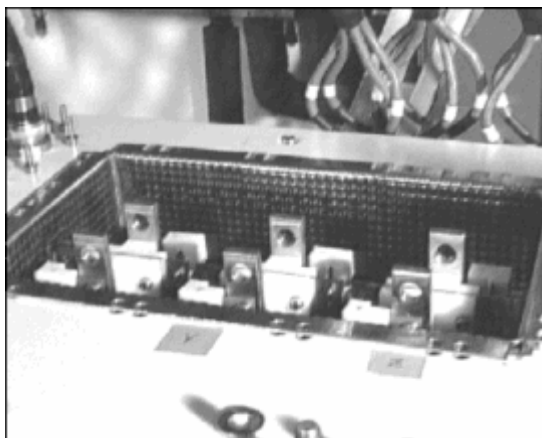


You are now able to access the lower cable connections of the SENS (→ Fig. 51 Page 57).

- Use a crescent wrench and Allen key to open the screw connections (M10) of the gradient cables.
Use a socket Allen. It will make it easier to access and remove the screws.

2.6.3.4 Removing the SENS Assembly

Fig. 52: SENS component in the top plate of the GPA cabinet

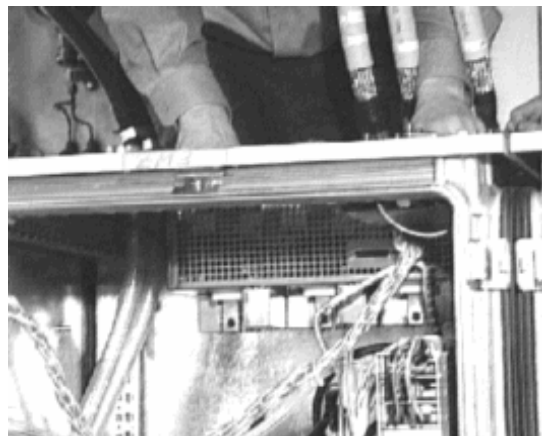


- Remove the screws for the sensor assembly on top of the GPA cabinet (→ 1/ Fig. 52 Page 57).

You should mark the positions of the screws of the SENS assembly with a waterproof marker pen for easier reassembly.

Look for existing RF-seals.

Fig. 53: Removing the SENS assembly



- Take out the entire SENS rack through the top of the GPA cabinet (→ Fig. 53 Page 57).

NOTICE

If a metallic part (e.g. nut, washer) drops into a Power Stage it must be removed prior to switch-on!

Otherwise, risk of severe damage to the Power Stage!

- » Turn the Power Stage upside down and shake it carefully until the missing part falls out.



Metal dust can disturb the current sensor function!

After fastening all gradient cables, carefully clean the entire SENS housing and remove any residual metal dust.

2.6.3.5 Assembly Procedure



Tighten the gradient cable lugs as much as possible!

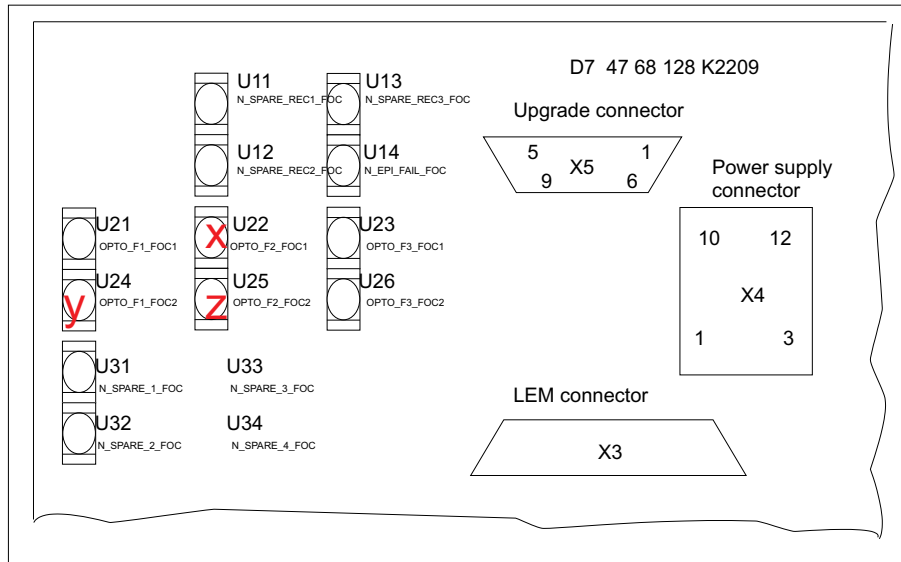
Once fastened, they should not move any more.

Perform the installation in reverse order of removal:

- Insert and secure the new SENS assembly. Note the RF-seals.
- Connect the lower gradient cables inside the GPA cabinet (start with the cables in the back).
- Insert and secure the GSSU (→ Fig. 50 Page 56).
- Plug D7 into the backplane.
- Secure D7 with two screws (→ 1/ Fig. 48 Page 55). Connect X1 into the SENS assembly. Connect the fiber optic cable for the driver current supply at U21/D7.

- For the new current sensor type IT600S & LA205-S (material number 11 41 311), connect the three additional FOC's from the (x,y,z)-current sensors to the D7 board as shown below (X->U22, Y->U24, Z->U25).

Fig. 54: Position of connections of FOC's from SENS on D7



- Make sure, the correct type of D2 regulator board is installed (47 69 084 required for the new current sensor type IT600S & LA205-S)!
- Reconnect the cables to D1 and D6/GSSU.
- Attach the safety cover at the upper right (→ 1/ Fig. 47 Page 55).
- Connect the upper gradient cables at GPA top. Tighten as much as possible!
- Attach and secure the cover. Note the RF-seals.
- Attach the strain relief for the gradient cables above the cover.
- Start up the MR system and perform the necessary measurements. Refer to the (→ Final Steps / Page 59).

2.6.4 Final Steps

- Offset adjustment on D2
(refer to (→ GSSU Components / Page 21))

Run TuneUp:

- Regulator Adjustment
- Phantom Shim
- Grad Delay
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check
- Phantom Shim Check

- CTC
- ECC
- Grad Delay Check
- Calc Artefacts
- Image Orientation



For replacing GPA components, perform the adjustments and checks as stated in the following descriptions:

3.1 GSSU Components

3.1.1 Safety Information



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/4A (GSSU) in the CCA.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the following procedures.

3.1.2 Required Tools

- Grounding kit (ESD)
- Set of standard tools
- Digital voltmeter

3.1.3 Common Procedure for all GSSU Components



For the exchange of boards of the small signal unit, refer to the section of ESD rules included in (→ General Safety Notes / TD00-000.860.01)

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/4A (GSSU) in the CCA.
- Remove the cover plate in front of the GSSU.
- Remove the front plate of the GSSU.
- Remove the corresponding connectors. Replace the defective board after checking the existing jumpers and DIP switch positions on the new board.

As an aid to pull the boards you should insert a cable tie or a wire forming a loop through holes at the front edge of the board.
- Reconnect all relevant connectors (FOC's, signal/power lines).

- Switch-on F3 circuit breaker F3 / 80A mains box first, then switch-on circuit breaker F14 in the CCA.

**CAUTION**

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

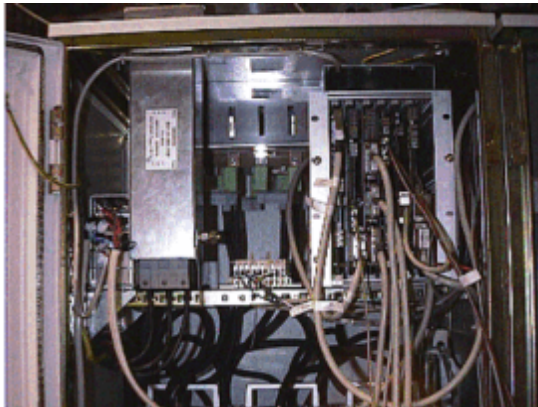
- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

- Switch-on the contactor K2 by pressing S402 on D16/GSSU, reset the scanner and perform the required adjustments.
- Ensure that the fan at E4A and GSSU is functioning properly.

3.1.4 Parts Location

3.1.4.1 FRU's in the upper part of the GPA Cabinet

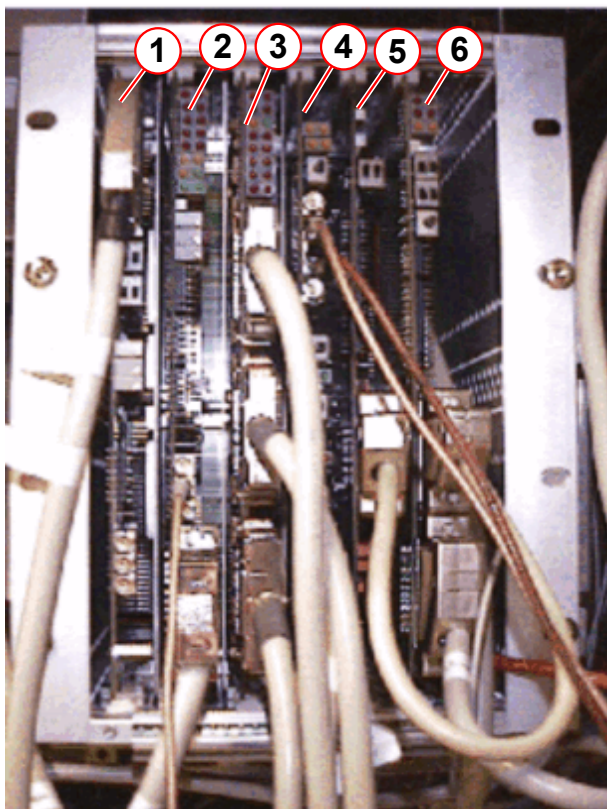
Fig. 55:



Components from left to right

- Power Supply E4A
- Line input filter
- Connector terminal X100
- GSSU
- In the background: Current sensor assembly

Fig. 56: Gradient small signal unit (GSSU) K2217



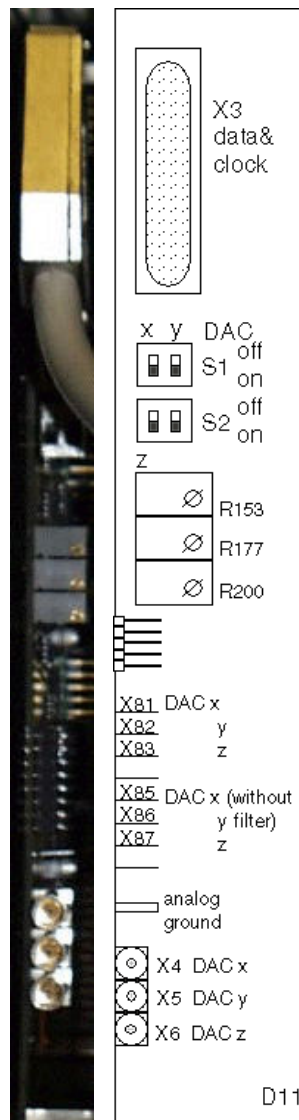
- (1) D112
- (2) D12
- (3) D14
- (4) D16
- (5) D13
- (6) D17

GSSU boards from left to right:

- DAC D112 (replaces D111, D11)
- Regulator D12 (D124 for GPA light)
- Modulator D14 (D142 for GPA light)
- CAN-Service D16
- Stimulation Monitor D13 (not used with NUM4)
- I/O Board D17

3.1.5 DAC D112/111/11

Fig. 57: Front of DAC board D11



Descriptions:

X3:

18 bit parallel gradient data and 4MHz clock signal from MMC

S1/S2:

The DAC output can be switched on/off for every axis; S2/2 is not used.

Potentiometers for offset adjustment

DAC output = nominal value (0-> +/-10V) 10 V corresponds to the maximum output current.

Check Jumpers and Switch Positions:

- Jumper X16: 2-3 (sign of gradient data, 2-Complement)
- DIP switch S1: both in lower position
- DIP switch S2: both in lower position
- Rotary switch S3: position 3
- Rotary switch S4: position 6 (this configuration results in a clock signal FSYNC = 20 kHz)

Offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes.



Prerequisite is the Regulator Adjustment (Tune-up).

- Set switches S1/2 to up position ("DAC output zero")
- The corresponding modulator must be on: S1/2 on D17 in down position.
- Connect a DVM to the desired actual value output X11,12,13 on D12.
Use the QLA-BNC service cable.
- Check the offset: Offset < +/- 0.05 mV
- Use the corresponding R153, 177, 200 on DAC-board D11 to adjust the offset for the X, Y and Z axes if necessary.
 - » If the offset was changed, perform the Tune-up steps Phantom Shim and Gradient Sensitivity again

Run TuneUp:

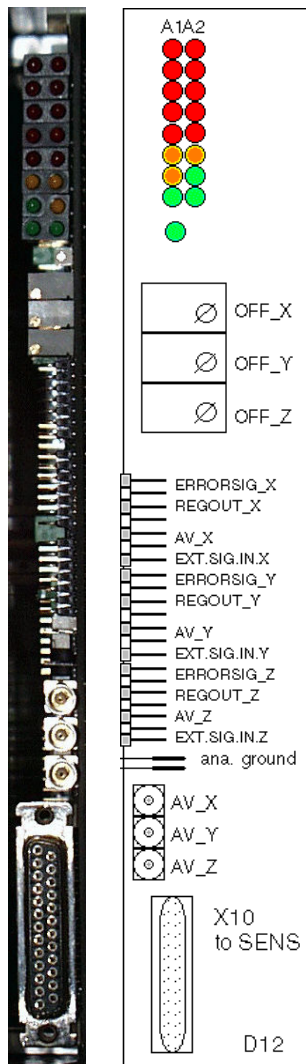
- Phantom Shim
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Phantom Shim Check
- Calc Artefacts

3.1.6 Regulator D12/123/124

Fig. 58: Front of Regulator D12



Descriptions:

LED's (A1 left / A2 right):

V130: IMAX_X / IMAX_Y

V131: IMAX_Z / CAV_ERR_X

V132: CAV_ERR_Y / CAV_ERR_Z

V133: DUCY_X / DUCY_Y

V134: DUCY_Z / OHMLOSS

V135: LOOP_ERR_X / LOOP_ERR_Y

V136: LOOP_ERR_Z / REG_ON_X

V137: REG_ON_Y / REG_ON_Z

V159: current flow

Offset pots

X: R 155

Y: R 189

Z: R 223

Test points

X11,12,13: Actual current value X,Y,Z

QLA (1V=50A)

X10 SENS connection

Default Regulator Setting by DIP Switches:

- Check the DIP switches according to the following table:

DIP-switch	Pos1	Pos2	Pos3	Pos4	Pos5	Pos6	Pos7	Pos8
S1,S2,S4 S5,S7,S8	ON	off	off	off	off	off	off	off
S3,S6,S9	ON	ON	ON	off	off	off	off	off

Check Jumpers:

- Check the jumper setting according to the table below:

Jumper	Setting	Function
X16, X18, X20	1-2	Nominal value X,Y,Z from D11

X17, X19, X21	1-2	Current regulation normally
X22/X23	1-2	Processor controls DAC gain (= GPA current configuration)
X24	1-2	Switching depending from current direction
X25	18 - 19, 21 - 22, 24 - 25	

SENS Offset measurement/adjustment:

- Gradient cabinet has been on for at least 10 minutes and the regulators are adjusted (see Regulator Adjust/Check below)
- Connect the DVM to X11,12,13 ("actual value") on the regulator board. Use the QLA-BNC service cable.
- Set the switch, corresponding to the axis, on the I/O board D17 in upward position ("Modulator off").
- Check the offset: Offset < +/- 0.05 mV; use offset pots to adjust the offset if necessary.
 - » If the offset was changed, perform the Tune-up steps regulator adjustment, Phantom Shim, Grad Delay and Grad Sens

Run TuneUp:

- Regulator Adjustment
- Phantom Shim
- Grad Delay
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check
- Phantom Shim Check
- CTC
- ECC
- Grad Delay Check
- Calc Artefacts

3.1.7 Modulator D14/142

Check Jumpers:

- Check the jumper setting according to the table below:

Jumper	Setting
X9	1 - 2
X10	1 - 2

X11	6 - 7, 18 - 19
X12	1 - 2
X13	1 - 2

Perform Quality Assurance:

- Coil Check (SN DIP)

3.1.8 CAN-Service D16

Check Jumper Settings:

- Jumper X3: 1 - 2
- Jumper X601: 1- 2

Perform Quality Assurance:

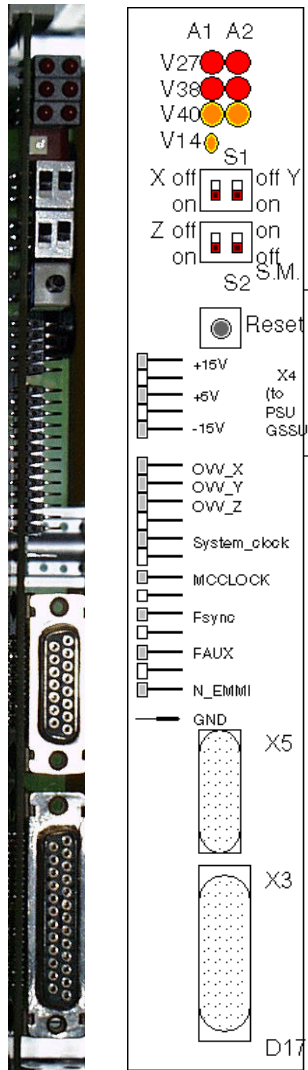
- Coil Check (SN DIP)
- Regulator Check

3.1.9 I/O D17



The I/O board D17 is the interface of GSSU to the switch-on assembly D31, as well as the SENS (synchronization via three FOC's) and supervision assemblies (e.g. D9, coil supervision) and the stimulation monitor. Electric signals are converted into FOC signals and vice versa.

Fig. 59: Front of I/O board D17

**Descriptions:**

LED's (A1 left / A2 right):

V27: Rectifier -Temp. / Transformer-Temp.

V38: Coilerr / Filterfan

V40: EMMI / DB_WDG_ERR (stim.mon.)

V14: Yellow LED: Service mode

Switches for modulator on/off and service mode on/off

Error Reset

Test Pins:

+/-15V, +5V

Overvoltage X,Y,Z

System clock 4 MHz

Modulator clock 20 kHz

Synchron.clock 20 kHz

Auxiliary clock 50 kHz (not used)

X5 Connector to Software stimulation monitor CCA/D21/X4

X3 Connector to switch-on assembly D31

Check Jumper and Switch Settings:

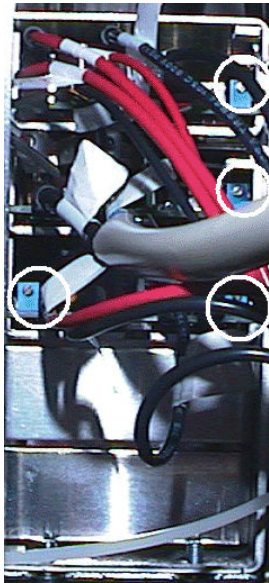
- Jumper X9: 11 - 12, 19 - 20, 21 - 22
- Switch S1: both down
- Switch S2: both down

Perform Quality Assurance:

- Coil Check (SN DIP)

3.1.10 Small Signal Power Supply E4A

Fig. 60: Adjustment of supply voltages



Adjustment of Supply Voltages:

- Adjust the following voltages:

Slot 1: +5,075 +/- 0,025 VDC

- Measure the voltage on D17 (→ Fig. 59 Page 70).

Slot 2: +16,5VDC, supply voltage for drivers at D21 on the Power Stages and fan on top of GSSU:

- Measure the voltage at the terminals of E4A (→ Fig. 60 Page 71).

Slot 3: +/- 15,05 +/- 0,025VDC

- Measure the voltage on D17 (→ Fig. 59 Page 70).
 - » Adjust the voltages with switched off modulators using the marked potentiometers (→ Fig. 60 Page 71).

Slot	Terminal at E4A
Slot 1	A = 5V, B = GND, C = 5V (Sense), D = GND(Sense)
Slot 2	E = +Blower, F = GND Blower, G = +Driver supply, H = GND Driver supply
Slot 3	I = +15V, J = GND, K = GND, L = -15V

Perform Quality Assurance:

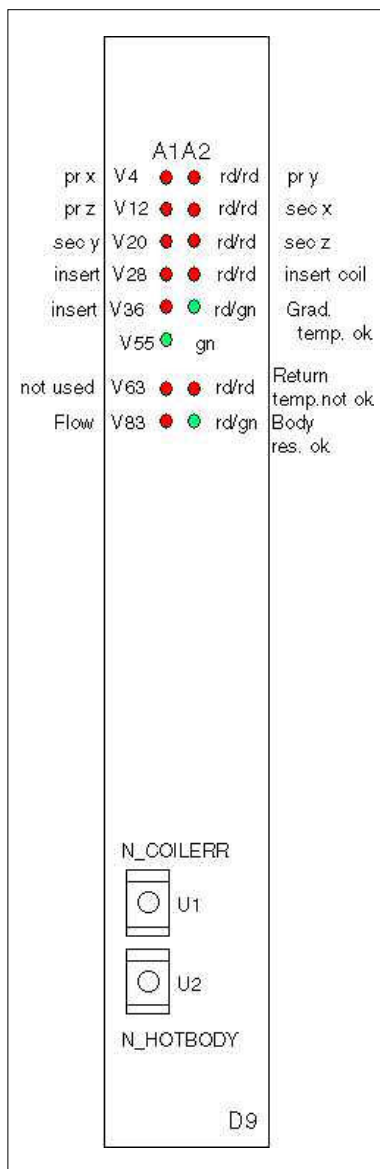
- Coil Check (SN DIP)

3.2 Supervision D9



The coil supervision board D9 is located in the patient table control electronics rack, at left-side.

Fig. 61: Front of Supervision D9



Descriptions:

LED's:

pr x,y,z:

NTC-sensor status at primary side of the gradient coil

sec x,y,z:

NTC-sensor status at secondary side of the gradient coil

V55gn: 22V supply voltage ok

V63 rd right:

Return temperature status of cooling water for the body resonator (Temperature too high)

V83 rd left:

Water flow supervision for the body resonator (Flow not ok).

V83 gn right: Body coil ok

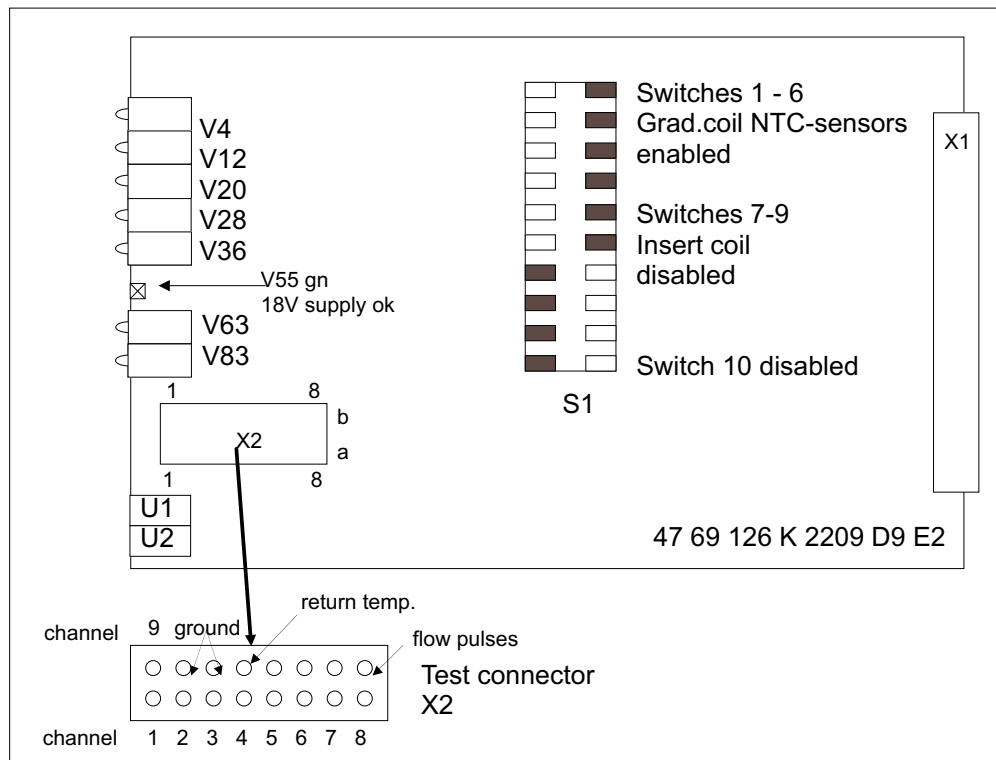
N_COILERR:

FOC connection to D7/U41 (Light off or glowing weakly: => Error)

N_HOTBODY

FOC connection to BTB1/U5 (Light off => Error/ Body Coil not ok)

Fig. 62: Position of DIP switches at D9

**Check Jumpers and Switch Positions:**

- Switches 1-6 = ON
 - Switches 7-10 = off
- (Switch 1 => pr x, switch 2 => pr y, a.s.o.)

Perform Quality Assurance:

- Coil Check (SN DIP)

3.3 Power Stage Supply Components

3.3.1 Safety Information



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

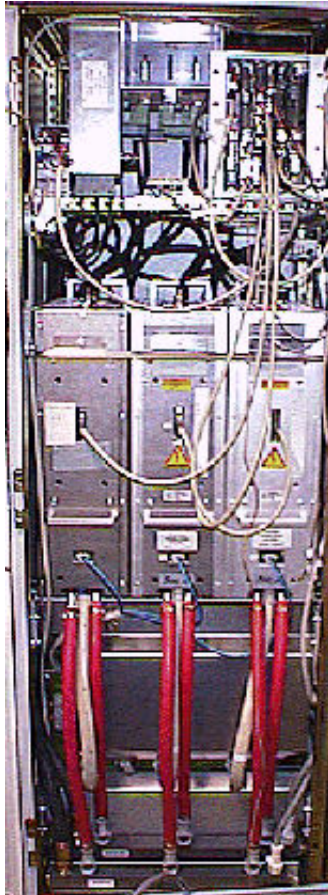
- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/4A (GSSU) in the CCA.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the following procedures.

3.3.2 Required Tools

- Set of standard tools

3.3.3 Parts Location

Fig. 63: Front side of GPA_059 cabinet



Components

(from left to right)

Temperature sensor (above E4A)

Power Supply E4A & Fan

Line filter

SENS (in the back)

GSSU & Fan

Cabling, FOC's

Power Stages X,Y,Z-axis

NF filters (in the back)

5 fans

Transformer T1 (in the back)

Main contactor K2

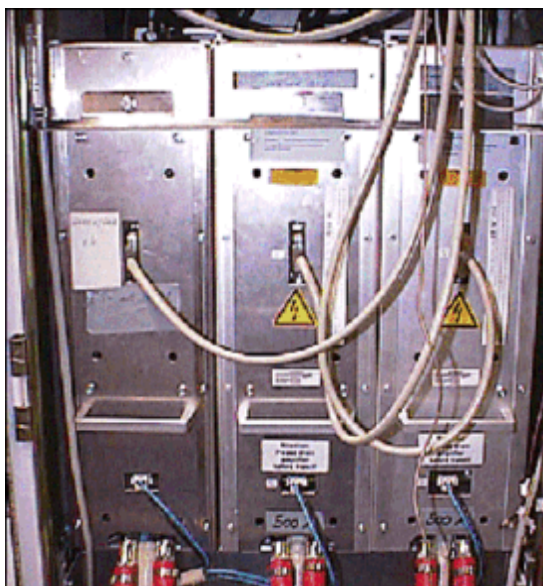
D31, Soft start unit

D32, Rectifier board

Cooling water distributor

3.3.3.1 FRU's in the middle part of the GPA Cabinet

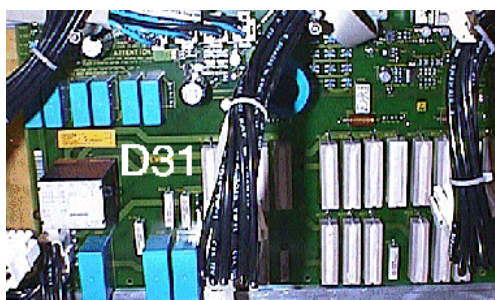
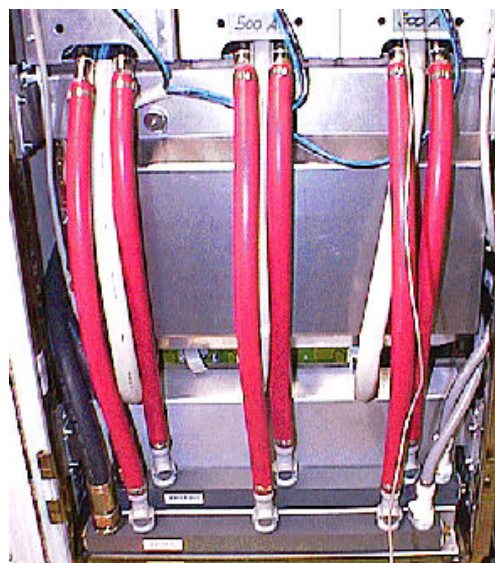
Fig. 64: Middle part of GPA cabinet



X,Y,Z-Power Stage with cabling and water hoses.

3.3.3.2 FRU's in the lower part of the GPA Cabinet

Fig. 65: Lower part of GPA cabinet (with D32 partially shown left and D31 right and main contactor K2)



Fans below the Power Stages
Water distributor with water hoses

Behind cover plates:

- Rectifier board D32 in front
- Soft start board D31 behind D32
- Main contactor K2
- 25A fuses (15fuses, slow-blow)

3.3.4 Soft Start Unit D31

(material number 57 26 414)

3.3.4.1 Safety Information



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

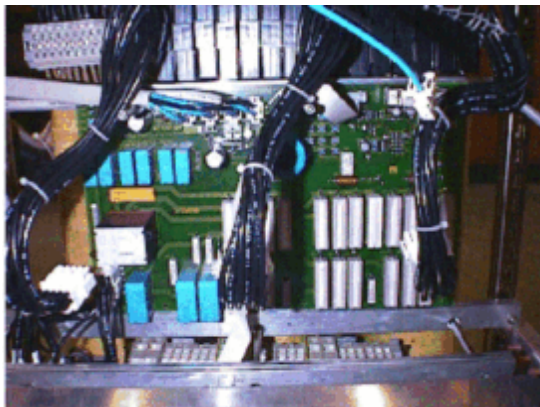
Otherwise, severe damage to the D32 and the Power Stages may occur.

- » **Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.**

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/4A (GSSU) in the CCA.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.
- Wait at least 20 minutes until the capacitors are discharged.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the D31 replacement procedure.

3.3.4.2 Procedure

Fig. 66: View at D31 (D32 removed)



- Remove D32 first; (→ Rectifier D32 / Page 78).
 - Remove connector X1 to GSSU from D31.
 - Remove connectors X10 to GSSU power supply unit from D31.
 - Remove connectors X11,12,13 to Power Stages from D31.
 - Remove flat cable connector X8 to D32 from D31.
 - Remove connector X2 to main contactor K2 from D31.
- Loosen the four fixing screws/nuts at the upper edge of D31.
 - Remove D31.
 - For the installation of D31 proceed in reversed order.



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

Perform Quality Assurance:

- Coil Check (SN DIP)

3.3.5 Rectifier D32

(material number 57 27 016)

3.3.5.1 Safety Information



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.
- Wait at least 20 minutes until the capacitors are discharged.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the D32 replacement procedure.

3.3.5.2 Procedure

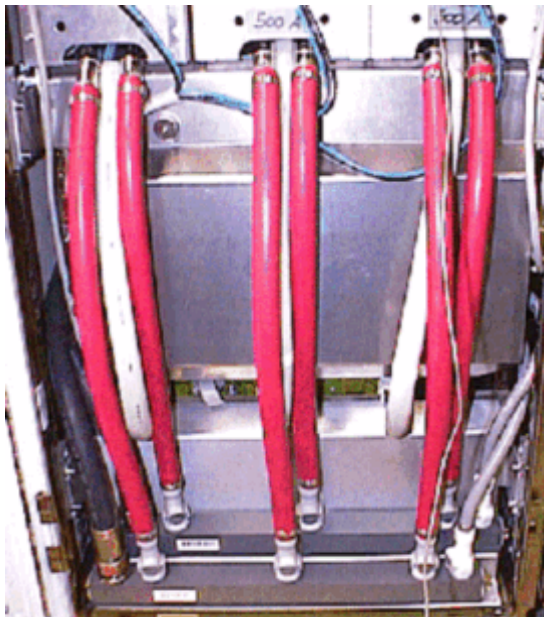
- Disconnect all quick connectors of water hoses from the water distributor.

NOTICE

The water connections shut off automatically.

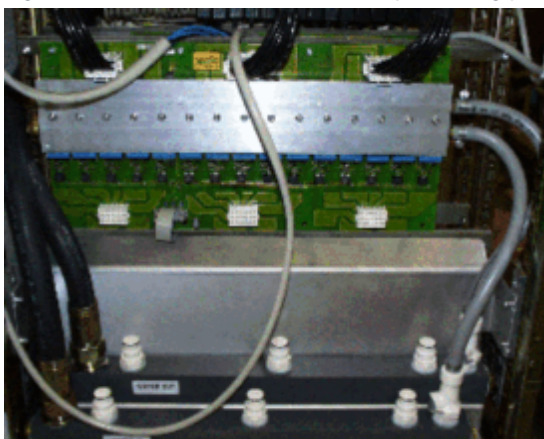
- » However, ensure that there are no water spills. Wipe up any spilled water with a cloth.

Fig. 67: Lower part of the GPA with protecting plates in front of D31



- Disconnect the DCV supply connectors to Power Stages X12, X22 and X32 from D32 (lower row of connectors).
- Disassemble the protection cover by loosening the two nuts shown in (→ Fig. 67 Page 79).

Fig. 68: View on D32 with removed protecting plate



- Disconnect the ACV supply connectors X 11, X21 and X31 (upper row of connectors of D32).
- Remove flat cable connector X1 to D31 from D32.

- Loosen the four fixing screws/nuts at the upper edge of D32.
- Remove the rectifier board D32.
- For the installation of D32 proceed in reversed order.



CAUTION

Never switch-on F3 (80A cutter switch in the mains box on top of CCA, if the GSSU is switched on.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.
- » Other wise severe damage to the D32 and the Power Stages may occur.

Perform Quality Assurance:

- Coil Check (SN DIP)



Before returning the replaced rectifier board D32, drain the water from the pipes by opening both ends.

3.3.6 Line Filter of GPA K2217

(material number 57 25 671)

3.3.6.1 Safety Information



CAUTION

Never switch-on circuit breaker F3 / 80A (at mains box, CCA roof-top), if the GSSU is switched on.

Otherwise, severe damage to the D32 and the Power Stages may occur.

- » Before switching on F3, switch-off the GSSU with circuit breaker F14 in the CCA.

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/4A (GSSU) in the CCA.
- Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.
- Wait at least 20 minutes until the capacitors are discharged.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the replacement procedure.

3.3.6.2 Procedure

Fig. 69: Line filter mounted in the upper part of the GPA cabinet



- Remove the line cables at the connectors at the bottom of the filter.
- Remove the cover on top of the GPA cabinet; (→ Fig. 70 Page 82).
- Remove the input line cables on top of the filter; (→ Fig. 71 Page 82).
- Loosen the four screws on top of the GPA cabinet; (→ Fig. 71 Page 82).
- Remove the nuts at the right side which clamp the filter housing to the GPA cabinet (via a current rail); (→ Fig. 69 Page 81).
- Take out the line filter.
- For the installation of the line filter proceed in reversed order.

Fig. 70: Cover of line filter connections on top of GPA cabinet

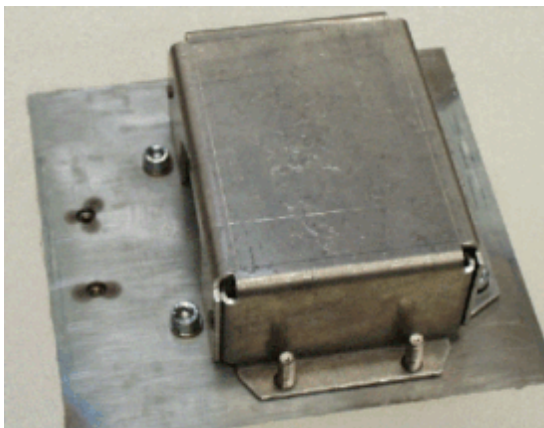
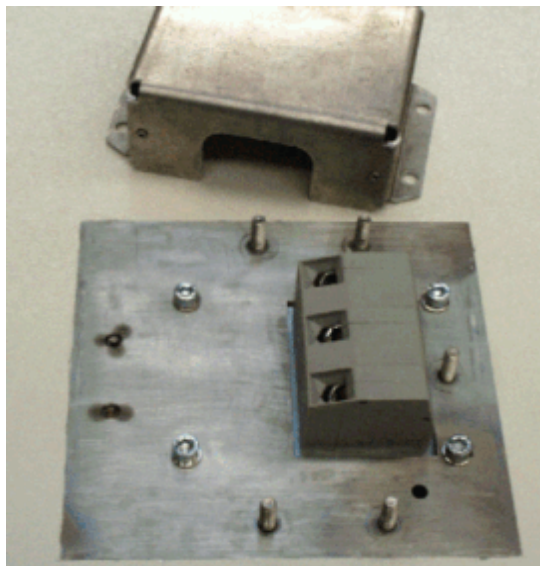


Fig. 71: Line cable connection on top of line filter



Perform Quality Assurance:

- Coil Check (SN DIP)

3.3.7 Main Contactor K2

(material number 46 73 653)

3.3.7.1 Safety Information

WARNING

Dangerous high voltage may exist across the capacitors of the Power Stage inputs after the GPA is switched off via cutter switch F3/Mainsbox CCA.

Risk of electrical shock!

- » Wait at least 20 minutes until the voltage drops to a harmless level, before working on the Power Stages.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy the Power Stages!

- » Use a towel to catch any spilled water when handling the water quick connectors.

- Check the green LED's behind the six holes in the PS front side; refer to (→ Fig. 79 Page 89).
 - » If the LED's are on, the capacitors are charged, i.e. the supply voltage 400 VDC from D32 is connected to the PS.

3.3.7.2 Preparation

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Wait at least 20 minutes until the capacitors are discharged.

3.3.7.3 Procedure



The following description refers to the worst case, i.e. the left side of the GPA cabinet is not accessible.

If the GPA cabinet is standing free in the computer room, the replacement can be facilitated by removing the left side cover of the gradient cabinet. Then you can remove the mounting plate together with the K2 by loosening the two Torx 25 screws, shown in (→ Fig. 74 Page 84).

- Remove first D32, (→ Rectifier D32 / Page 78).
- Remove the water distributor, (→ GPA Water Distributor / Page 112). The water hoses could remain connected.
- Remove connector X2 from D31; (→ Fig. 73 Page 84).
- Loosen the screws (3 mm Allen), which fix the K2 on the mounting plate; (→ Fig. 75 Page 84) and (→ Fig. 76 Page 85).
- Remove the power cables (to transformer T1, bottom side and line filter, top side) from the K2.
- Remove the cables to X2/D31 from K2.
- Remove K2.
- For the installation of the K2 proceed in reversed order.

Fig. 72: View from open left side of GPA cabinet on D31, D32 and main contactor K2

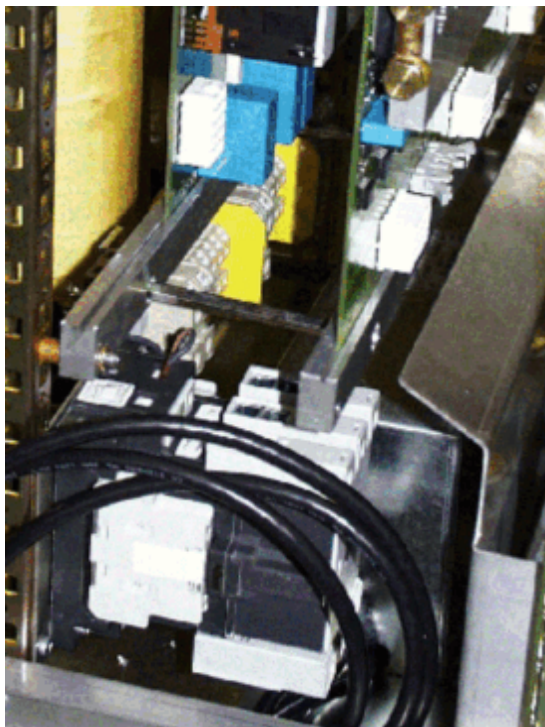


Fig. 73: Connector X2 on D31 and main contactor K2

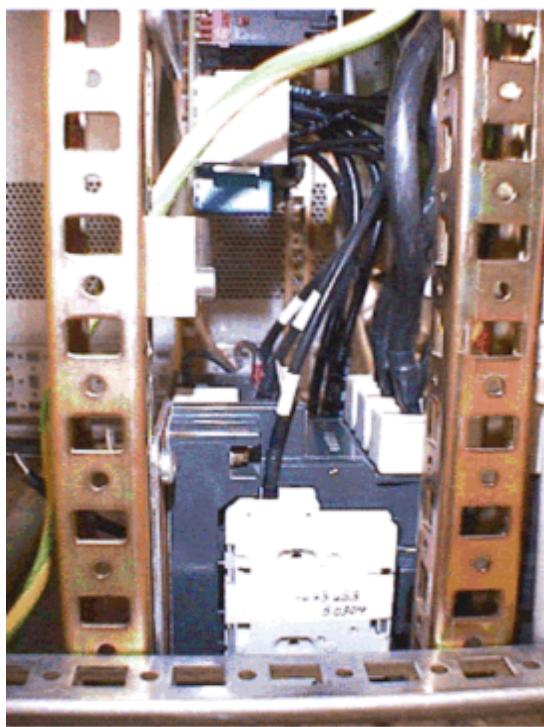


Fig. 74: Mounting plate of K2 fixed with two Torx 25 screws

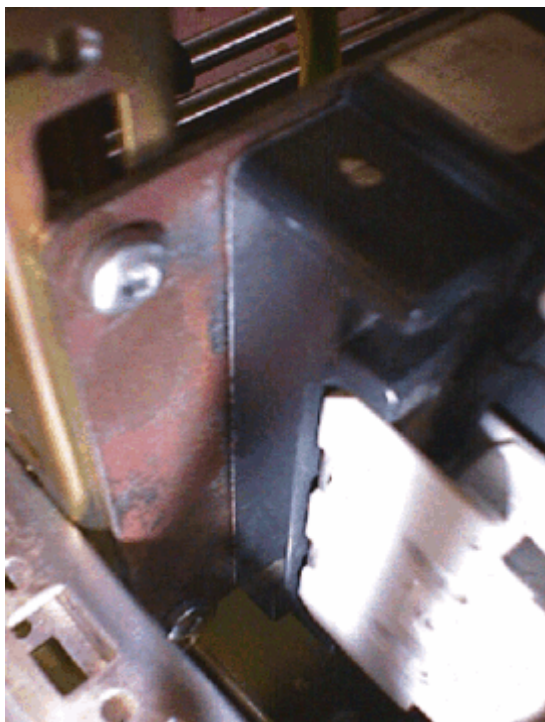


Fig. 75: Fixing screw (3 mm Allen) on lower left of main contactor

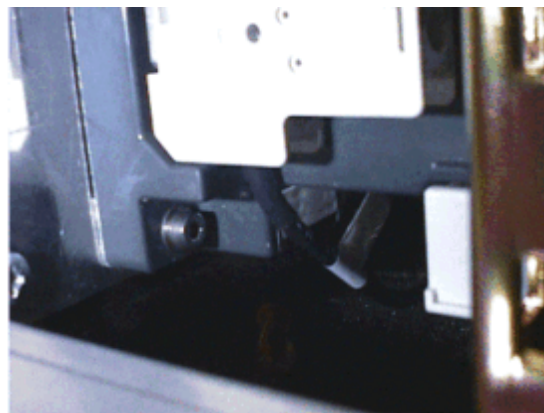
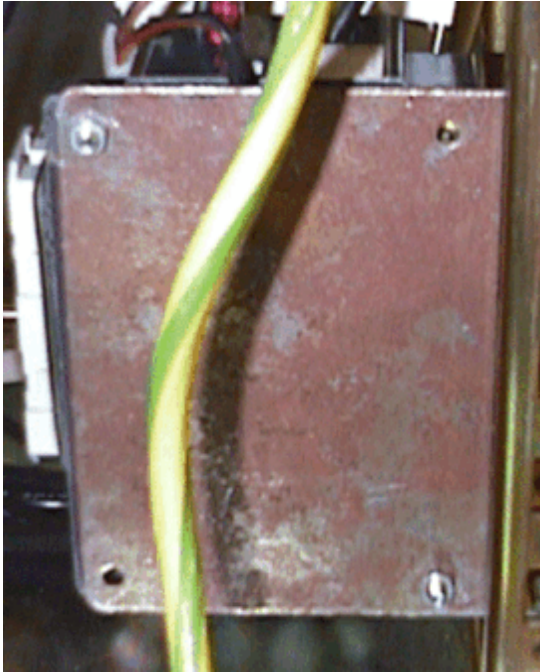


Fig. 76: Backside of K2 mounting contactor



Perform Quality Assurance:

- Coil Check (SN DIP)

3.4 Power Stages

(material number 59 38 142/ 59 38 720 "Light")

3.4.1 Safety Information

**WARNING**

Dangerous high voltage may exist across the capacitors of the Power Stage inputs after the GPA is switched off via cutter switch F3/Mainsbox CCA.

Risk of electrical shock!

- » Wait at least 20 minutes until the voltage drops to a harmless level, before working on the Power Stages.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy the Power Stages!

- » Use a towel to catch any spilled water when handling the water quick connectors.

- Check the green LED's behind the six holes in the PS front side; refer to (→ Fig. 77 Page 87).
 - » If the LED's are on, the capacitors are charged, i.e.the supply voltage 400 VDC from D32 is connected to the PS.
 - » If the green LED's behind a single hole are off during normal operation, the corresponding part of the Power Stage is probably defective.

LED X1 corresponds to the partial stage X1 of X-Power Stage; the lower to LED's (e.g. X3) belong to the same partial stage. LED's X,Y,Z2 and X,Y,Z4 are missing on the "Light"-version of PS.	LED's at PS (X,Y,Z)					
	X1	X5	Y1	Y5	Z1	Z5
	X2	X4	Y2	Y4	Z2	Z4
	X3	X3	Y3	Y3	Z3	Z3

- It is also possible, that after a single PS breakdown one of the corresponding 5 fuses between T1(sec) and D32 is blown. (→ Reassembly Procedure / Page 90).

3.4.2 Required Tools

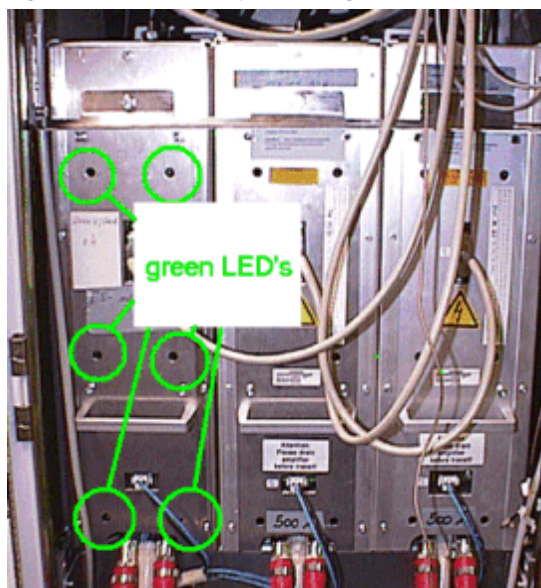
- Set of standard tools

3.4.3 Preparation

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Wait at least 20 minutes until the capacitors are discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.
- Wait at least 20 minutes until the capacitors are discharged.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the PS replacement procedure.

3.4.4 Removal Procedure

Fig. 77: View of GPA power stages

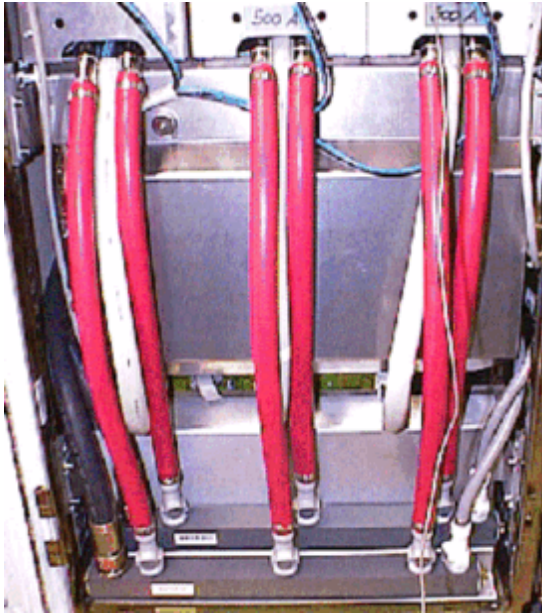


The configuration of the Power Stages from left to right:

- X - Y - Z.

3.4.4.1 Removing the Water Hoses

Fig. 78: Water distributor and water pipes to PS and Rectifier D32



- Remove the quick connections of the water hoses from the inlet and outlet sides. (→ Fig. 78 Page 88)

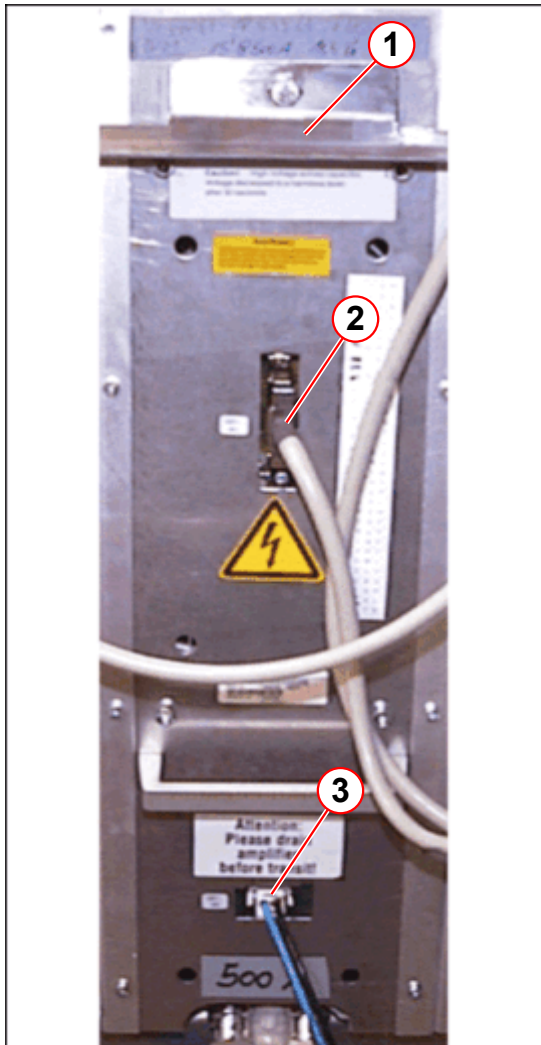


The water connections shut off automatically.

However, ensure that there are no water spills and that splashes of water are cleaned up with a cloth.

3.4.4.2 Removing the Connectors and Cabling at PS and D32

Fig. 79: Connectors at PS

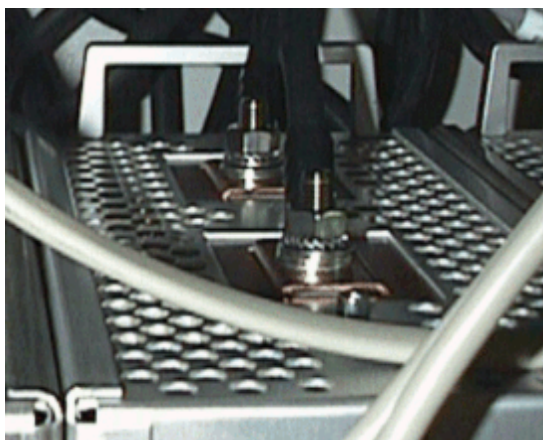


- (1) Ground rail
- (2) X1
- (3) X2

- Disconnect the corresponding DCV supply connectors X12,22 or 32 from D32.
- Disconnect the driver supply connector X2 from the Power Stage; (→ 3/Fig. 79 Page 89).
- Disconnect the 50-pin connector X1 from the Power Stage; (→ 2/Fig. 79 Page 89).
- Loosen the gradient cables on top of the Power Stage (open-end wrench SW 19); (→ Fig. 80 Page 90).

3.4.4.3 Removing the Power Stage

Fig. 80: Power cable connection on top of power stage



- Remove the ground rail across the three Power Stages; (→ Fig. 79 Page 89).
- Remove the Power Stage with the help of the handles on top and front side. (Weight of the PS is approximately 30 kg).

3.4.5 Reassembly Procedure

Fig. 81: All power stages removed



- Slide in the replacement Power Stage. The guide cones shown in (→ Fig. 81 Page 90), engaged in a hole at the back of the PS, hold the Power Stage in position.
- Attach the securing rail across the three Power Stages; (→ Fig. 79 Page 89).



To prevent damage to the Power Stage by faulty control of the IGBT switches, inspect the "SCSI-like" control cable from D14 to the Power Stage.

Check for damages and broken connector pins.

Replace defective cables!

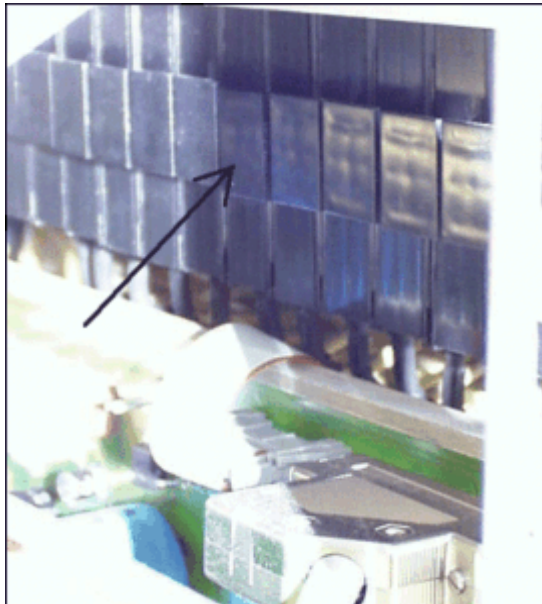
- Reconnect the gradient cables and all other connectors
- Reconnect the DCV supply connector at D32.
- Connect the water hoses of the Power Stage with the water distributor as shown in (→ Fig. 78 Page 88).
 - » The longer, left water hose of the Power Stage is connected with the inlet of the water distributor.

3.4.6 Additional Fuse Check

If one or more Power Stages has blown off, it is very likely that at least one of the corresponding 15 fuses (25A slow) between T1(sec) and D32 has been blown and/or that the D32 rectifier board is damaged on the corresponding axis:

- Get access to the fuse holders by removing the protection cover in front of D32, shown in (→ Fig. 67 Page 79).
- Pull the five fuse holders of the corresponding PS and check the fuses.
Replace defective fuses (material number: 30 77 476).
The configuration of fuses is from left to right:
X1...X5, Y1...Y5, Z1...Z5 (according to the partial Power Stages). (→ Power Stages / Page 86).
- In case of blown fuses, replace the D32 too. (→ Rectifier D32 / Page 78).

Fig. 82: Fuse holders above D32 (behind the cover)



3.4.7 Final Steps

Run TuneUp:

- Regulator Adjustment
- Phantom Shim
- Grad Delay
- Grad Sens

Perform Quality Assurance:

- Coil Check (SN DIP)
- Regulator Check
- Phantom Shim Check
- Grad Delay Check

- Calc Artefacts
- Image Orientation



Before returning the replaced Power Stage, drain the water from the pipes by opening both ends.

3.5 Current Sensor Assembly

(Old material number 57 25 655)

(New Current converter unit 059: 10743890)

3.5.1 Safety Information



WARNING

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch off the gradient cabinet with circuit breakers. (Both GPA Power Stage supply F3/ mains box and GSSU Power Supply F14/ CCA)



WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.



Check the green LED's behind the six holes in the PS front side (→ Fig. 77 Page 87). If the LED's are on, the capacitors are charged! Newer final stages models may not have the LEDs fitted any more!

3.5.2 Estimated Completion Time and Number of Persons

Tab. 6 Completion time

Work steps	Time	Persons
Preparations	30 min.	1
Replacement / Installation	210 min.	1
Startup/TuneUp/QA	105 min.	1
Final steps	n.a.	1

3.5.3 Aids or Tools

- SW19 open-ended wrench with extension and ratchet
- Torque wrench 20-50 Nm (material number 8395803) for 25 Nm
- 10 mm, 4-mm Allen wrenches
- Torx 25 bit
- Standard tool set
- Special tools for gradient connection (material number 10563214)

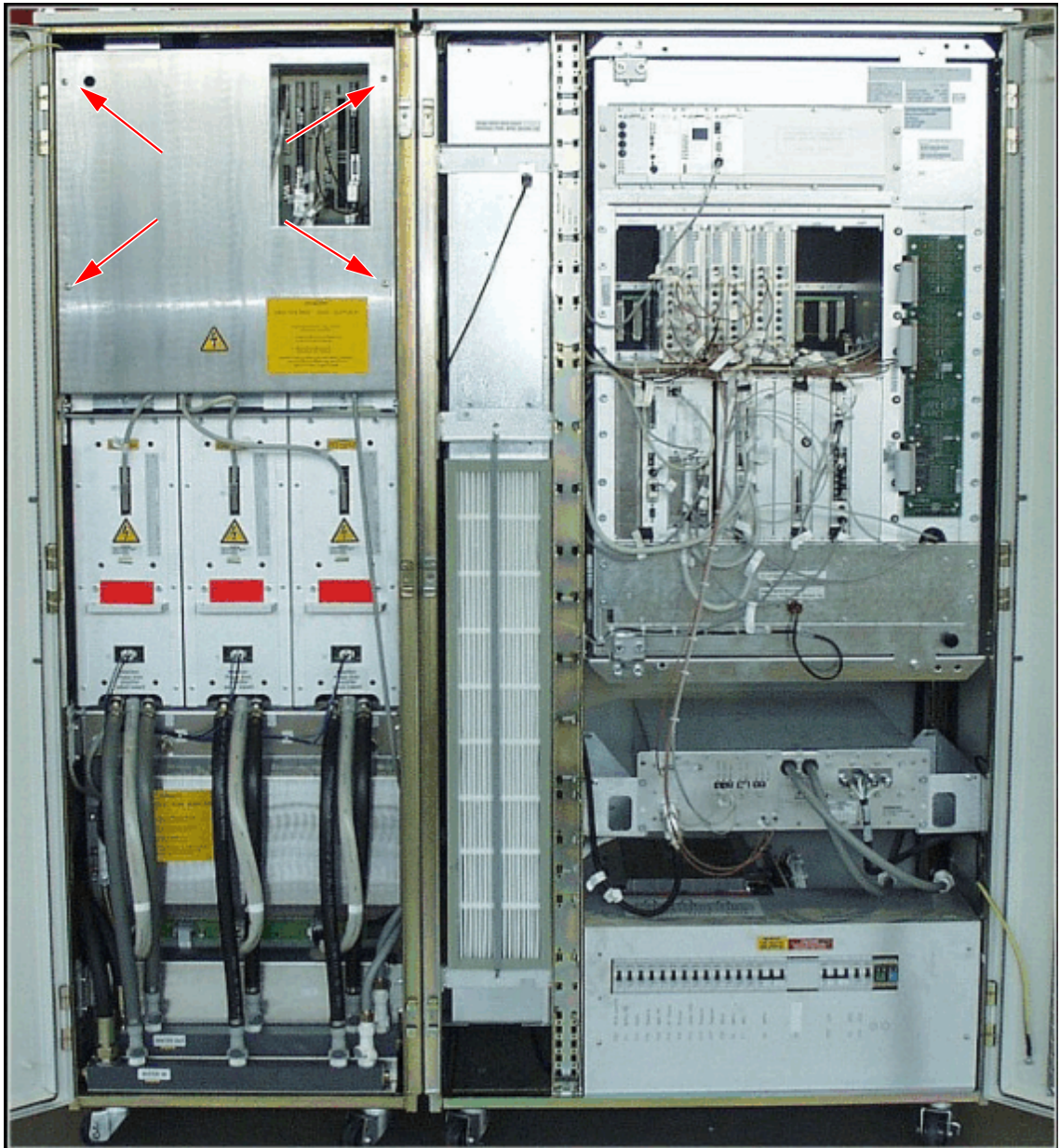
3.5.4 Preparation

- Switch-off the transformer T1 by pressing S403 at the D16/GSSU.
- Switch-off circuit breaker F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Wait 20 minutes until the capacitors in the Power Stages are fully discharged.

3.5.5 Removing the Current Sensor Assembly

- Remove the cover plate in front of the GSSU; 4 quick lock screws.

Fig. 83: GSSU cover plate

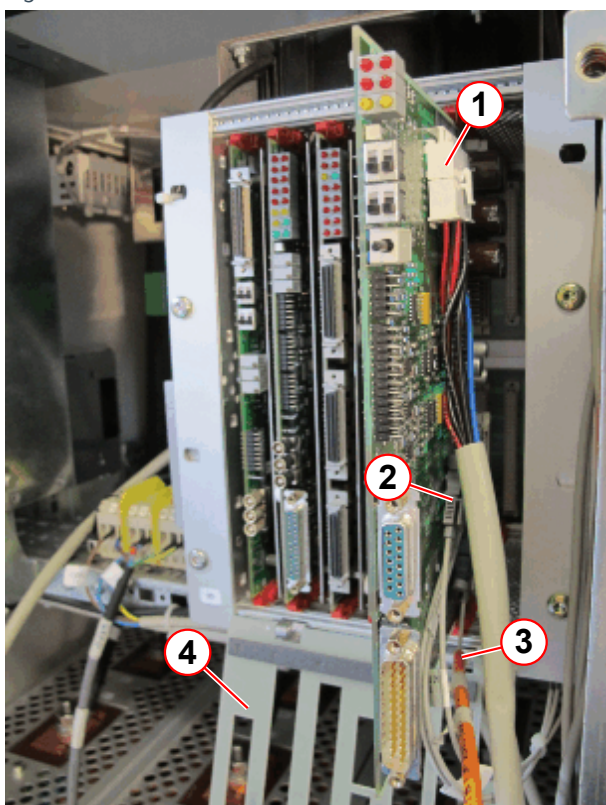


3.5.5.1 Removing the GSSU



For GSSU removal, you do not have to pull out any boards (except D17- for removing connector X4 and FOC's). Use the ESD kit.

Fig. 84: D17 in the GSSU



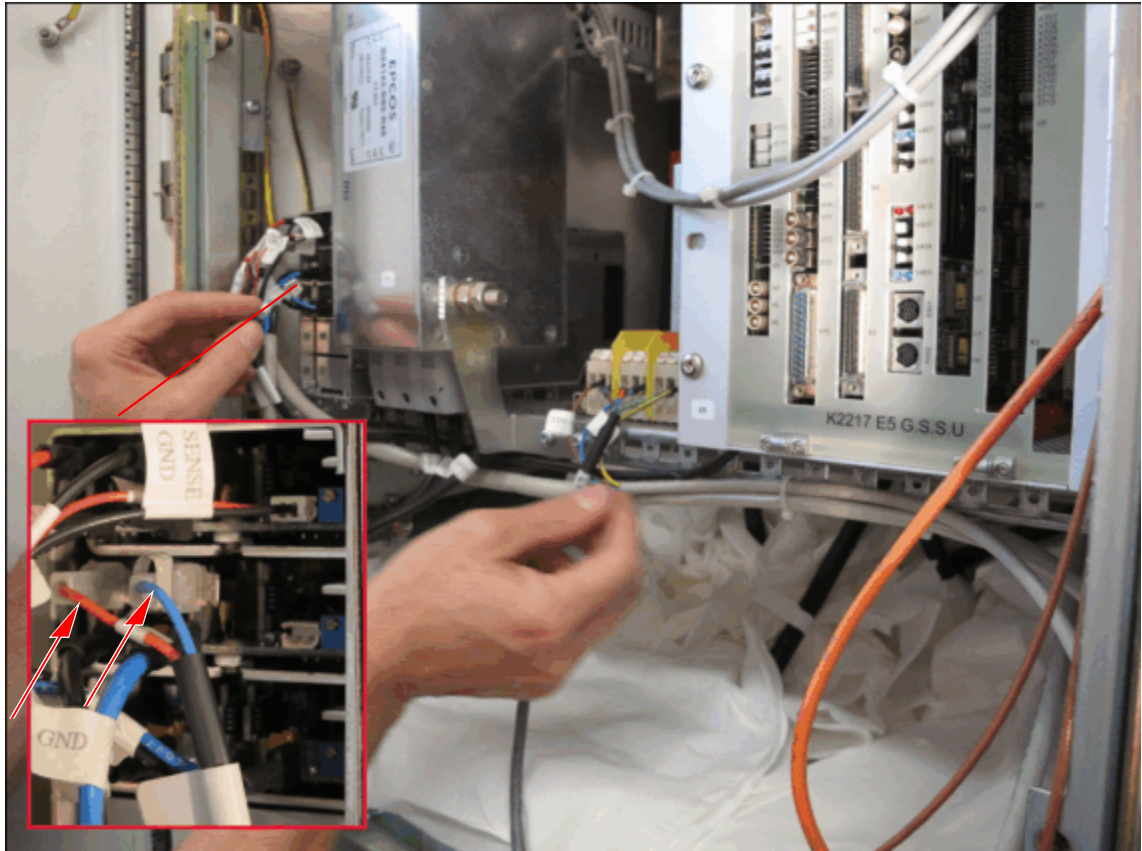
- (1) X4
- (2) FOC's U1, U2, U3
- (3) U7
- (4) Cover plate GSSU

- Remove all signal cables and FOCs from the GSSU boards
- Remove the cover plate of the GSSU (Pos. 4).
- Remove the board D17 carefully and disconnect X4 (Pos. 1) and FOC's U1,2,3 (Pos. 2) to the current sensor assembly and U7 (Pos. 3)

- The GSSU fan must be disconnected from the GSSU power supply unit
- Cut the cable ties of the fan power cable.

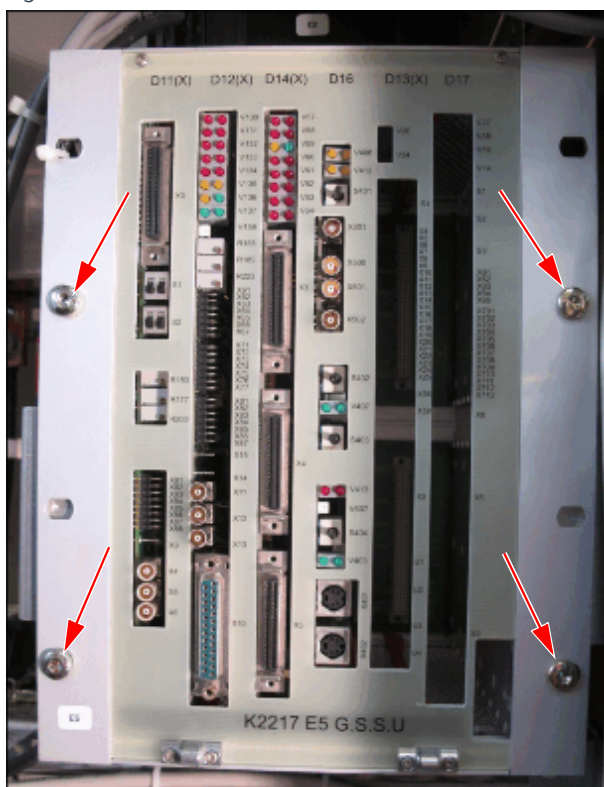
- Disconnect the fan cable from the GSSU power supply unit (→ Fig. 85 Page 97)

Fig. 85: Fan power connection



(1) Power lines fan

Fig. 86: GSSU disconnect



(1) quick lock screws

Remove the GSSU completely to access the cable connections to the current sensor assembly..

- Loosen the quick lock screws.(Pos. 1)
- Remove the complete GSSU rack (together with the fan) and store it safely.

3.5.5.2 Connection at GPA roof-top



The type of gradient cable and cable cover depends on the installation position of the GPA with respect to the RF-filter panel.



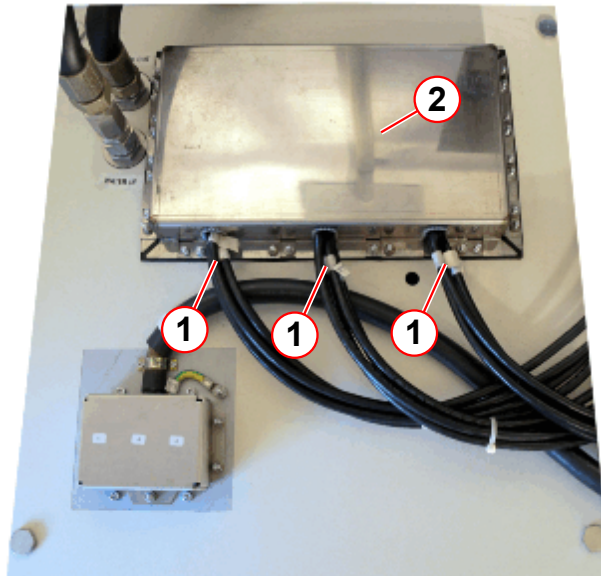
For systems with the short cable set, the gradient connection screws at the GPA roof are difficult to reach.

Use an extra short-head 10-mm Allen key. Special tools for this connection can be also ordered from SPC.

These tools are fully magnetic and must be only used in the equipment room.

Shorter Cable Set 2.5 m - horizontal cable outlet unshielded version.

Fig. 87: gradient cable on top of the GPA cabinet (short cable set)



- (1) Gradient cables
- (2) cover of the gradient cable

- Remove the nuts M6 for the gradient cover to get access to the gradient connection. Use a socket wrench.

Fig. 88: Gradient connection on top of the GPA (short cable set)



- Loosen the gradient connection screws with a short-head 10- mm Allen key (included in special tools for gradient connection, material number 10563214). .
- Remove the nuts M6 nuts of the current sensor unit.

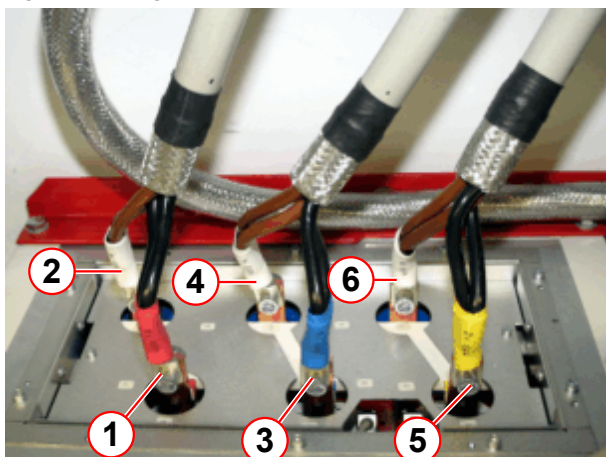
Long cable Set- vertical cable outlet

Fig. 89: Gradient cables on top of the GPA cabinet (long cable set)



- Remove the GPA top cover (including the cable clamps).

Fig. 90: GPA gradient cable connection



- Remove the gradient cable connection screws.

- (1) X+
- (2) X-
- (3) Y+
- (4) Y-
- (5) Z+
- (6) Z-

3.5.5.3 Disconnecting the Gradient Cables in the GPA



Before performing the next steps, cover the Power Stages with a piece of cardboard or paper to avoid screws dropping into the cabinet.

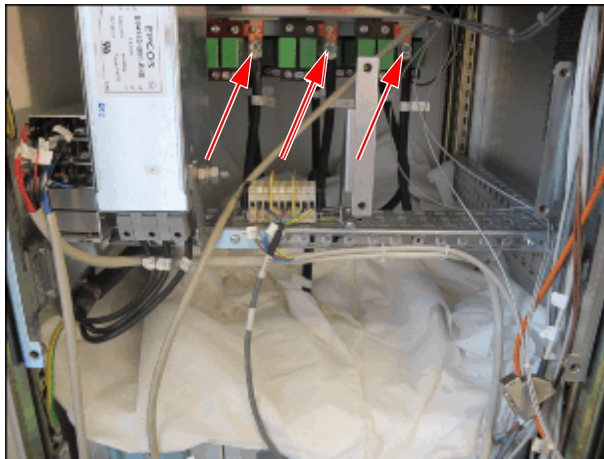
NOTICE

If a metallic part (e.g. nut, washer) drops into a Power Stage it must be removed prior to switch-on!

Otherwise, risk of severe damage to the Power Stage!

- » Turn the Power Stage upside down and shake it carefully until the missing part falls out.

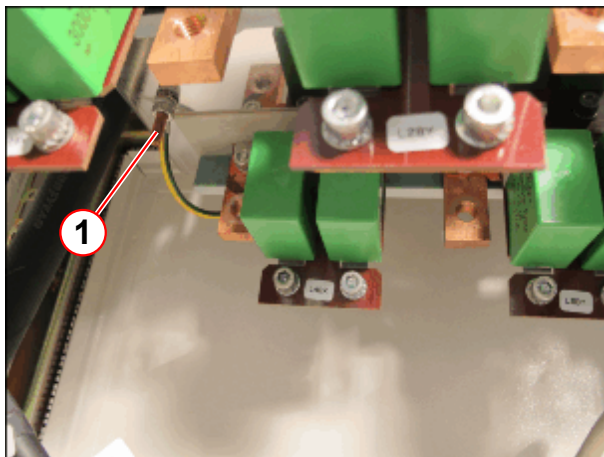
Fig. 91: Protected power stages



(1) Front gradient connection

- Loosen the screw connections (M10 x 8-mm hex socket head) of the gradient cables.

Fig. 92: Grounding of the Current sensor unit



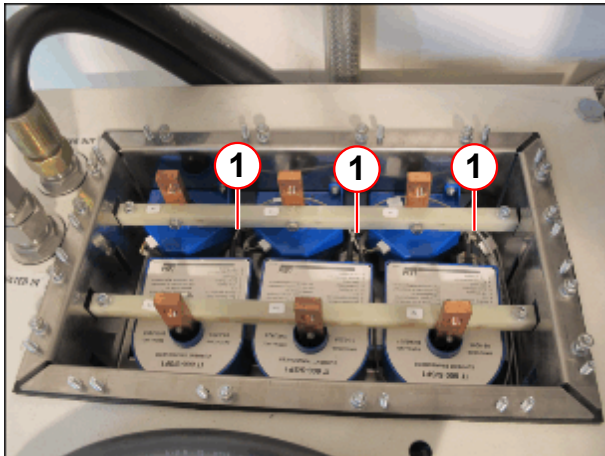
(1) Grounding bolt

- Loosen the ground wire at the current sensor unit inside the GPA cabinet. (Pos. 1)



Disconnect the 3 FOCs at the current sensor. These FOCs are not used for LEM current sensor material number 10743890.

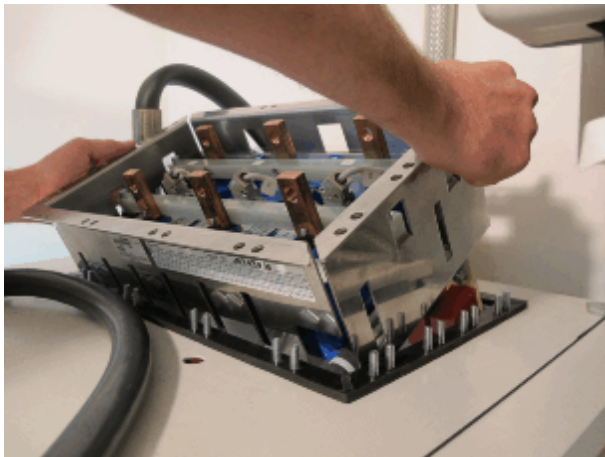
Fig. 93: LEM with FOC's



(1) FOC

- Remove the 3 FOC's (Pos. 1)

Fig. 94: Removing the current sensor unit



- Pull out the entire current sensor assembly from the top of the GPA cabinet.
- Tilt the rack (like shown in the picture) to protect the cable to the CCU.

3.5.6 Installation

Proceed in reverse order of removal:

- Insert and secure the new current sensor assembly



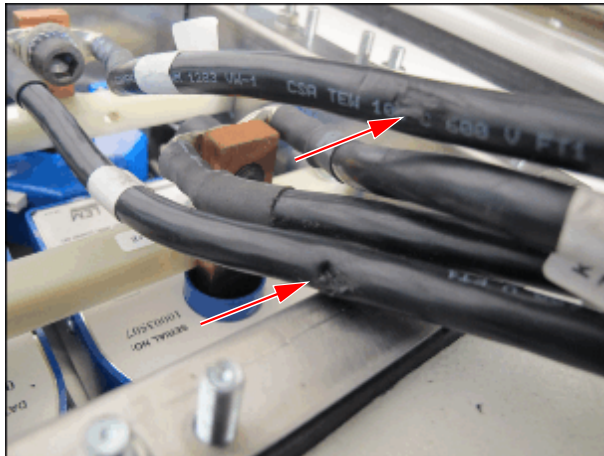
CAUTION

Risk of existing damaged gradient cable insulation (caused during former installation).

Carefully check the gradient cables (visual control) for existing damages.

- » In case of damage, repair the defect temporarily and replace the defective cable.

Fig. 95: Cable damage



- Take care not to damage the insulation when fitting the cover over the cables.

Fig. 96: Heat shrink



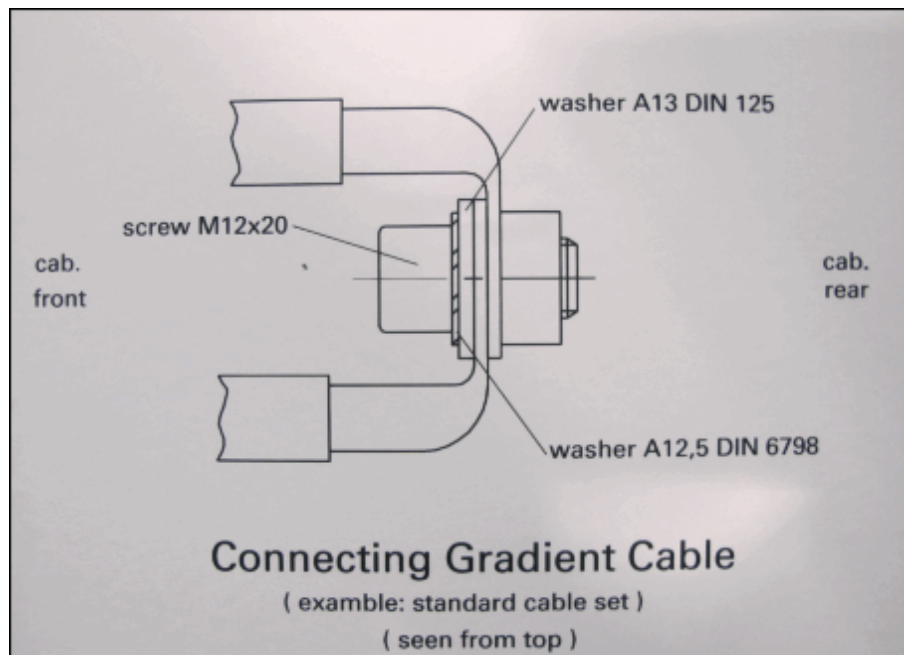
- In case of damaged insulation, repair the defect with heat shrink tube temporarily.
- Push the shrink tube over the cable from the RF-filter plate side.
- Shrink the tube with a heat gun.
- Replace the defect cable as soon as possible.

(1) Heat shrink



Ensure that the screws, washers and nuts are assembled in the correct sequence. See also the sticker at cabinet door.

Fig. 97: Connecting gradient cable



- Re-install the ground cable (only for shielded cable)



The cable clamps must fasten the shield only.
Ensure that the insulation is not damaged.

- Tighten the screws at GPA roof- top.



Metal dust can disturb the current sensor function!
After fastening all gradient cables, carefully clean the entire SENS housing and remove any residual metal dust.

- Install the GPA cover
- Reinstall the cables at the bottom of the current sensor assembly.



- Fasten the screws with a torque wrench.

» Apply a torque of 25 ± 1 Nm.

- Reinstall the GSSU and all cable connections.
- Reinstall the front cover of the GSSU.

- Switch-on the system



CAUTION

Switching on the GPA circuit breakers in wrong sequence can damage GPA.

Circuit breaker F3/ mains box **MUST** be switched on **BEFORE** powering up the GSSU with F14/ CCA.

Otherwise severe damage to the Power Stage Supply and/or Power Stages is possible!

- » First switch-on circuit breaker F3 / mains box, then F14/ CCA.

3.5.7 Check the Protective Earth Connection

- Measure the protective earth connection resistance ≤ 100 mOhm with a ground test meter (Fluke 1503, or equivalent) between the protective earth connector on the front cover and the protective earth wire connected to the GPA-frame.

3.5.8 Final Steps

- Offset adjustment on D12
(refer to (→ Regulator D12/123/124 / Page 67))

Perform the following **TuneUp steps** after replacing the current sensor unit

- Regulator Adjust
- Table Adjustment
- Phantom Shim
- CTC
- ECC
- Grad Delay
- Grad Sens

Perform the following **QA steps** after replacing the current sensor unit

- Image Orientation Check
- Regulator Check
- Shim Check
- CTC Check
- ECC Check
- Gradient Delay Check
- Grad Sens Check
- Calc Artifacts
- Spike Check
- Stability Check

- FatSat Check

3.6 Chokes (LF-Filters)

(material number 57 25 887)

3.6.1 Safety Information



WARNING

Dangerous high voltage may exist across the capacitors of the Power Stage inputs after the GPA is switched off via cutter switch F3/Mainsbox CCA.

Risk of electrical shock!

- » Wait at least 20 minutes until the voltage drops to a harmless level, before working on the Power Stages.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy the Power Stages!

- » Use a towel to catch any spilled water when handling the water quick connectors.

- Check the green LED's behind the six holes in the PS front side; refer to (→ Fig. 79 Page 89).
 - » If the LED's are on, the capacitors are charged, i.e. the supply voltage 400 VDC from D32 is connected to the PS.
 - » If the green LED's behind a single hole are off during normal operation, the corresponding part of the Power Stage is probably defective.

LED X1 corresponds to the partial stage X1 of X-Power Stage; the lower to LED's (e.g. X3) belong to the same partial stage. LED's X,Y,Z2 and X,Y,Z4 are missing on the "Light"-version of PS.	LED's at PS (X,Y,Z)					
	X1	X5	Y1	Y5	Z1	Z5
	X2	X4	Y2	Y4	Z2	Z4
	X3	X3	Y3	Y3	Z3	Z3

- It is also possible, that after a single PS breakdown one of the corresponding 5 fuses between T1(sec) and D32 is blown. (→ Reassembly Procedure / Page 90).

3.6.2 Required Tools

- Set of standard tools

3.6.3 Preparation

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Wait at least 20 minutes until the capacitors are discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.
- Wait at least 20 minutes until the capacitors are discharged.
- Adhere to (→ General Safety Notes / TD00-000.860.01) in the safety chapter, when performing the replacement procedure.

3.6.4 Procedure

Fig. 98: The six LF-filters in the back of GPA cabinet - PS's are removed



- You have to remove all Power Stages first; (→ Power Stages / Page 86).
- For disconnecting the cables from the current sensors, proceed as described on (→ Disconnecting the Gradient Cables in the GPA / Page 100).
- Loosen the screws with hold the supporting rails; (→ Fig. 98 Page 108).

NOTICE

Always both LF-filters of a gradient axis circuit have to be replaced.

- » Replace the corresponding Power Stage too.

Fig. 99: Temperature switch at one of the upper LF-filters

- Remove the relevant LF-filter.



The upper three filters might have built in temperature switches
Disconnect the temperature-switch cables before LF-filter removal (→ Fig. 99 Page 109).

- For the installation of the LF-filters proceed in reverse order.

Perform Quality Assurance:

- Image Orientation

3.7 Fans

(material number 30 63 476)

3.7.1 Safety Information



WARNING

Dangerous high voltage may exist across the capacitors of the Power Stage inputs after the GPA is switched off via cutter switch F3/Mainsbox CCA.

Risk of electrical shock!

- » Wait at least 20 minutes until the voltage drops to a harmless level, before working on the Power Stages.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy the Power Stages!

- » Use a towel to catch any spilled water when handling the water quick connectors.

- Check the green LED's behind the six holes in the PS front side; refer to (→ Fig. 79 Page 89).
 - » If the LED's are on, the capacitors are charged, i.e. the supply voltage 400 VDC from D32 is connected to the PS.

3.7.2 Required Tools

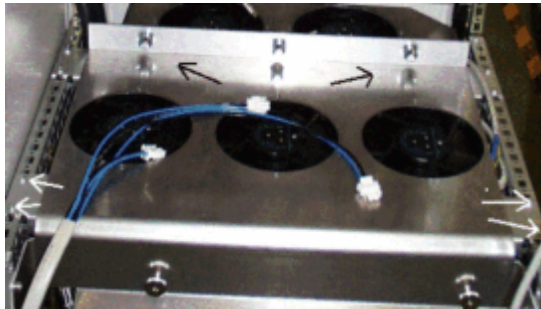
- Set of standard tools

3.7.3 Preparation

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Wait at least 20 minutes until the capacitors are discharged.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.

3.7.4 Procedure

Fig. 100: Blower assembly (five blowers)



- You have to remove all Power Stages first; (→ Power Stages / Page 86).
 - Loosen the two fixing screws at right and left side of the fan mounting plate and the two screws at back of the fan mounting plate, shown in (→ Fig. 100 Page 111).
 - Remove the mounting plate.
 - Disconnect the cables from the fan connectors.
-
- For the installation of the fans proceed in reversed order.
 - Check the function of the fans.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Image Orientation

3.8 GPA Water Distributor

(material number 57 25 960)

3.8.1 Safety Information



WARNING

Dangerous high voltage may exist across the capacitors of the Power Stage inputs after the GPA is switched off via cutter switch F3/Mainsbox CCA.

Risk of electrical shock!

- » Wait at least 20 minutes until the voltage drops to a harmless level, before working on the Power Stages.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

NOTICE

Take precautions to prevent water from dripping into the cabinet.

Drip water will destroy the Power Stages!

- » Use a towel to catch any spilled water when handling the water quick connectors.

- Check the green LED's behind the six holes in the PS front side; refer to (→ Fig. 79 Page 89).
 - » If the LED's are on, the capacitors are charged, i.e. the supply voltage 400 VDC from D32 is connected to the PS.

3.8.2 Required Tools

- Set of standard tools

3.8.3 Preparation

- Switch-off the transformer T1 by pressing S403 on D16/GSSU.
- Switch-off the cutter switch F3/3*80A in the CCA mains box (top of CCA) first, then F14/GSSU (4A) in the CCA.
- Switch-off the water pump via the circuit breaker F1 in the RCA cabinet.
- Wait at least 20 minutes until the capacitors are discharged.

3.8.4 Procedure

- Disconnect all quick connectors of water hoses from the water distributor.
- Loosen the two screws, which hold the cover plate on the GPA cabinet frame at left and right side; (→ Fig. 101 Page 113).
- Take the water distributor out of the GPA cabinet; (→ Fig. 102 Page 113).
- For emptying the water circuit open carefully the lateral seals at the right side of the distributors, held over a water bucket; first at inlet side then at outlet side.
- Remove the two water connections for in- and outlet (black water hoses).
- Refill the secondary water cooling circuit in the RCA with the water, which was drained before.
- Check tightness of cooling water connections and search for leaks

Fig. 101: Water distributor - all hoses are connected



Fig. 102: Disassembled water distributor



Perform Quality Assurance:

- Coil Check (SN DIP)

4.1 Gradient Coil Water Manifolds



Carefully read the entire section prior to commencing any work.

4.1.1 For systems with air-cooled RF Body Coil only

4.1.1.1 Explanation

Fig. 103: Rear of magnet exposed



The four small, brass water manifolds (circled in (→ Fig. 103 Page 114) for supply and return of the gradient coil water cooling circuit are each mounted to the gradient coil body with one M6 Allen bolt.

The screws are fastened to a screw thread insert which is embedded into the gradient coil body (→ Fig. 105 Page 115).

Over time, vibrations from the gradient coil body may cause the screw thread insert to shake loose and the water manifold to potentially fall off (→ Fig. 104 Page 115).

Potted screw thread inserts cannot be replaced in the field.

Loose water manifolds produce spikes.

To prevent time-consuming and expensive gradient coil replacements in the field, we have developed a solution that can be handled by local service personnel within a reasonable period of time (4-5 hr) at a relatively low price while the magnet remains at field

This procedure is described in the following paragraphs.

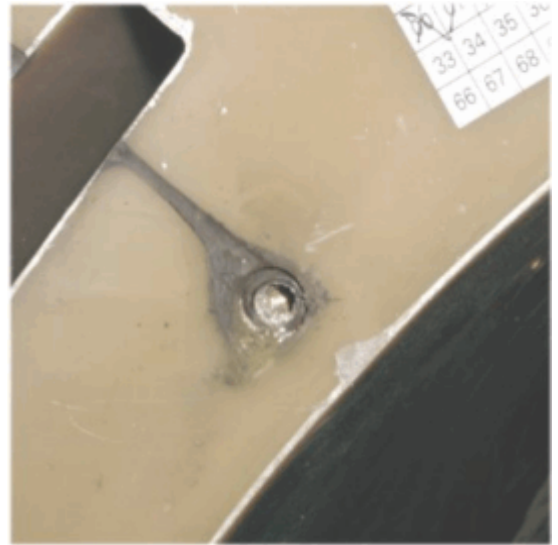


This procedure can also be applied when replacing the “manifold and hose assembly” without relocating the four small water manifolds.

Fig. 104: Loose manifold



Fig. 105: Damaged screw thread insert



4.1.1.2 Contents of the Repair Kit

The replacement kit 10048056 contains the parts listed in the table below.

Tab. 7 Contents of the repair kit 10048056

Item	Material number	Quantity	Description
1	10048057	1	Bakelite board
2	8119997	4	Brackets for small manifold
3	10048059	1	Manifold and hose assembly
4	10048058	2	holder
5	3417797	8	Countersunk screw M6X20
6	3359700	8	hex nut M6 CU2
7	7461812	8	Lock washer M6
8	3379724	4	Countersunk screw M4X10 CU2
9	3539236	10	sticky pads 25.4 X 25.4
10	n/a	n/a	Cable ties
11	5137596	2	ceramic knife
12	n/a	2	Label

Fig. 106: Replacement kit 10048056



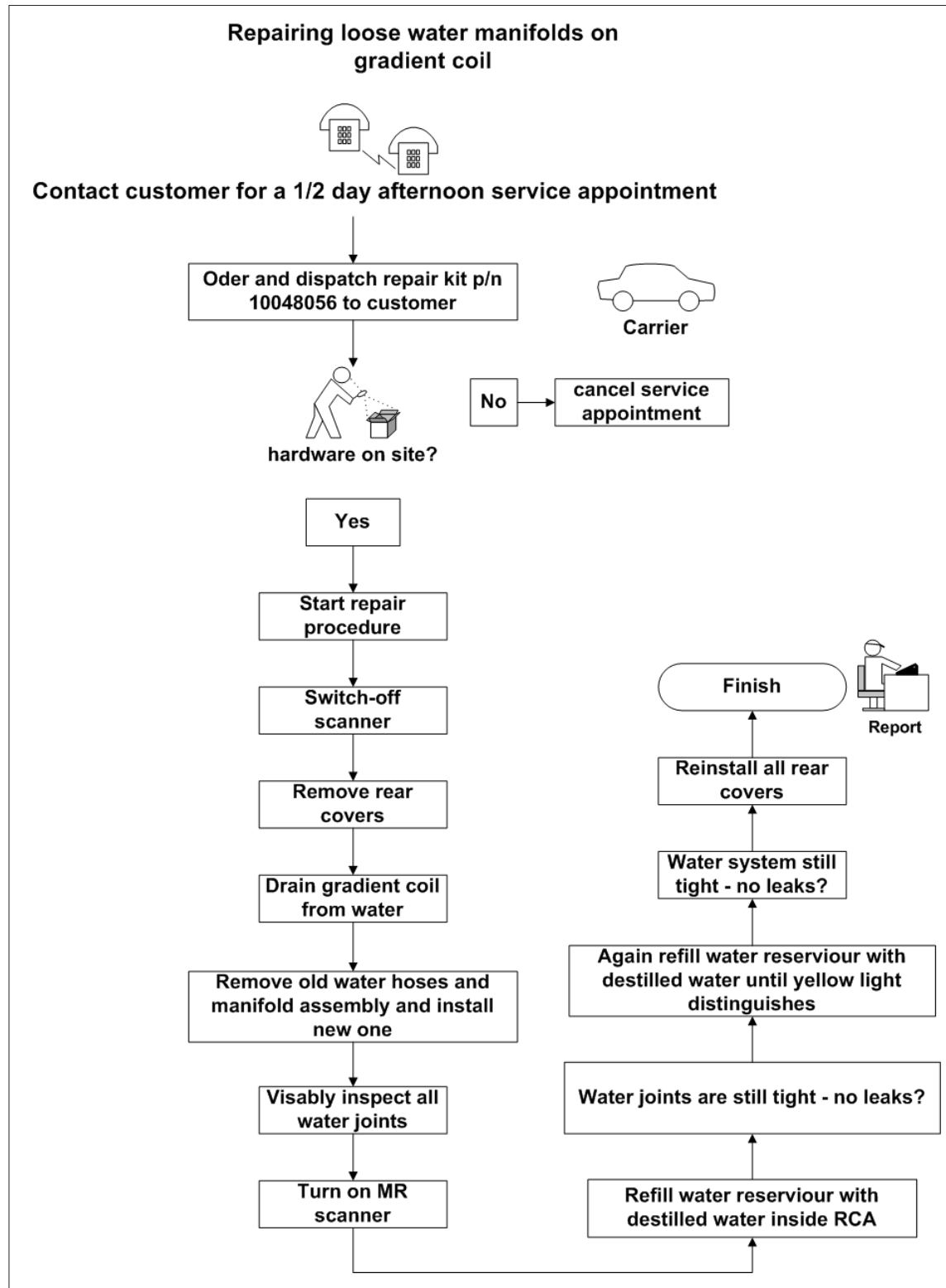
The kit can be ordered from spares.

4.1.1.3 Tools and additional materials required

- Non-magnetic tools
- Water bucket (10 liters)
- Plenty of paper towels
- Distilled water, at least 10 liters.
- Alcohol or similar cleaning agent (125 ml)
- Water-soluble lubricant (purchase locally, e.g. drugstores)

4.1.1.4 Flowchart

Fig. 107: Flowchart



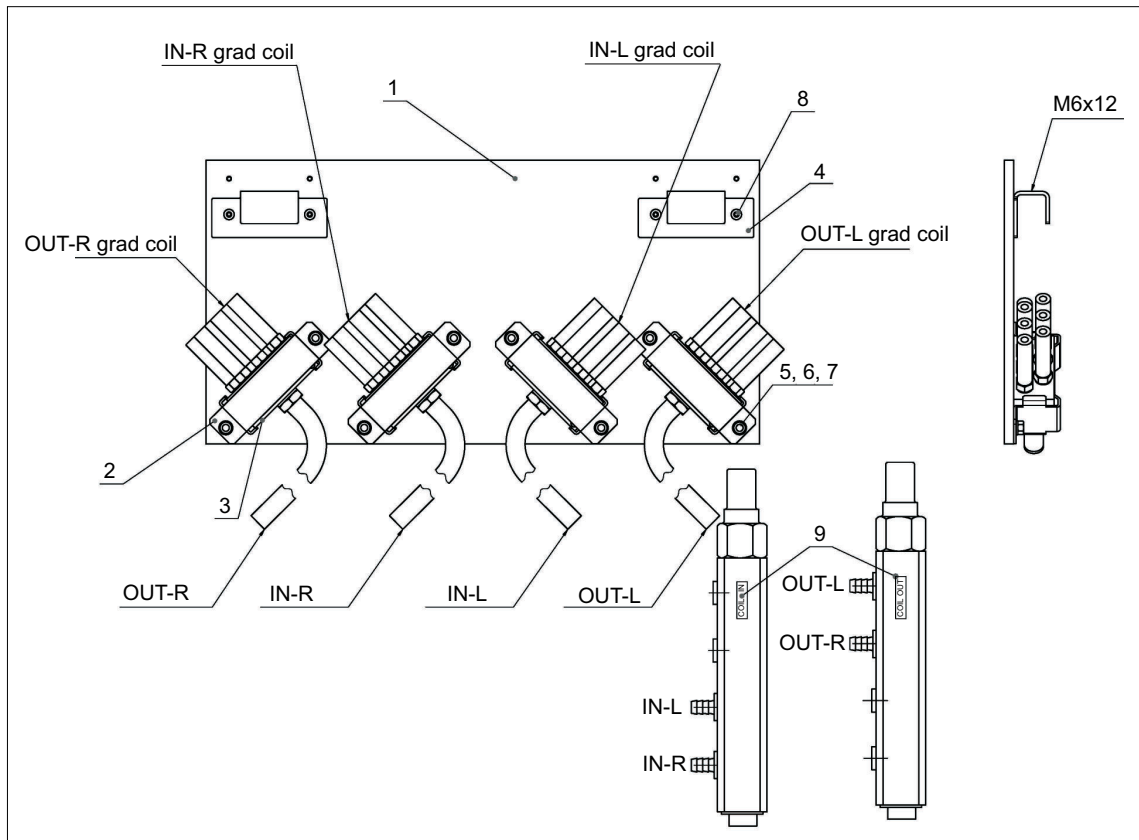
4.1.1.5 System Performance Check

To verify the system performance status prior to modification, perform a "Standard S/N, Spike Check and Phantom Shim"

4.1.1.6 Preparatory work

- Fasten the four small manifolds with the brackets, screws, hex nuts and twist lock washers onto the bakelite board, see next figure (→ 1-8/ Fig. 108 Page 118)

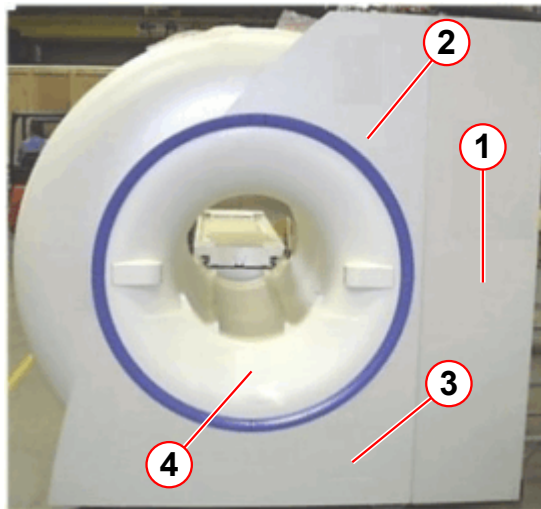
Fig. 108: Drawing of manifold board



- To avoid confusion at a later date, mark the four water hoses with OUT-R, IN-R, OUT-L and IN-L
- Cut off the four water hoses IN-L, IN-R, OUT-L, OUT-R (→ 9/ Fig. 108 Page 118) at the main manifolds (this allows for easier routing of the water hoses).
- Remove the hose stumps from the main manifolds, see paragraph (→ Removing the Water Hoses from the Gradient Coil Nozzles / Page 123).

4.1.1.7 Start

Fig. 109: Covers in the back

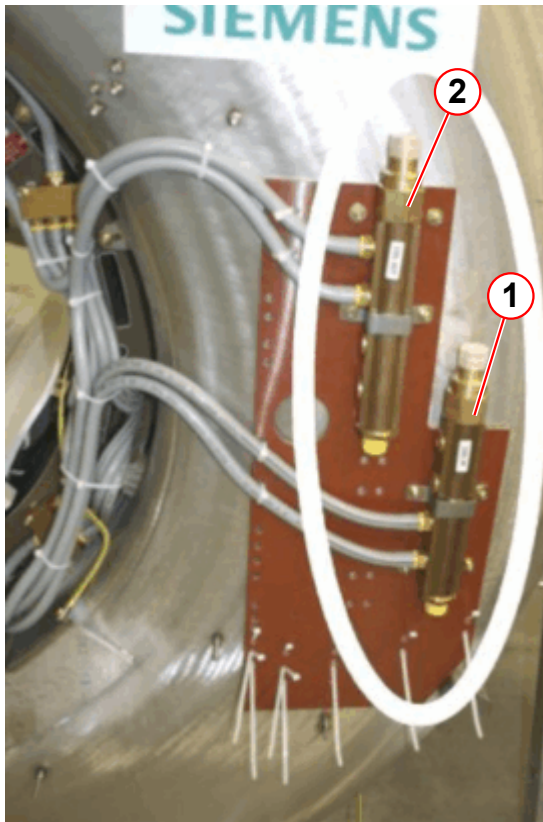


- (1) Cover 1
- (2) Cover 2
- (3) Cover 3
- (4) Cover 4

- Switch off the MR scanner (water pump may not run) and remove the four covers from the back, (→ 1, 2, 3, 4/Fig. 109 Page 119)

4.1.1.8 Draining the Gradient Coil and Water Hoses

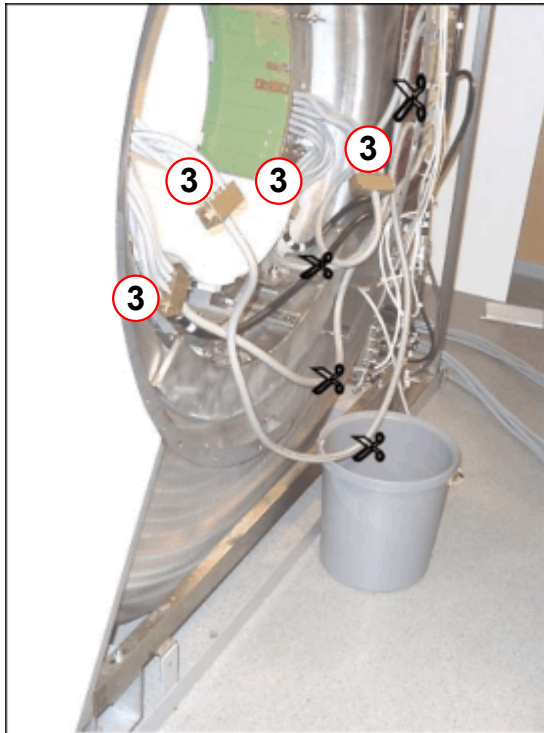
Fig. 110: Main manifolds (hoses disconnected in picture)



(1) IN
(2) OUT

- Quickly disconnect the two main hoses for supply and return (→ 1 and 2/ Fig. 110 Page 120) each from the main manifolds.
- Cut all cable ties that fix the water hoses to the magnet and gradient coil

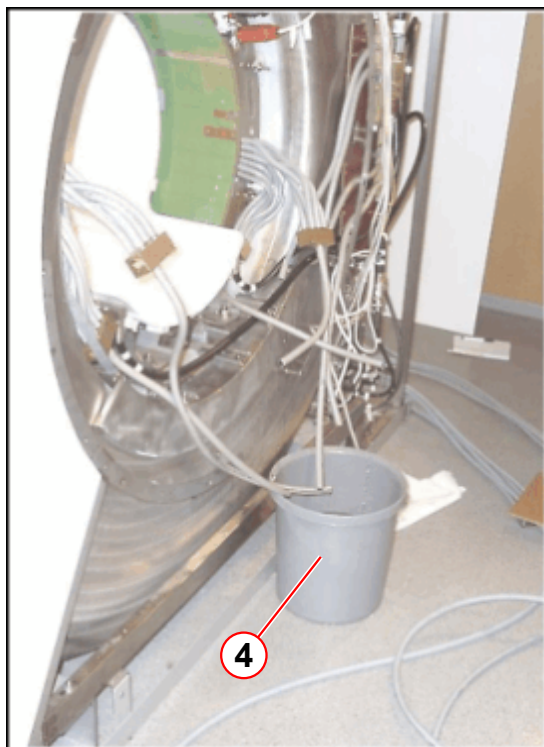
Fig. 111: Cutting the water hoses



(3) Small manifolds

- Detach the four small manifolds (→ 3/Fig. 111 Page 121) from the gradient coil.
- Use the ceramic knife and cut (scissors in (→ Fig. 111 Page 121)) the four water hoses one at a time (supply and return) halfway between the four gradient coil manifolds and the two main manifolds.
 - » Attention: Water will spill as soon as you cut the hoses. Therefore ensure that the old hoses are positioned over the pail

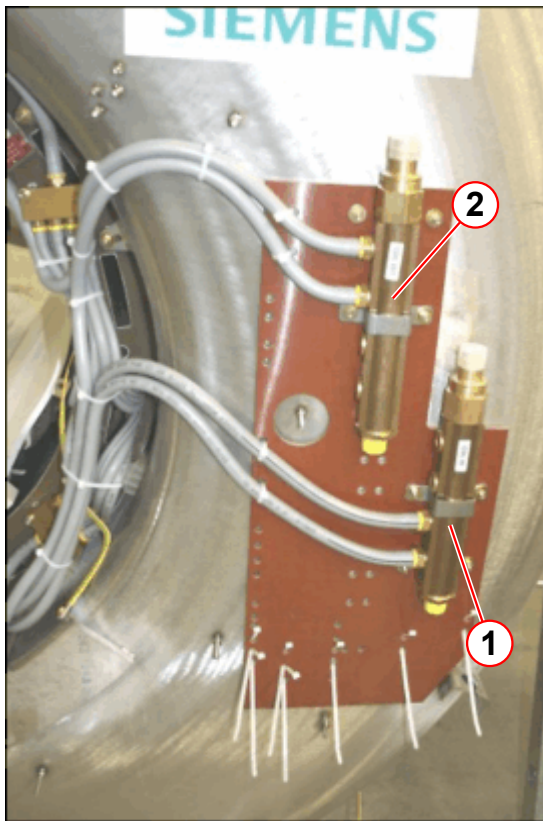
Fig. 112: Draining the gradient coil



(4) Water pail

- Drain the water into the pail (→ 4/Fig. 112 Page 121).

Fig. 113: Main manifolds



- (1) Main supply
(2) Main return

- Detach the two main manifolds for supply and return (→ 1 and 2/Fig. 113 Page 122) from the bakelite board.

4.1.1.9 Removing the four small Manifolds

All 24 gray water hoses (12 to the left and 12 to the right) must be cut between the 4 smaller manifolds and the gradient coil hose nipples

Leave 30 cm of the old hose connected to the gradient coil (→ Fig. 116 Page 123).

Fig. 114: Cutting the water hoses at the grad. coil



- Cut all 24 water hoses (→ Fig. 114 Page 122) with the ceramic knife and remove the old manifold and hose assembly (→ Fig. 115 Page 123).

Fig. 115: Old assembly

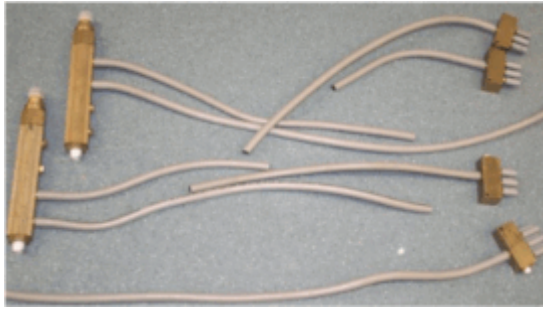


Fig. 116: All 24 water hoses are cut



Once the water hose and manifold (→ Fig. 115 Page 123) have been removed, the remaining 24 hose pieces (→ Fig. 116 Page 123) must be removed from the nipples on the gradient coil (→ Removing the Water Hoses from the Gradient Coil Nozzles / Page 123).

4.1.1.10 Removing the Water Hoses from the Gradient Coil Nozzles

Fig. 117: Perform horizontal cut



The water hoses must be cutoff in a horizontal direction, as demonstrated in (→ Fig. 117 Page 123) and (→ Fig. 118 Page 124).

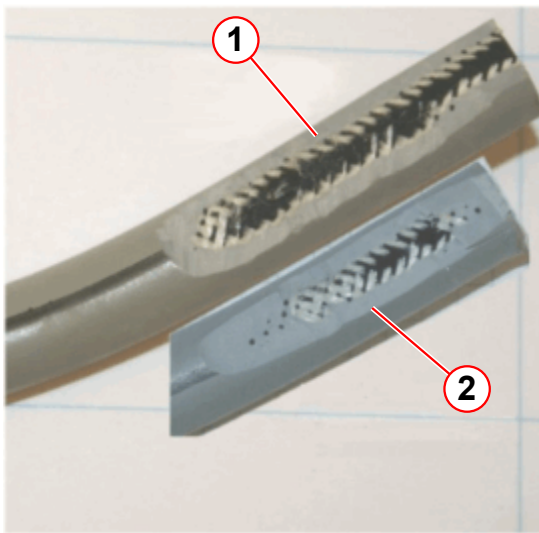
To be able to pull the water hose off the barbed nipples, cut through the Kevlar braiding completely as in (→ Fig. 119 Page 124) and (→ Fig. 120 Page 124). Otherwise, the water hose cannot be pulled off the barbed hose nipple.

Fig. 118: Cutting across the hose



- Carefully guide the ceramic blade across the sleeve on the water hose (→ Fig. 118 Page 124).
- Perform a clean horizontal cut - avoid cutting the brass nipple.
 - » The brass nipple is a delicate component. It must not be scratched or damaged.

Fig. 119: Kevlar braiding



- (1) Good
(2) Not Good

- » It is important to cut through the Kevlar braiding (→ 1 and 2/ Fig. 119 Page 124).

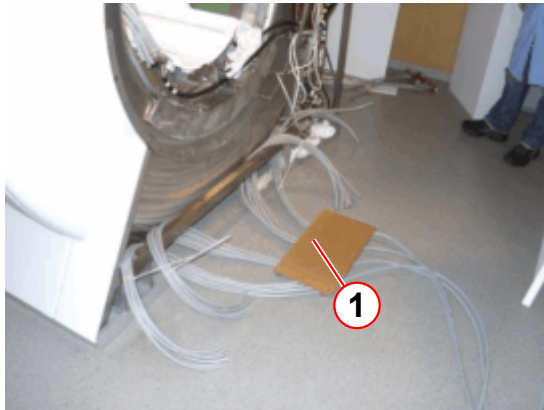
Fig. 120: Kevlar braiding cut through



Once the Kevlar braiding has been cut through, the water hose can be pulled off with ease (→ Fig. 120 Page 124).

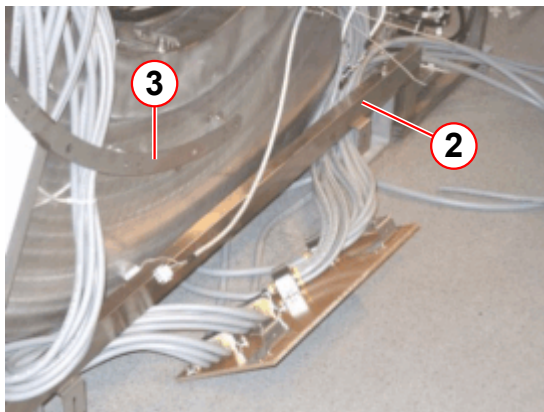
4.1.1.11 Installing the new Manifold

Fig. 121: New manifold assembly



- Spread out the new assembly on the floor(→ 1/Fig. 121 Page 125).

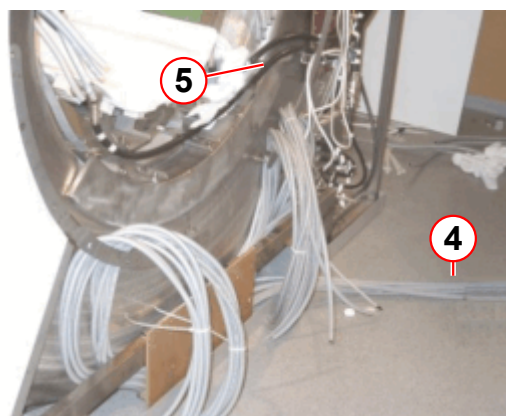
Fig. 122: Mounting board



- Route the new 24 water hoses behind the crossbeam (→ 2/Fig. 122 Page 125) and between the funnel-mounting sword (→ 3/Fig. 122 Page 125) and the magnet to the gradient coil water nipples.
- Simply hang the new mounting board with the four manifolds centered on the crossbar (→ Fig. 122 Page 125) and (→ Fig. 123 Page 126).

(2) Cross beam
(3) Funnel mounting sword

Fig. 123:



- (4) Supply and return hoses
(5) RF cables

- Route the 4 supply and return hoses (→ 4/Fig. 123 Page 126) behind the crossbar, the funnel mounting sword and the RF cables (→ 5/Fig. 123 Page 126) up to the two large supply and return manifolds
 - » It is important that the **new** hoses leading to the 2 main manifolds are routed between the magnet and the body coil RF cables for forward and reverse.

The **new** four hoses are labeled with in and out

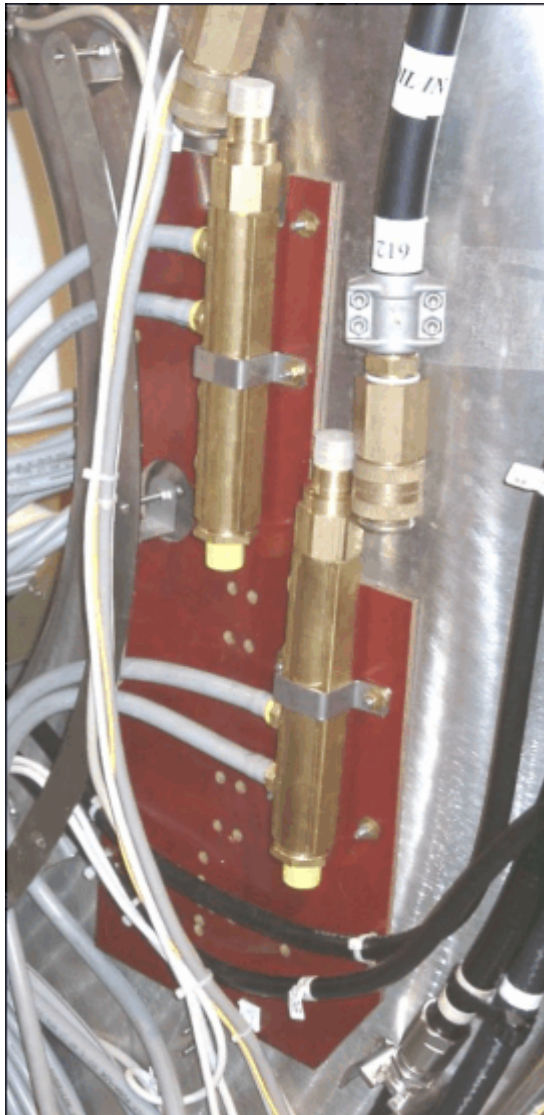
- » At this point, the two new main manifolds should not yet be mounted onto the magnet.

Fig. 124: Lubricated barbed hose nipple



- Lubricate the main manifold hose nipples (→ Fig. 124 Page 126) and connect the water hoses for IN (supply) and OUT (return) by pushing the four hoses over the hose nipples on the manifolds.
 - » Do not mix up the hoses for supply and return

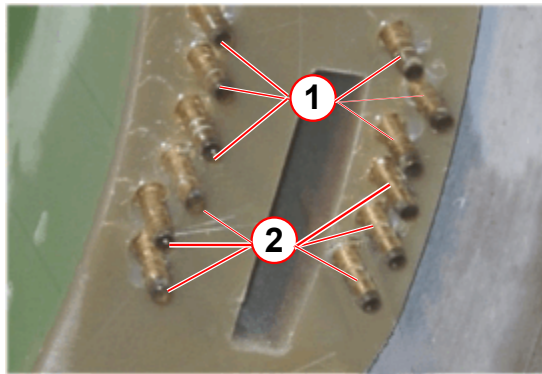
Fig. 125: Main manifolds



- Now fasten the two main manifolds with the U-brackets to the brown bakelite board (→ Fig. 125 Page 127).

4.1.1.12 Fitting the 24 Water Hoses to the Water Nozzles at the Gradient Coil

Fig. 126: Hose nipples on grad. coil



(1) IN
(2) OUT

The method for connecting the hoses to the nozzles is always the same:

Apply lubricant to the barbed hose nipple as in (→ Fig. 124 Page 126) and push the water hose over the barbed brass nipple in one firm motion.

- » The 6 IN water supply hoses go on top and the 6 OUT water return hoses to the bottom.
(→ 1 and 2/Fig. 126 Page 128)

4.1.1.13 Left-hand side at Magnet rear

Fig. 127:



Fig. 128:



- Repeat this procedure for all 12 water hoses for **IN** and **OUT** for the left area.
- The 6 water **IN** hoses should go on top and the 6 water **OUT** hoses go to the bottom.
- Once all 12 hoses have been mated, use cable ties and sticky pads and secure the water hoses to the magnet.
 - » Before attaching the sticky pads, clean the surface with alcohol or similar cleaning agent to remove grease.

The aim is to route the water hoses so that the funnel does not come into contact with water hoses when it is installed later (→ Fig. 128 Page 128) and (→ Fig. 129 Page 129).

Finished with left side

4.1.1.14 Right-hand side at Magnet rear

The procedure is virtually the same as for the left-hand side of the magnet.

Ensure that all water hoses are routed individually and arranged adjacently. Use plenty of sticky pads and cable ties.

- » In the case of water hoses, the top is cold and bottom is warm.

Fig. 129:

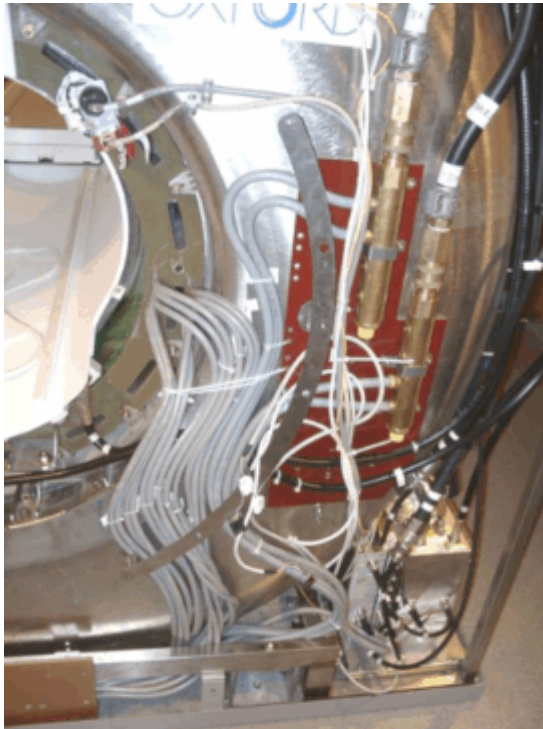
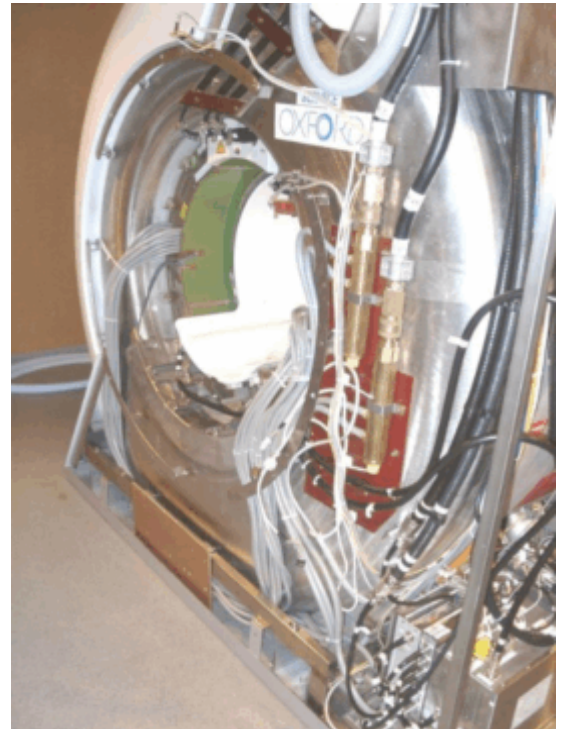


Fig. 130: Modification finished



Arranged water hoses in (→ Fig. 130 Page 129).

- Quick connect the main water hoses for supply and return at the main manifolds.

4.1.1.15 Filling up the Water Reservoir with distilled water

The filling port for the system's cooling water reservoir is on top of the RCA.

System must be switched on

- Remove the plug and fill up the reservoir with distilled water until the yellow light on the Satchwell controller extinguishes.

4.1.1.16 Final Checks

- Go to the back of the magnet and check the entire new water hose section for possible leaks
- All connections should be watertight and dry.

- Finish off your system - put on all covers, close cabinet doors, clean up your trash and tools.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Phantom Shim Check
- Spike Check

4.2 Gradient Cables (Harmony, Symphony, Sonata)

4.2.1 General



This procedure is only valid for Harmony syngo MR, Symphony syngo MR and Sonata syngo MR.



Should the gradient cable kit between the gradient coil and the connection plate need replacing, the Sonata cables (50 mm²) must be used on all systems.

4.2.1.1 Contents of the Cable Repair Kit

The replacement kit 1141923 contains the parts listed in the table below.

Tab. 8 Contents of the cable kit 1141923

Item	Material number	Quantity	Description
1	5940502	1	Gradient cable kit Sonata 50 mm ²
2	5940577	12	Cable clamp 50 mm ²
3	3424793	12	Screw M4x30
4	3342052	12	Washer A4.3
5	3417631	12	Nut M4
6	3432911	6	Screw M10x20
7	7063969	13	NORD-Lock washer M10
8	7063761	6	NORD-Lock washer M12
9	5938670	1	Protective cover
10	3539236	1	Loctite_211

4.2.1.2 Tools and additional materials required

- Non-magnetic standard tool set (7757706)
- Non-magnetic torque wrench ranging from 5 - 20 Nm (8395829)
- Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803)
- Non-magnetic socket assortment (8396116)

4.2.2 Safety

! WARNING

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch off the gradient cabinet
- » Old gradient cabinet K2209: The circuit breakers F9/3*50A and F14 in the CCA must be switched off.
- » New gradient cabinet K2217: The cutter switch F3/3*80A in the CCA mains box or mains box 2 and F14/CCA must be switched off.

! WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.

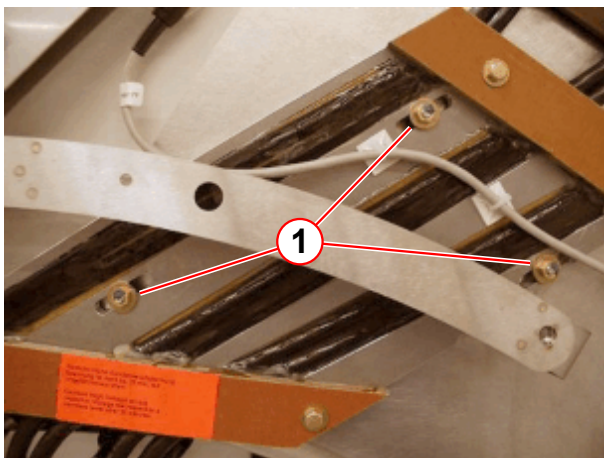


Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

4.2.3 Procedure

- Remove the rear cover to gain access to the connections
- Remove 3 x M6 (10 mm open wrench) nuts to the magnet (→ 1/ Fig. 131 Page 132)
- Remove 6 x M12 (19 mm open wrench) nuts (→ Fig. 132 Page 132)

Fig. 131: Gradient plate attachment



(1) 3 of the 10 mm nuts

Fig. 132: Connection plate

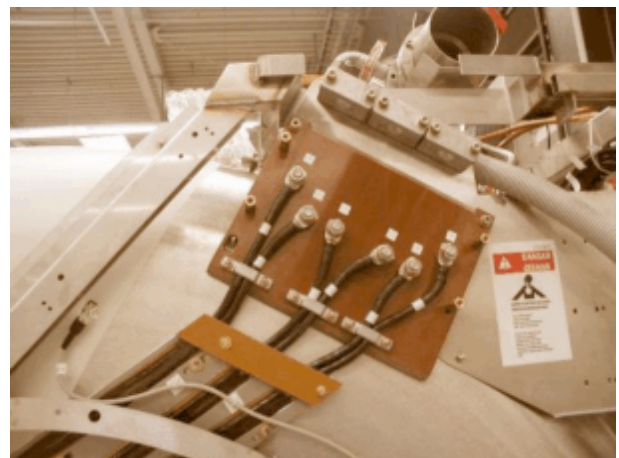


Fig. 133: Strain relief



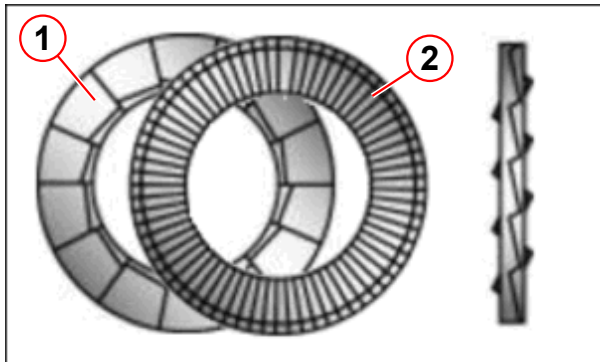
- Remove 3 x metal strain relief clamps at the connection plate (→ Fig. 133 Page 133).
- Remove the gradient plate.

- Replace the gradient plate with the Sonata version (50 mm²) and attach it to the magnet in reverse order using the 10 mm nuts (only tighten by hand at first).



Before connecting the gradient cables, check the cable insulation for defects to prevent short circuits. Make absolutely sure that the cables are not touching any conductive materials.

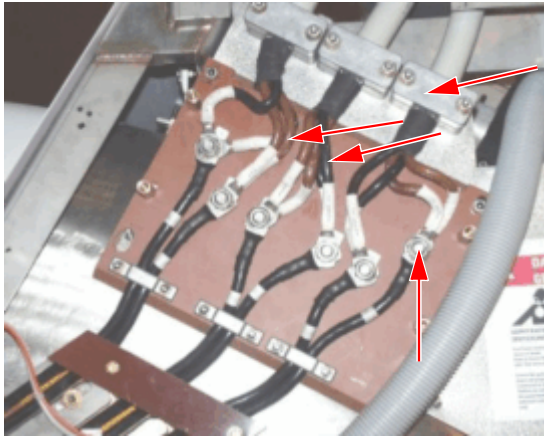
Fig. 134: NORD-LOCK washer



- (1) Cam side
(2) Ridged side

A set of NORD-Lock washers must be placed between the cable lug and nut when fixing the cables at the gradient coil connection. The correct order of the washers (→ Fig. 134 Page 133) is important for a proper securing function.

Fig. 135: Gradient connection plate



- Connect the gradient cables at the connection plate (→ Fig. 135 Page 134) using 6 of the NORD-Lock washers (M12) delivered with the kit.
- Check that the gradient cables are correctly clamped.
- Tighten the nuts with a non-magnetic torque wrench to 35 ± 2 Nm.

- Repeat the checks and seek confirmation of the torque and polarity from a second person.
- Check that the torque is 35 ± 2 Nm.
- Mark the connections (nuts/bolts) with a waterproof pen to prove they have been tightened with a torque wrench.
- Attach the cables at the gradient coil according to the Wiring Diagrams (→ Fig. 136 Page 134) and using the M10 NORD-Lock washers in the correct order (→ Fig. 137 Page 134).

Fig. 136: Wiring at the gradient coil

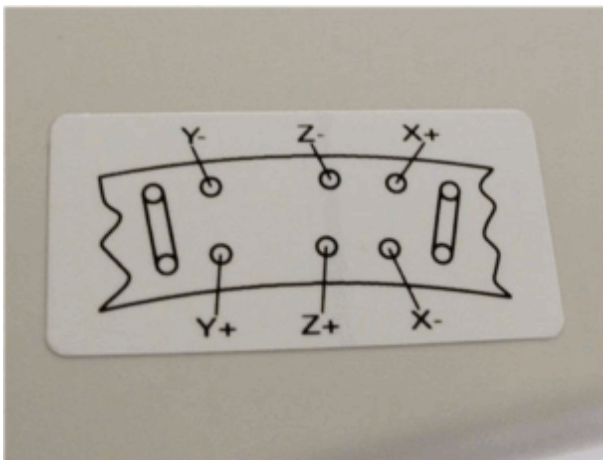
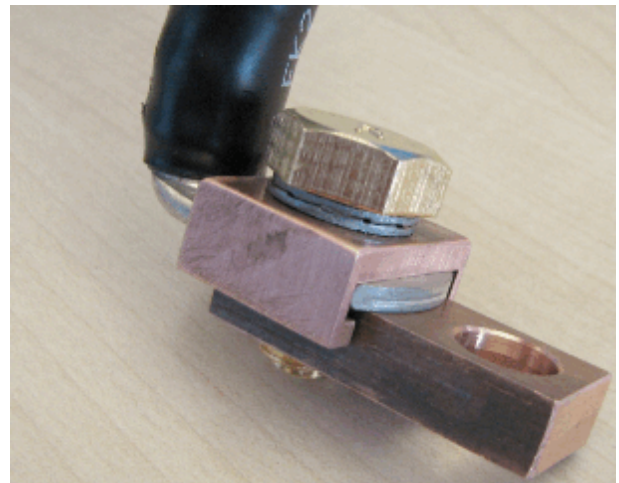


Fig. 137: Correct coil connection arrangement





Route the cables so that the lugs are even with the copper retaining bolts at the coil in order to achieve a good level of contact. The leads should be routed in parallel to each other (→ Fig. 138 Page 135).



Ensure that the cable lugs do not come into contact while tightening the M10 x 20 hex brass screws.



Newly delivered NORD-Lock washers are to be used on all M10x20 screws at the gradient coil (13 in total).

Fig. 138: Cables routed correctly

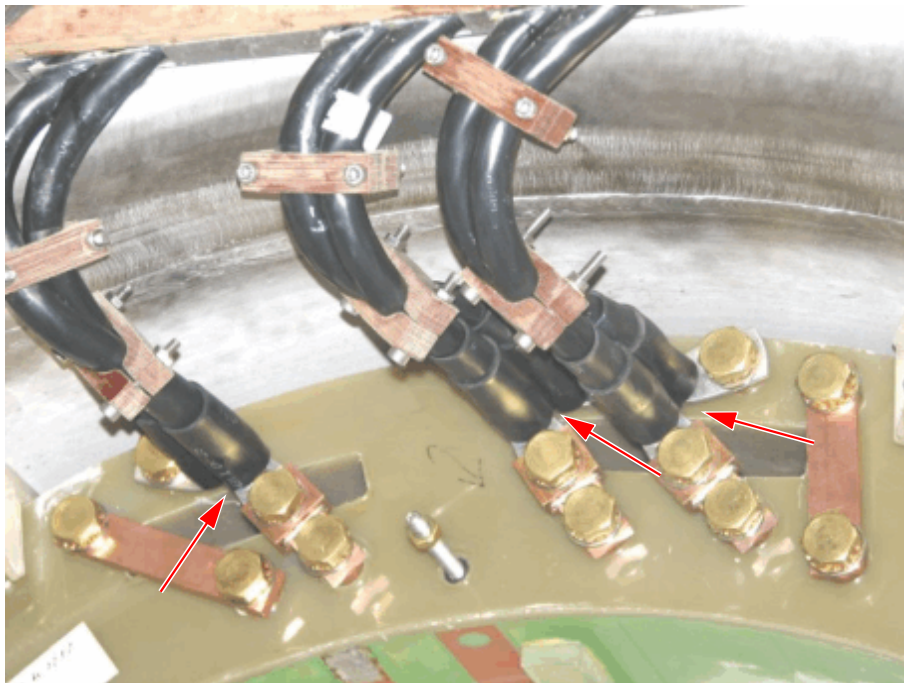
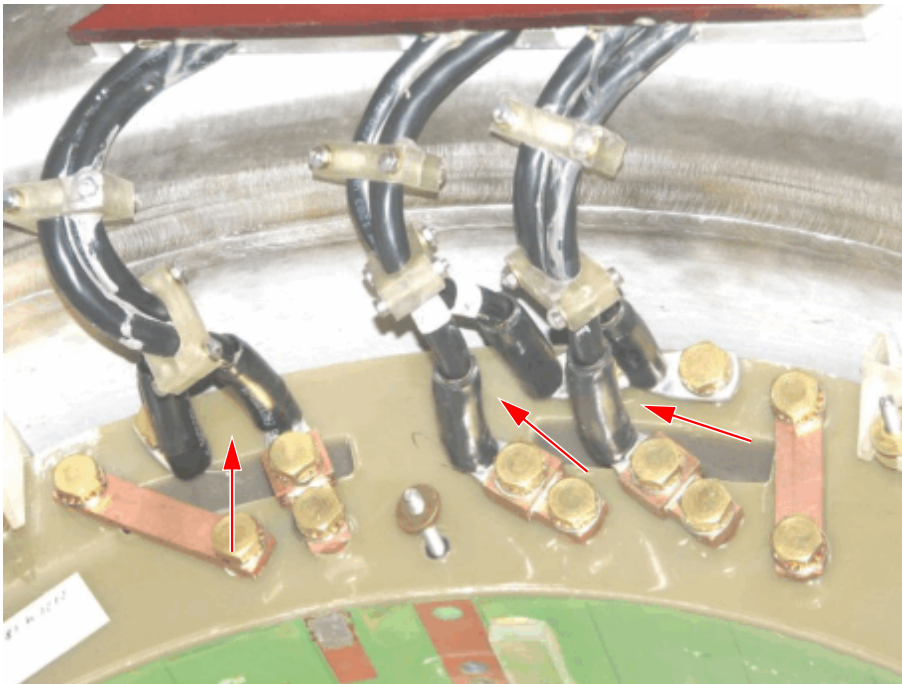


Fig. 139: Cables routed incorrectly

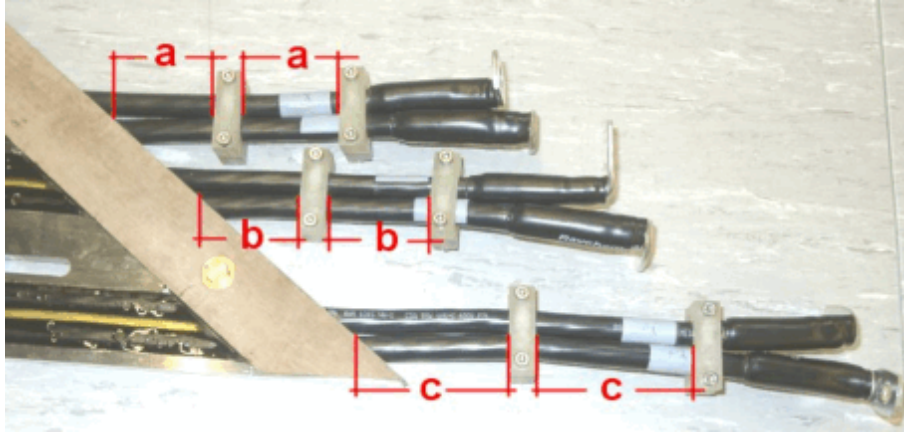


- Use a **non-magnetic** torque wrench to tighten the M10x20 hex brass screws at the gradient coil.
 - » **Torque 22 ± 1 Nm**
- Now fully tighten the 3 x M6 nuts holding the gradient plate in place (→ Fig. 131 Page 132).
- Fasten the cable clamps, secure them with Loctite and tighten them to **5 Nm**.
 - » The first clamp should be fastened as tightly as possible at the heat shrink tubing (→ Fig. 140 Page 137).
 - » The clamps should be equally positioned along the length of the leads (→ Fig. 140 Page 137).



Cable ties are not permitted.

Fig. 140: Position of the cable clamps

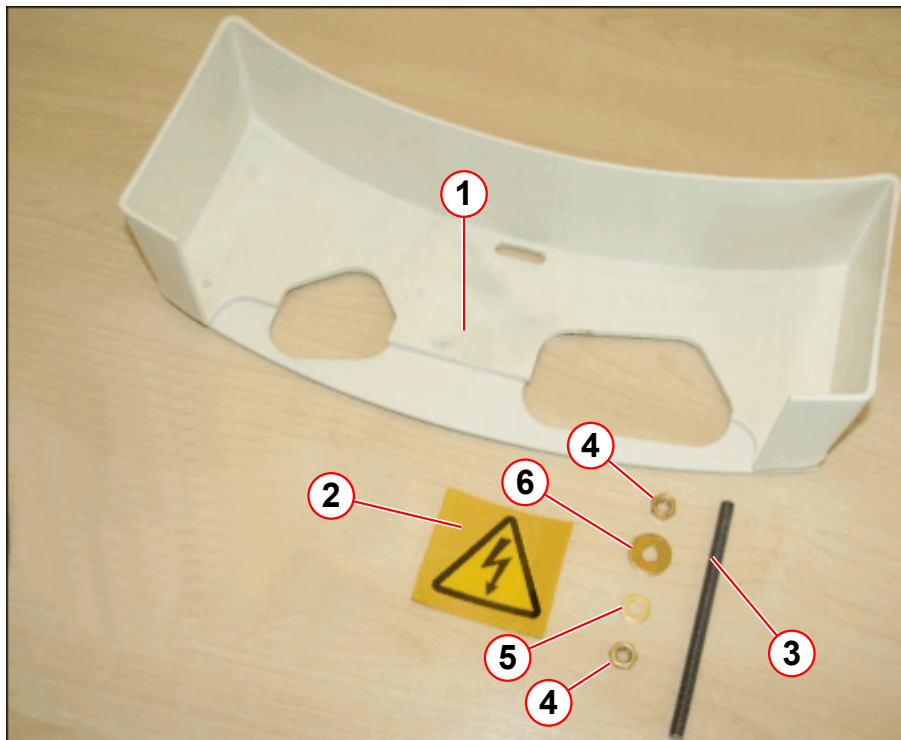


4.2.3.1 New Protective Cover at Gradient Coil Terminals



If the previous style gradient coil cover was installed, it must be replaced with the new version (→ 1/ Fig. 141 Page 138), provided in the spare parts kit.

Fig. 141: New gradient coil cover

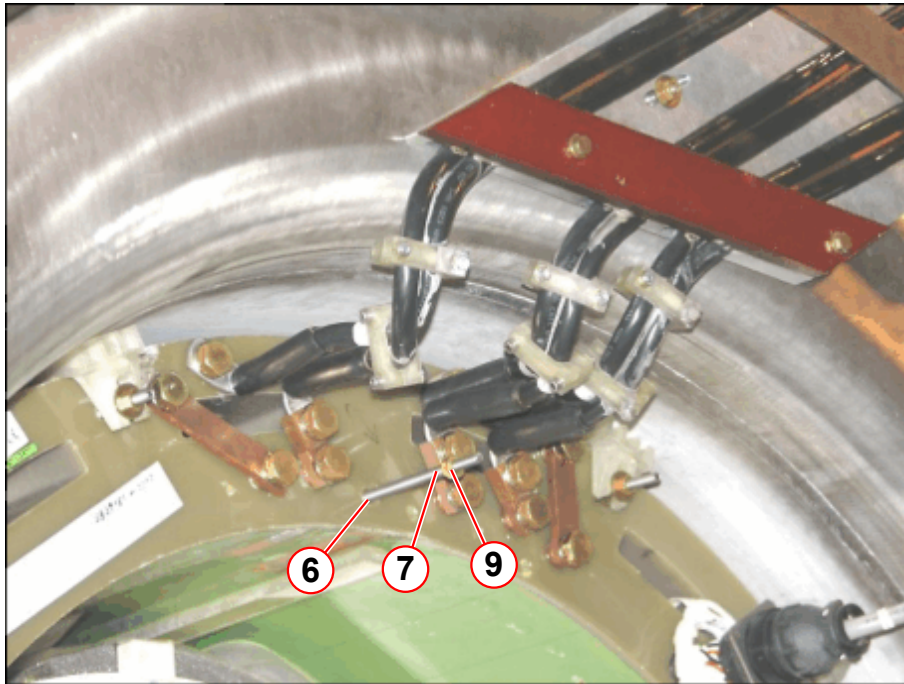


- (1) Protective cover
- (2) Warning label
- (3) Threaded rod, M6*100
- (4) Hex nut M6
- (5) Disk washer, 6.4
- (6) Disk washer, 6.2*18*2

1. Replace the 60 mm long threaded rod with the new 100 mm long threaded rod (→ 3/Fig. 141 Page 138) from the spare parts kit. Apply a drop of Loctite to the thread to secure it.
2. Screw the threaded rod to the gradient coil until it stops (→ 6/Fig. 142 Page 139).

3. Attach an M6 brass nut (using a little Loctite) and new smaller disk washer (item 9 in the kit) to the threaded rod (→ Fig. 142 Page 139) and (→ Fig. 143 Page 139)

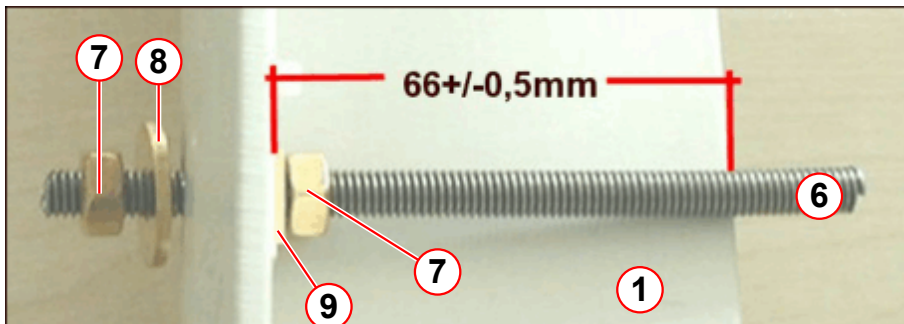
Fig. 142: Threaded rod



- (6) 100 mm threaded rod
- (7) Hex nut M6
- (9) Disk washer 6,2 mm

4. Turn the M6 hex nut (→ 7/ Fig. 143 Page 139) to achieve a distance of 66 +/- 0.5 mm.

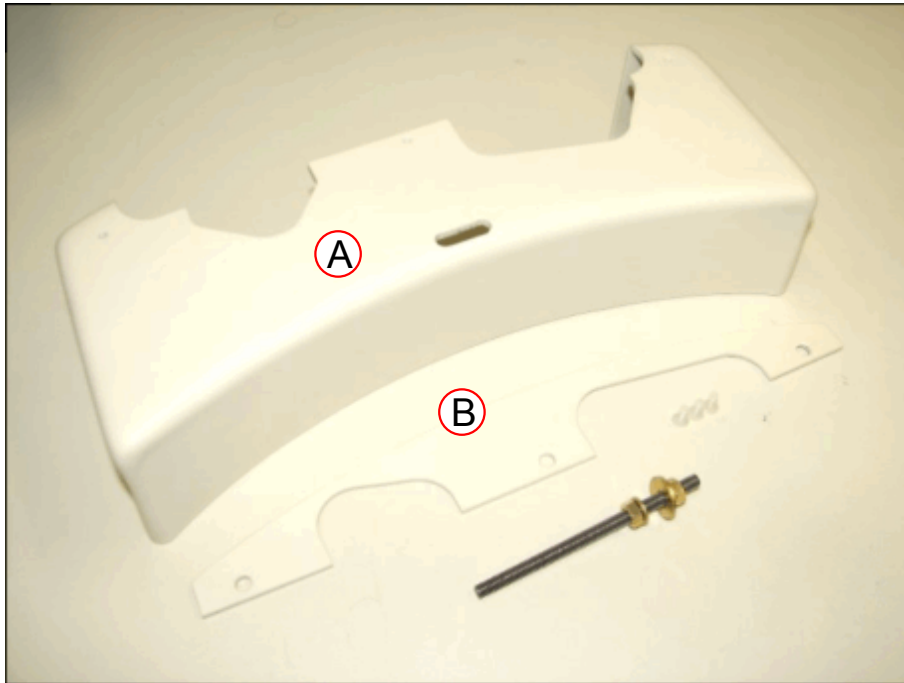
Fig. 143: New threaded rod with protection cover



- (1) Protection cover
- (6) M6*100 mm threaded rod
- (7) M6 hex nut
- (8) Disk washer 6,2*18
- (9) Disk washer 6,2

5. If not done previously, disassemble (3 slotted nylon cylinder head screws) the new protective cover (→ Fig. 144 Page 140). Extra nylon screws have been provided in the kit.

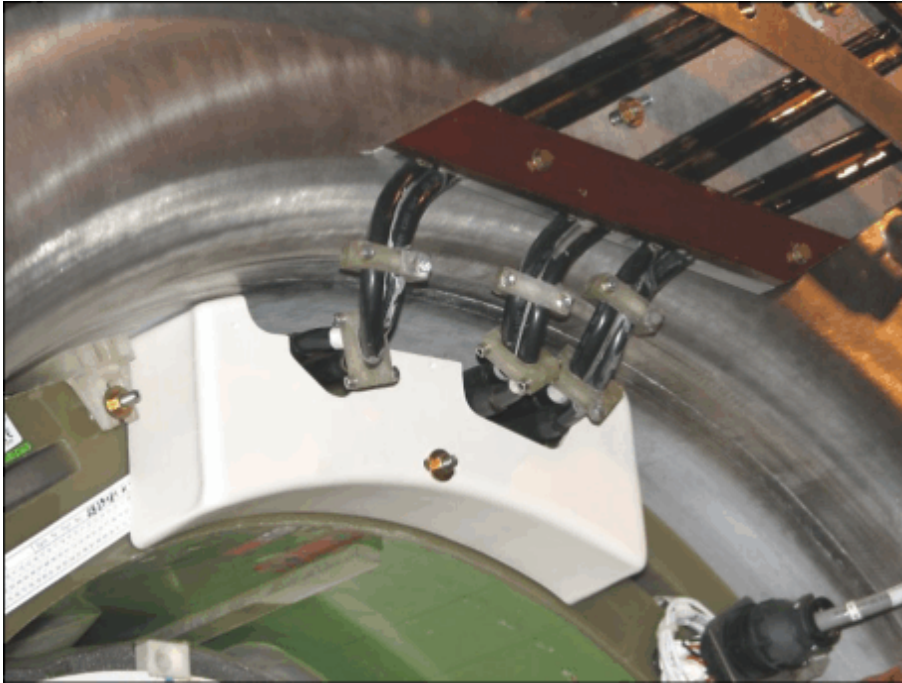
Fig. 144: Lid



- (1) Part A
- (2) Part B

6. Fasten the new protective cover (→ A/ Fig. 144 Page 140) with the M6 hex nut (→ 7/ Fig. 143 Page 139) and washer (→ 8/ Fig. 143 Page 139) to the 100 mm threaded rod (→ Fig. 145 Page 141).

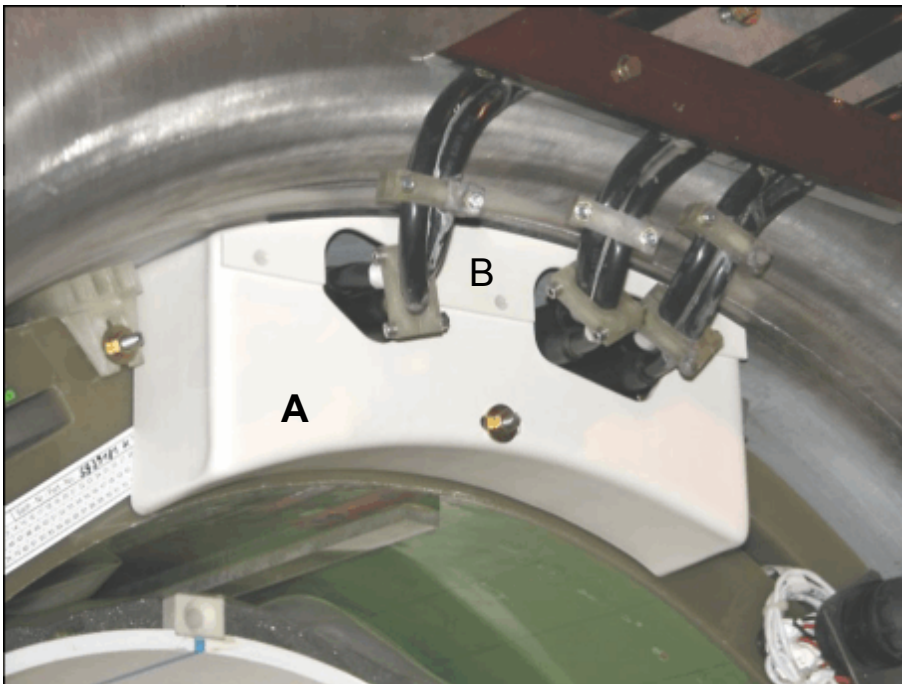
Fig. 145: Lid part A fastened



(1) Part A

7. Fasten part B (→ Fig. 146 Page 141).

Fig. 146: Lid completed

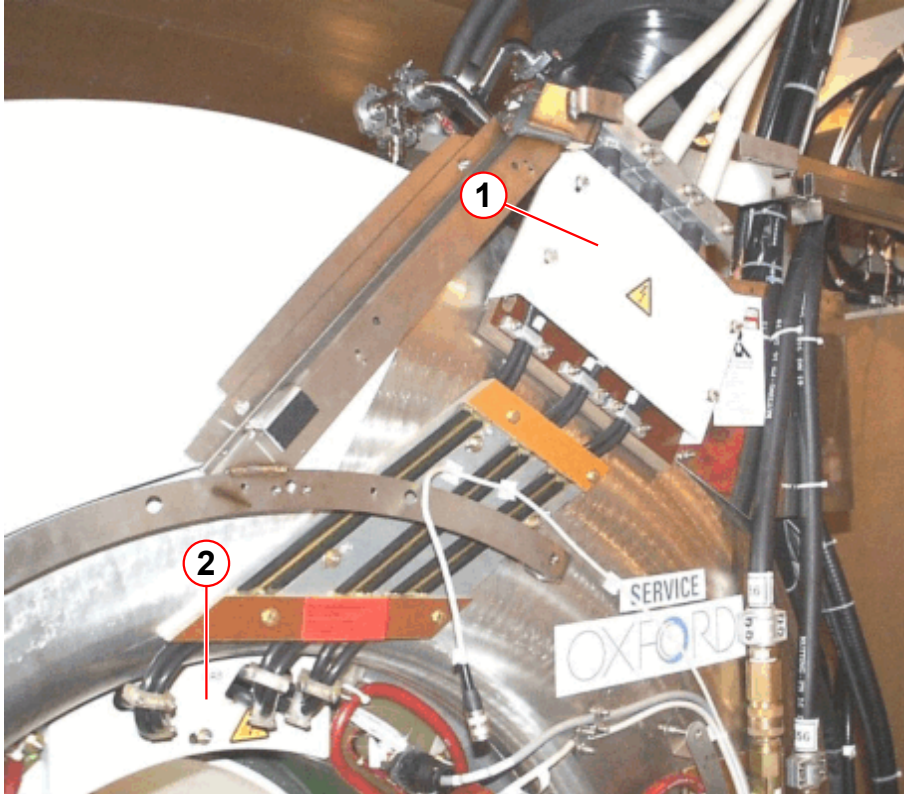


(1) Part A

(2) Part B

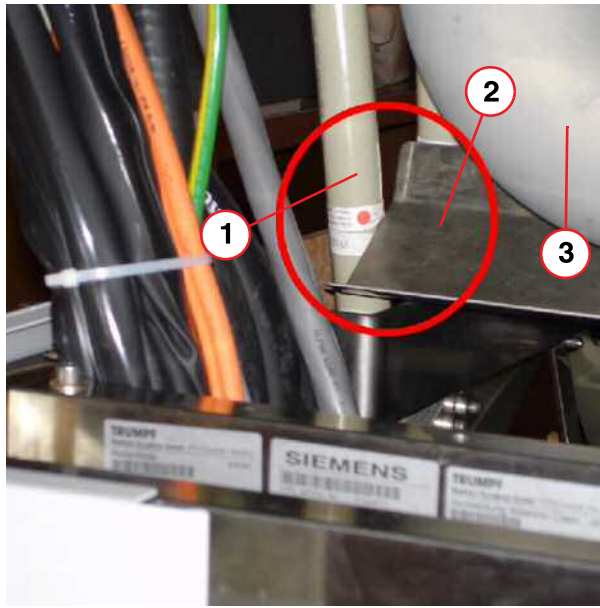
8. If you have not already done so, affix the warning label (→ 2/Fig. 141 Page 138) to the new lid.
- Fit lid for the external gradient cables (→ 1/Fig. 147 Page 142).

Fig. 147: Two covers installed



- (1) Connection plate cover
- (2) Gradient coil cover (previous style shown in this image)

Fig. 148: Cable routing



- (1) Gradient cable
- (2) Trip tray
- (3) Quench pipe

- Check the routing of the gradient cable.
- The gradient cable should not touch the drip tray.

4.2.4 Final Work Steps

- Reinstall the magnet covers in reverse order.
- Set the circuit breakers for GPA in the CCA to ON.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Spike Check
- Image Orientation

4.3 Gradient Coil Connection Plate (Vision Upgrades)

4.3.1 General



This procedure is only valid for Vision to Symphony syngo MR and Vision to Sonata syngo MR upgrades.



Should the gradient cable kit between the gradient coil and the connection plate need replacing, the Sonata cables (50 mm²) must be used on all systems.

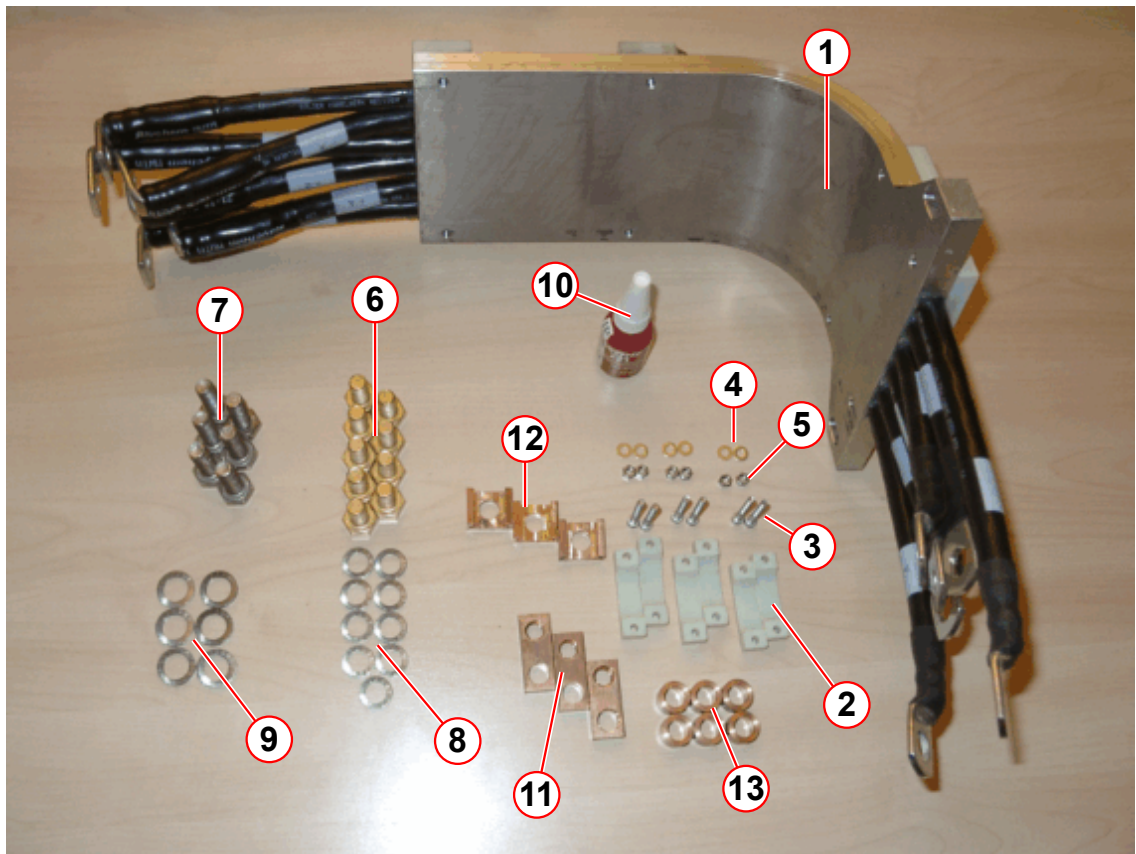
4.3.1.1 Contents of the Cable Repair Kit

The replacement kit 7365815 contains the parts listed in the table below.

Tab. 9 Contents of the set of gradient cables 089/Son 7365815

Item	Material number	Quantity	Description
1	7365807	1	Gradient cable kit 089 Sonata 50 mm ²
2	5940577	6	Cable clamp 50 mm ²
3	3424793	12	Screw M4x30
4	3342052	12	Washer A4.3
5	3417631	12	Nut M4
6	3432911	6	Screw M10x20
7	10358808	6	Screw M10x30
8	7063969	9	NORD-Lock washer M10
9	7063761	6	NORD-Lock washer M12
10	2048874	1	Loctite 221
11	4767559	3	Gradient connector (copper)
12	4768730	3	Connector (copper)
13		6	10 mm copper bush

Fig. 149: Content of the 089 gradient cable kit



- (1) Gradient cable kit 089 Sonata 50 mm²
- (2) Cable clamp 50 mm²
- (3) Screw M4x30
- (4) Washer A4.3
- (5) Nut M4
- (6) Screw M10x20
- (7) Screw M10x30
- (8) Nordlock washer M10
- (9) Nordlock washer M12
- (10) Loctite 221
- (11) Gradient connector (copper)
- (12) Connector (copper)
- (13) 10 mm copper bush

4.3.1.2 Tools and additional materials required

- Non-magnetic standard tool set (7757706)
- Non-magnetic torque wrench ranging from 5 - 20 Nm (8395829)
- Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803)
- Non-magnetic socket assortment (8396116)

4.3.2 Safety

! WARNING

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch off the gradient cabinet
- » Old gradient cabinet K2209: The circuit breakers F9/3*50A and F14 in the CCA must be switched off.
- » New gradient cabinet K2217: The cutter switch F3/3*80A in the CCA mains box or mains box 2 and F14/CCA must be switched off.

! WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

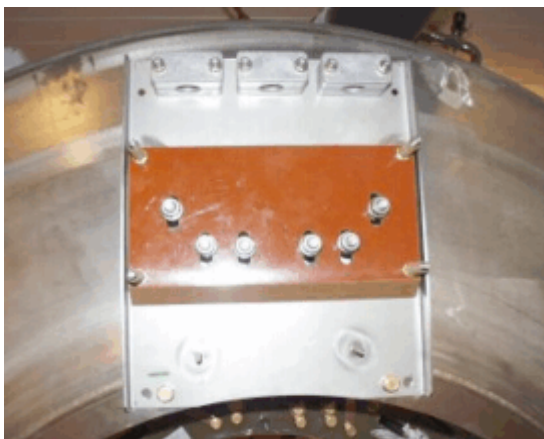
- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

4.3.3 Procedure

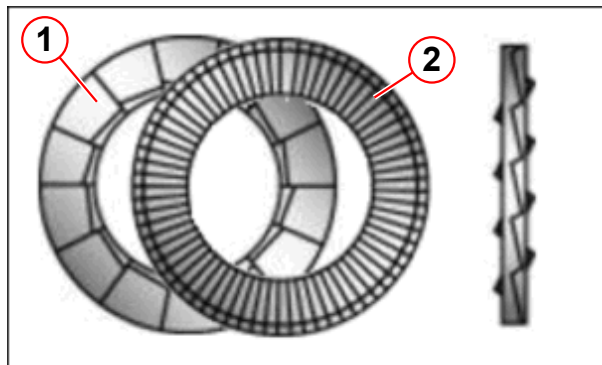
Fig. 150: Connection plate



- The cable clamps can be adjusted vertically.
- Install the gradient cable connection plate.
- The connections can be adjusted vertically.

Note: The left-hand photograph shows the connection plate without the gradient cables going to the RF-filter plate (which are usually installed).

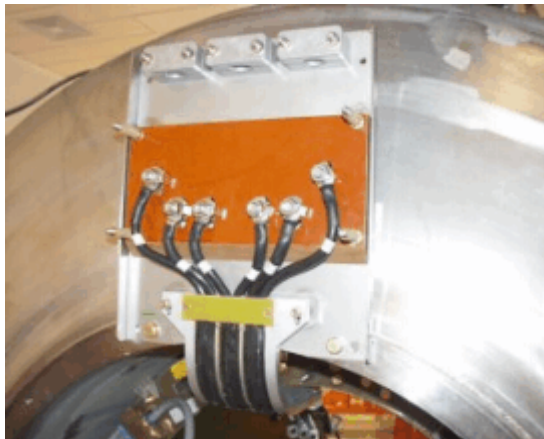
Fig. 151: NORD-LOCK washer



- (1) Cam side
(2) Ridged side

A set of NORD-Lock washers must be placed between the cable lug and nut when fixing the cables at the gradient coil connection. The correct order of the washers (→ Fig. 151 Page 147) is important for a proper securing function.

Fig. 152: Connection adapter



- Install the connection adapter to the gradient cable connection plate.
- Connect the cables to the gradient coil.

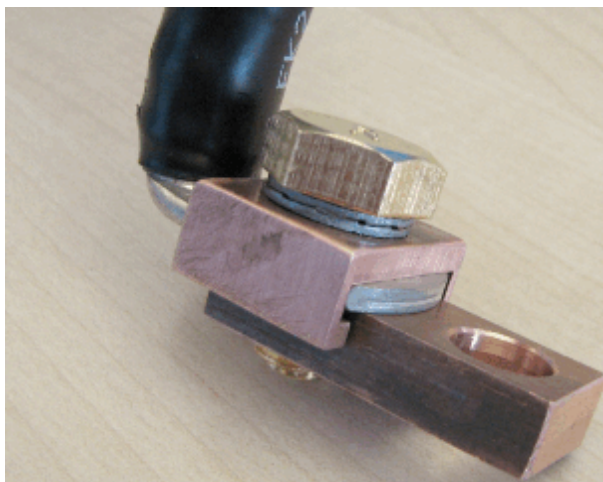
Note: The left-hand photograph shows the connection plate without the gradient cables going to the RF-filter plate (which are usually installed).



Use a torque wrench for tightening the gradient cables.
The torque is 22 ± 1 Nm.

- Prepare the gradient cable kit by assembling the copper connectors as shown in

Fig. 153: Correct copper connector arrangement



- Refer to for the correct order of parts for installation.

Fig. 154: Gradient cables assembled with copper connectors



- Connect the cables to the gradient coil. Watch out for correct polarity and orientation as shown in (→ Fig. 156 Page 148).

Fig. 155: Torque

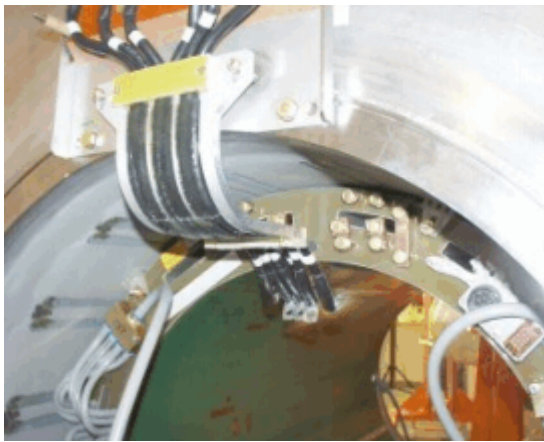
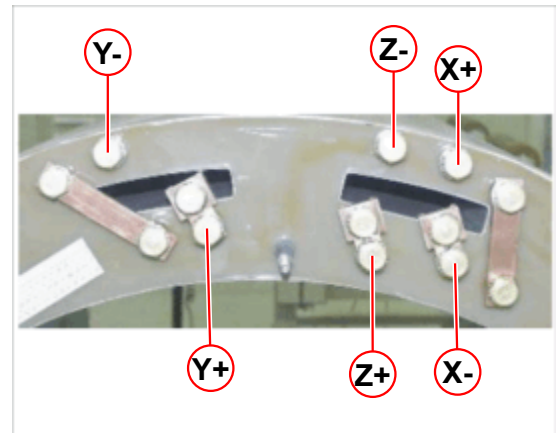
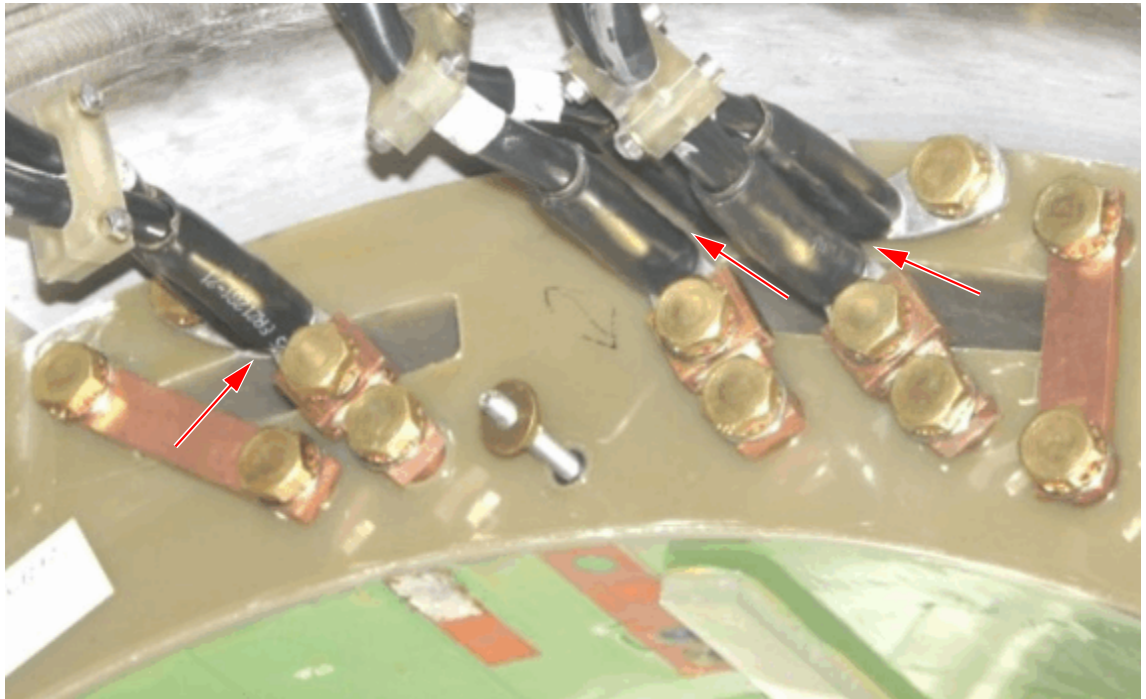


Fig. 156: Coil connection



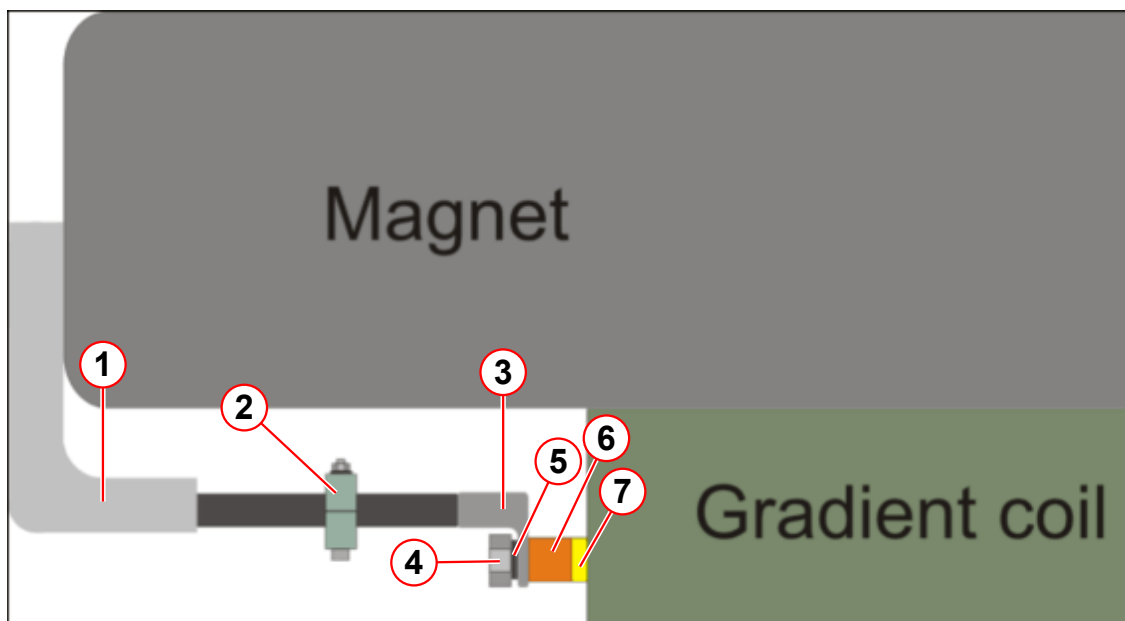
Route the cables so that the lugs are even with the copper retaining bolts at the coil in order to achieve a good level of contact. The leads should be routed in parallel to each other (→ Fig. 157 Page 149).

Fig. 157: Cables routed correctly



If strain is asserted on the cable when trying to fix it at the gradient coil insert, the 10 mm² copper bush as shown in (→ 6/ Fig. 158 Page 150) and (→ Fig. 160 Page 150) can be used in between.

Fig. 158: Overview of gradient cable arrangement



- (1) Gradient cable kit with mounting plate
- (2) Cable clamp 50 mm²
- (3) 90° cable shoe
- (4) Screw M10x20 or M10x30
- (5) Nordlock washer M10
- (6) 10 mm copper bush (optional)
- (7) M10 gradient coil insert

Fig. 159: Standard coil connection configuration

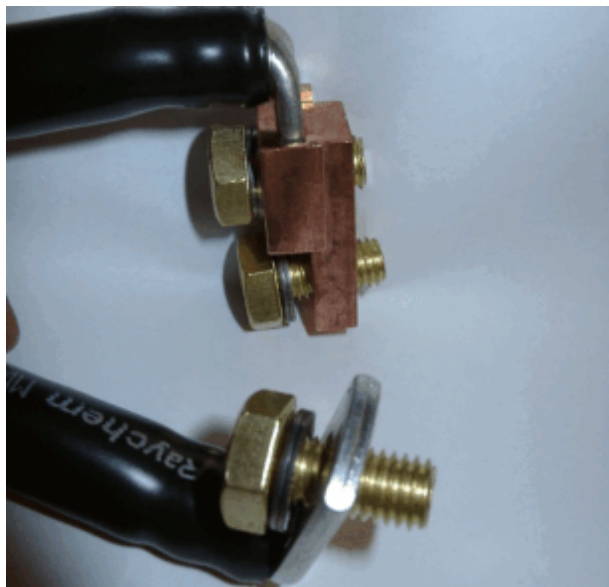
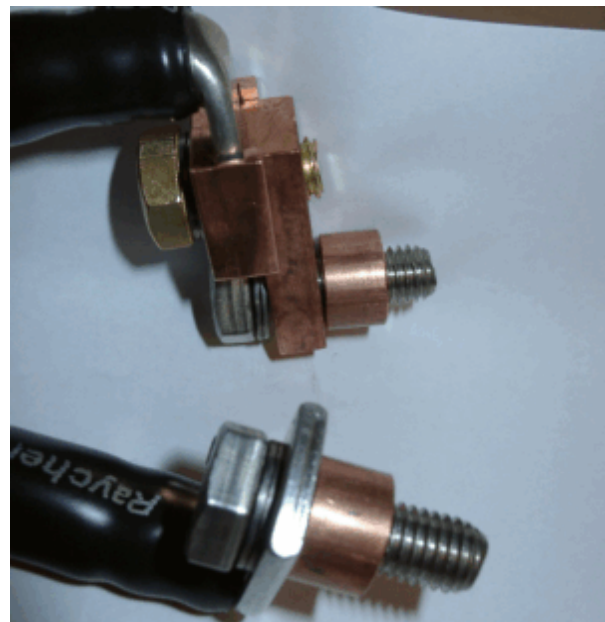


Fig. 160: Extended coil connection configuration with copper bushes (only if required)





The gradient coil connection has to be established with a non-magnetic torque wrench. The torque is 22 ± 1 Nm.



Ensure that the cable lugs do not come into contact while tightening the M10 x 20 hex brass screws.



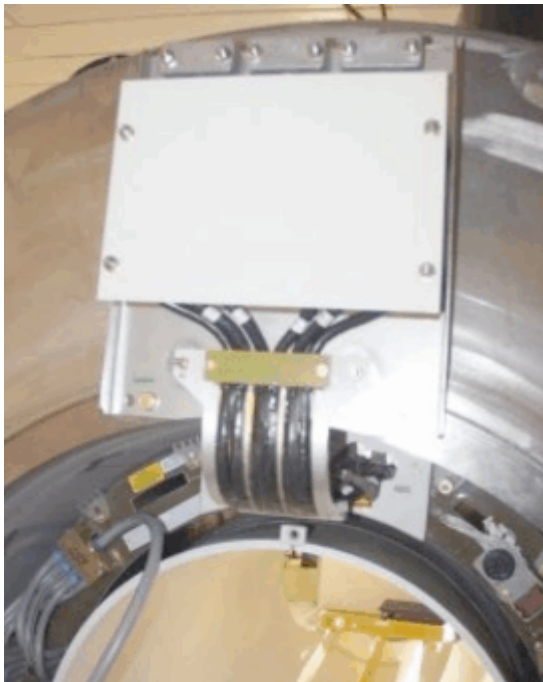
Newly delivered NORD-Lock washers are to be used on all M10x20 screws at the gradient coil (9 in total).

Fig. 161: Cover



- Install the cover for the cable connection at the gradient coil.

Fig. 162: Cover



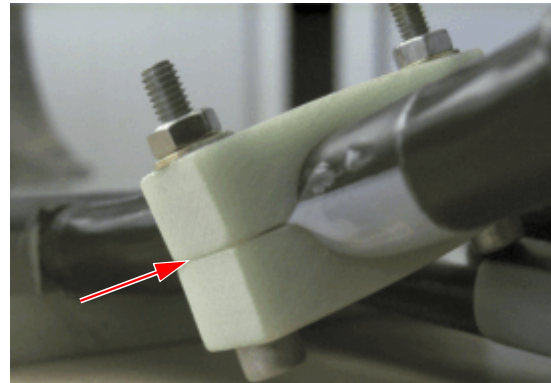
- Install the cover at the connection plate after connecting the gradient cables.

- Install the cable clamps (→ Fig. 163 Page 152), secure them with Loctite and tighten them to **5 Nm** with a non-magnetic torque wrench.
 - » There should be no clearance between the 2 clamp halves (→ Fig. 164 Page 152).

Fig. 163: Cable fixing



Fig. 164: Clamp tightened properly



To prevent damage to the gradient coil, due to broken cable lugs these cable fixings must be installed.



No cable ties are permitted

4.3.4 Final Work Steps

- Reinstall the magnet covers in reverse order.
- Set the circuit breakers for GPA in the CCA to ON.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Spike Check
- Image Orientation

4.4 Gradient Cables (Trio)

4.4.1 General



This procedure is only valid for Trio systems.

4.4.1.1 Contents of the Cable Repair Kit

Fig. 165: Gradient cable



The replacement kit 7367910 contains the parts listed in the table below.

Tab. 10 Contents of the set of gradient cables 089/Son 7367910

Item	Material number	Quantity	Description
1	n.a.	1	Gradient cable kit Trio
2	8119724	3	Copper bush
3	n.a.	1	Heat shrink
4	3432911	3	Screw M10x20
5	n.a.	3	Screw M10x80
6	3519295	6	Washer A10.5
7	3418050	6	Star washer A10.5
8	n.a.	3	Screw M12x20
9	n.a.	3	Washer A13
10	n.a.	3	Star washer A13

4.4.1.2 Tools and additional materials required

- Non-magnetic standard tool set (7757706) including a 32 mm open-ended wrench

- Non-magnetic torque wrench ranging from 20 - 50 Nm (8395803)
- Non-magnetic socket assortment (8396116)

4.4.2 Safety

! WARNING

High voltage!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » Switch off the gradient cabinet K2217: The cutter switch F3/3*80A in the CCA mains box or mains box 2 and F14/CCA must be switched off.

! WARNING

High voltage is still present on the final stage output terminals!!!

Contact with voltage conducting parts will lead to death or serious physical injury.

- » After switching off the gradient cabinet, wait at least 20 minutes before working on the gradient cables.



Protect the circuit breakers against unintentional switch-on, e.g. by locking the cabinet.

4.4.3 Procedure

- Unscrew and remove the old/defect cable from the gradient coil and from the connection plate.
- Clean all contact surfaces including the copper bush (-es) metallic bright, using Scotch-Brite.

Fig. 166: Cleaning the copper bush



Fig. 167: Cleaning the gradient coil insert surface



Fig. 168: Cleaning the copper lug



Fig. 169: Provisional fixing of the gradient cables at the connection plate

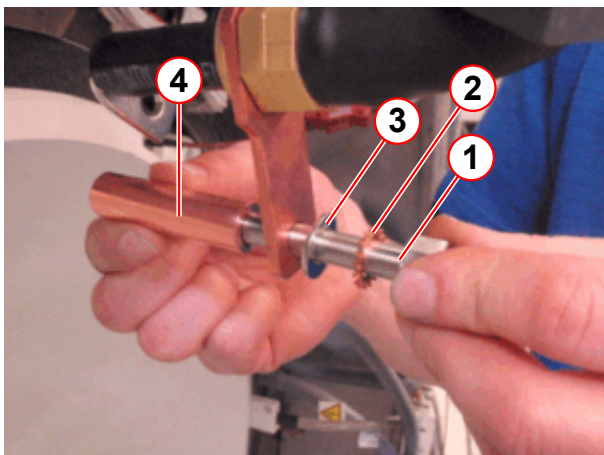


- Provisionally fix the gradient cables to the connection plate, leave the bolts loose and screw on the nuts rotating them a few times.

The left-hand photograph shows the connection plate without the gradient cables going to the RF-filter plate (which are usually installed).

- Prepare the gradient cable and bush using the fixing elements (→ Fig. 170 Page 155).

Fig. 170: Correct arrangement of fixing elements



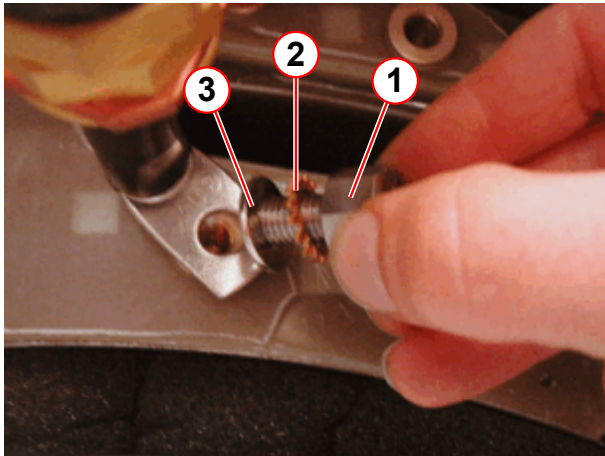
- (1) Screw M10x80
- (2) Star washer A10.5
- (3) Washer A10.5
- (4) Copper bush

Fig. 171: Mounting the gradient cable to the coil with the bush



- Fix the gradient cable bush screw to the coil insert - do not pull it tight yet.

Fig. 172: Mounting the 90° cable shoe to the gradient coil



- (1) Screw M10x20
- (2) Star washer A10.5
- (3) Washer A10.5

- Screw the M10x20 bolt into the 90° gradient cable shoe, do not pull it tight at this point.

- Align the gradient cable (-s) in order to tighten it (them) without strain.



Use a non-magnetic torque wrench for tightening the gradient cables at the gradient coil (→ Fig. 173 Page 156), (→ Fig. 174 Page 156) and at the joint (→ Fig. 175 Page 157).

The torque is 22 ± 1 Nm..

Fig. 173: Tightening the 90° gradient coil cable shoe (torque: 22 +/- 1 Nm)



Fig. 174: Tightening the copper bush to the gradient coil (torque: 22 +/- 1 Nm)



Fig. 175: Tightening the cable joint (torque: 22 +/- 1 Nm)

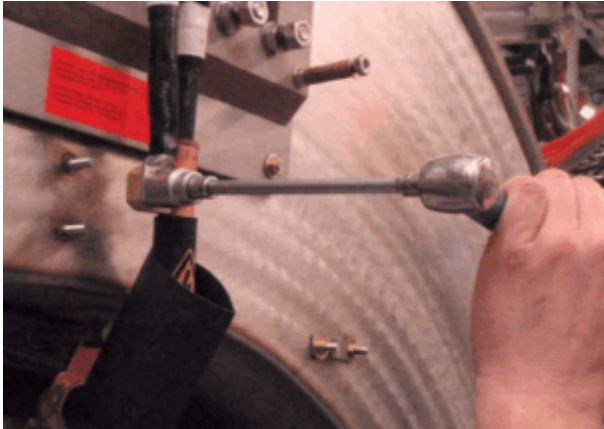
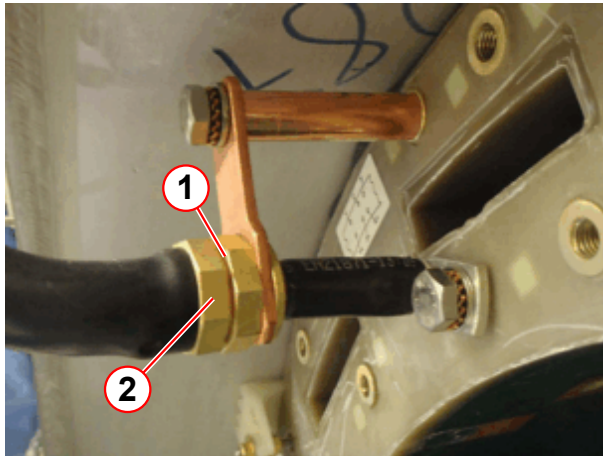


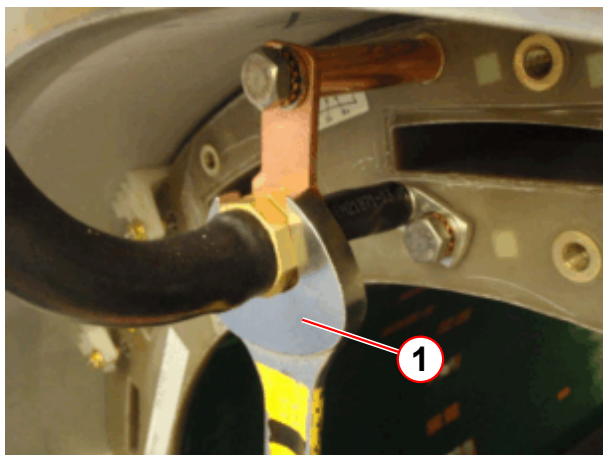
Fig. 176: Releasing the rear SW32 nut



- (1) Gap
- (2) Rear nut SW32

- Release the rear nut (SW32 (→ 2/Fig. 176 Page 157)) by a number of turns in order to gain a gap between both nuts.

Fig. 177: Tightening the front nut (SW32)



- (1) Non-magnetic open-ended wrench SW32

- Pull the front nut (SW32) tight, using a non-magnetic open-ended SW32 wrench (→ Fig. 177 Page 157).

Fig. 178: Securing the nuts and thread with Loctite 221



- Place some drops of Loctite 221 into the gap between both nuts (→ Fig. 178 Page 158).

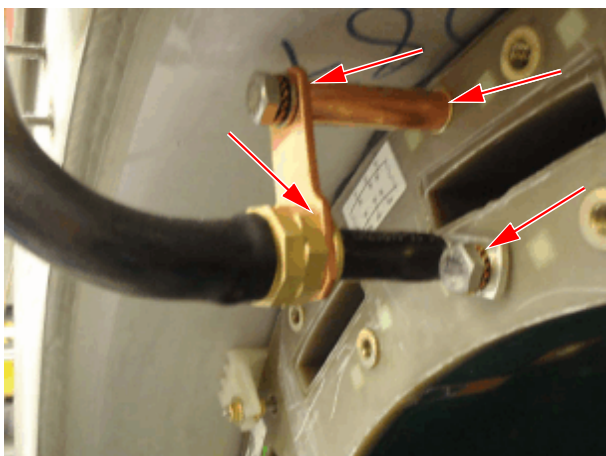
Fig. 179: Tightening the rear nut SW32



- Pull the rear nut (SW32) tight, using a non-magnetic open-ended SW32 wrench (→ Fig. 179 Page 158).

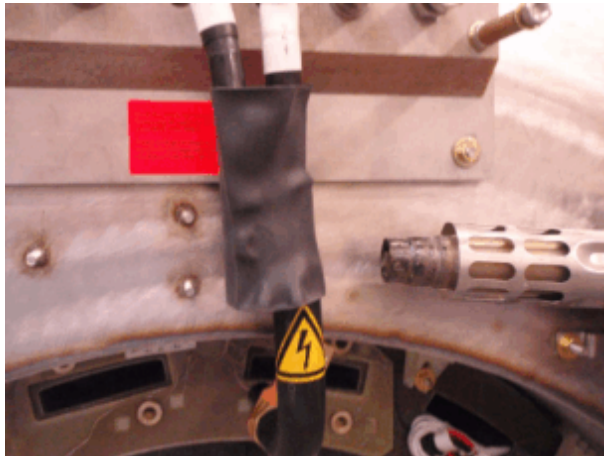
Note: There must not be a gap between the nuts.

Fig. 180: Checking for proper alignment and mounting



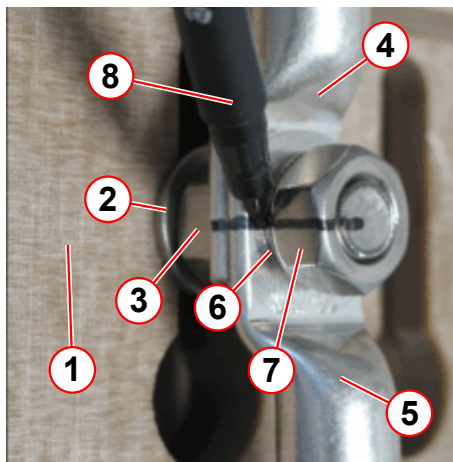
- Check all contact surfaces for proper alignment and mounting (→ Fig. 180 Page 158).
 - » There must not be a gap between the contacts.
 - » The cable connection on the gradient coil side is complete.

Fig. 181: Heat shrinking the gradient cable joint



- Move the heat shrink tube above the joint and shrink it using a heat gun (→ Fig. 181 Page 159).

Fig. 182: Gradient connection plate



- (1) Connection plate
- (2) Washer A13
- (3) Nut SW19
- (4) Upper cable lug (gradient cable to filter plate)
- (5) Lower cable lug (gradient cable to gradient coil)
- (6) Star washer A13
- (7) Nut SW19
- (8) Waterproof pen

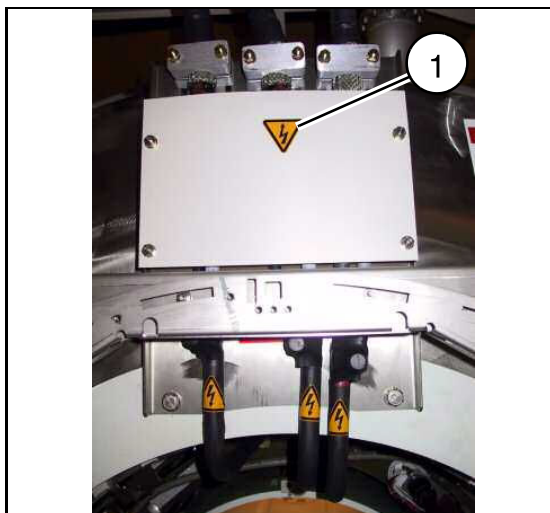
- Tighten the lower nut (→ 3/ Fig. 182 Page 159) with an open-ended wrench prior to tightening the upper nut (→ 7/ Fig. 182 Page 159).
- Arrange the fixing elements and the cable lugs as shown in (→ Fig. 182 Page 159)
- Tighten all nuts on the gradient connection plate to 35 ± 2 Nm using a non-magnetic torque wrench.
- Repeat the checks and seek confirmation of the torque and polarity from a second person.
Check the torque 35 ± 2 Nm
- Mark the connections with a waterproof pen (→ 8/ Fig. 182 Page 159).

Fig. 183: Gradient connection plate



- The left-hand photograph shows the cables finally installed at the connection plate.

Fig. 184: Cover



- Install the cover at the connection plate after connecting the gradient cables.

4.4.4 Final Work Steps

- Reinstall the magnet covers.
- Set the circuit breakers for GPA in the CCA to ON.

Perform Quality Assurance:

- Coil Check (SN DIP)
- Spike Check
- Image Orientation

Version 04

■ Chapter 1

Gradient Coil Components (→ Gradient Coil Components / Page 16) - time needed for replacing the gradient cables has been corrected.

Version 05

■ All chapters - editorial changes.

Version 06

■ Chapter 3

Current Sensor Assembly (→ Final Steps / Page 105): SENS Offset measurement/adjustment, added.

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